



# **IoT-Based Smart Home**

This project is Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Computer Science at the State University of Bangladesh

## **Submitted by**

Md. Rafiul Alam Durjoy

*UG02-47-18-041*

Md. Evan Khan Emon

*UG02-47-18-009*

Md. Mahmudul Islam

*UG02-47-18-004*

Mushiur Rahman Any

*UG02-47-18-019*

## **Supervised By**

Muhammad Masud Tarek

Associate Professor & Head

Department of Computer Science and Engineering

## **Co-Supervised By**

Muntasir Hasan Kanchan

Lecturer

Department of Computer Science and Engineering

**August 2022**

Course Code: CSE-0400

Course Title: Project Works

Project Title: IoT-Based Smart Home

Student IDs: UG02-47-18-009, UG02-47-18-004, UG02-47-18-041, UG02-47-18-019

| Knowledge Profile, Engineering Problems and Engineering Activities |                                                                   |                                                                                              | Remarks                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Knowledge Profile                                                  | <input checked="" type="checkbox"/> K1 (Natural Sciences)         | <input checked="" type="checkbox"/> K5 (Engineering Design)                                  | To get a beautiful overview of our project we do engineering design and practice as we design & implement the whole project personally. We research a lot about our project that goes under Natural Science and Engineering Fundamental. We do Mathematics to calculate the distance in the IR sensor for the parking system. |
|                                                                    | <input checked="" type="checkbox"/> K2 (Mathematics)              | <input checked="" type="checkbox"/> K6 (Engineering Practice)                                |                                                                                                                                                                                                                                                                                                                               |
|                                                                    | <input checked="" type="checkbox"/> K3 (Engineering Fundamentals) | <input type="checkbox"/> K7 (Comprehension)                                                  |                                                                                                                                                                                                                                                                                                                               |
|                                                                    | <input checked="" type="checkbox"/> K4 (Specialist Knowledge)     | <input type="checkbox"/> K8 (Research Literature)                                            |                                                                                                                                                                                                                                                                                                                               |
| Complex Engineering Problems                                       | <input checked="" type="checkbox"/> P1 (Depth of knowledge)       | <input type="checkbox"/> P5 (Extent of applicable codes)                                     | We had to do depth analysis because of the Interdependence between RFID and servo motor or flame sensor & starting buzzer sound, which are so tricky to handle.                                                                                                                                                               |
|                                                                    | <input type="checkbox"/> P2 Range of conflicting requirements)    | <input type="checkbox"/> P6 (Extent of stakeholder involvement and conflicting requirements) |                                                                                                                                                                                                                                                                                                                               |
|                                                                    | <input checked="" type="checkbox"/> P3 (Depth of analysis)        |                                                                                              |                                                                                                                                                                                                                                                                                                                               |
|                                                                    | <input checked="" type="checkbox"/> P4 (Familiarity of issues)    | <input checked="" type="checkbox"/> P7 (Interdependence)                                     |                                                                                                                                                                                                                                                                                                                               |
| Complex Engineering Activities                                     | <input checked="" type="checkbox"/> A1 (Range of resources)       | <input checked="" type="checkbox"/> A4 (Consequences to society and the environment)         | We collect the instrument from the local market and online Platform. We had to deal with the technical issue for implementing & testing our project. It can help society by reduction of mass work.                                                                                                                           |
|                                                                    | <input type="checkbox"/> A2 (Level of interactions)               |                                                                                              |                                                                                                                                                                                                                                                                                                                               |
|                                                                    | <input checked="" type="checkbox"/> A3 (Innovation)               | <input checked="" type="checkbox"/> A5 (Range of resources)                                  |                                                                                                                                                                                                                                                                                                                               |

# DECLARATION

We, therefore, declare that the work given in this project is the result of our study conducted under the direction of **Muhammad Masud Tarek**, Associate Professor & Head, Department of Computer Science and Engineering, State University of Bangladesh. We also certify that no part of this project or any of its components has been or will be submitted to another institution for the award of a degree or diploma.

## Countersigned By

## Candidate Signature

.....

.....

**Muhammad Masud Tarek**  
Associate Professor & Head  
Department of Computer Science and  
Engineering  
State University of Bangladesh  
Supervisor

**Md. Evan Khan Emon**

.....

**Md. Mahmudul Islam**

.....

.....

**Muntasir Hasan Kanchan**  
Lecturer  
Department of computer science and  
Engineering  
State University of Bangladesh  
Co-Supervisor

**Md. Rafiul Alam Durjoy**

.....

**Mushiur Rahman Any**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING STATE  
UNIVERSITY OF BANGLADESH**



## **CERTIFICATE OF APPROVAL**

The foregoing project is hereby accepted as a credible study of an engineering subject carried out and presented in a manner satisfactory to warrant its acceptance as a prerequisite to the degree for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein, but approve the thesis only for the purpose for which it is submitted.

### **Board of Examiners**

---

Muhammad Masud Tarek  
Associate Professor (Acting Head)  
Department of Computer Science and Engineering  
State University of Bangladesh

---

Muntasir Hasan Kanchan  
Lecturer  
Department of Computer Science and Engineering  
State University of Bangladesh

# ACKNOWLEDGEMENT

First of all, we would like to thank the almighty ALLAH. Today we are successful in completing our work with such ease because ALLAH gave us the ability, chance, and cooperating supervisor. We would like to take the opportunity to express our gratitude to **Muhammad Masud Tarek**, Associate Professor & Head, Department of Computer Science and Engineering, State University of Bangladesh, our respected supervisor and **Muntasir Hasan Kanchan** our respected co-supervisor.

Although they were always loaded with several other activities, they gave us more than enough time for this work. The door to their office was always open whenever we ran into a trouble spot or had a question about our research or writing. They not only gave us time but also proper guidance and valuable advice whenever we faced some difficulties. Their comments and guidance helped us in preparing our thesis report.

We would like to thank our respected teachers and friends who inspired us in every step and helped us in a number of ways by providing various resources and moral support.

Finally, we must express our very profound gratitude to our parents for providing us with unfailing support and continuous encouragement throughout the years of study and throughout the process of researching and writing the thesis. This accomplishment would not have been possible without them.

# ABSTRACT

Our country's technology is still far behind all the countries of the world. If we look at developed countries, we can see how advanced they are in terms of technology. The advancement of technology has significantly influenced daily life, offering solutions to enhance convenience, security, and efficiency. This project explores the potential of an **IoT-Based Smart Home**, integrating Internet of Things (IoT) technology to address common challenges in home automation, security, and parking management.

The project introduces several features, including automated control of electrical appliances through a mobile app, RFID-based door access for enhanced security, and a smart parking system that monitors slot availability and controls gate operations. These innovations aim to reduce energy wastage, improve safety, and address urban challenges such as traffic congestion caused by unmanaged parking.

The system leverages components like Arduino Uno, NodeMCU, RFID modules, sensors, and IoT cloud platforms for seamless operation. Users can remotely control lights, fans, and air conditioners, monitor room temperature, and receive alerts in case of fire. The parking system efficiently manages entry, exit, and slot availability, ensuring a user-friendly experience.

This project demonstrates the potential of IoT in creating cost-effective, scalable solutions for modern living. It highlights how such systems can improve quality of life by offering greater convenience, enhancing security, and promoting energy efficiency. The findings pave the way for future enhancements, including integrating advanced security measures, expanding automation features, and exploring renewable energy applications.

# Table of Contents

|                                                                                    |    |
|------------------------------------------------------------------------------------|----|
| ABSTRACT.....                                                                      | vi |
| Chapter 1: Introduction .....                                                      | 1  |
| 1.1 Motivation .....                                                               | 2  |
| 1.2 Objective .....                                                                | 2  |
| 1.3 Feature .....                                                                  | 3  |
| 1.4 Problem Statement .....                                                        | 3  |
| 1.5 Social Impact.....                                                             | 5  |
| 1.6 Ethics Behind This Project.....                                                | 6  |
| 1.7 Life-Long Learning .....                                                       | 6  |
| 1.8 Team-member Contribution to This Project .....                                 | 7  |
| 1.9 Project Management.....                                                        | 7  |
| 1.10 Report Layout.....                                                            | 8  |
| Chapter 2: Literature Review and Background Study .....                            | 10 |
| 2.1 Introduction .....                                                             | 11 |
| 2.2 Background Study .....                                                         | 11 |
| 2.3 Previous Work.....                                                             | 12 |
| 2.3.1 Low-Cost Smart Parking System for Smart Cities.....                          | 12 |
| 2.3.2 IoT Based Smart Parking System for Large Parking Lot.....                    | 13 |
| 2.3.3 Automatic Access Control prototype using Arduino and RFID.....               | 14 |
| 2.3.4 A Low-Cost Entry Door Using Database Based On RFID And Microcontroller. .... | 15 |
| 2.3.5 An IoT based Home Automation Using Android Application.....                  | 16 |
| 2.3.6 Wireless Home Automation System Using IoT and PaaS .....                     | 17 |
| Chapter 3: Methodology and Requirement Analysis.....                               | 19 |
| 3.1 Introduction .....                                                             | 20 |
| 3.2 Project Flow .....                                                             | 20 |
| 3.3 Arduino IoT Could .....                                                        | 21 |
| 3.3.1 What is Arduino IoT Cloud? .....                                             | 21 |
| 3.3.2 How does IoT connect to Arduino?.....                                        | 21 |
| 3.3.3 How to find Arduino IOT cloud? .....                                         | 21 |

|                                                        |    |
|--------------------------------------------------------|----|
| 3.3.4 How to create an account in the IoT cloud? ..... | 22 |
| 3.3.5 How to create things in the IoT cloud .....      | 25 |
| 3.4 Arduino Setup .....                                | 38 |
| 3.4.1 Arduino IDE .....                                | 38 |
| 3.4.2 Arduino Code Example .....                       | 38 |
| 3.4.3 How to Add Library in Arduino IDE .....          | 39 |
| 3.4.4 How to Compile Sketch.....                       | 42 |
| 3.5 Pictorial Explanation of all Requirements .....    | 44 |
| 3.5.1 ARDUINO UNO .....                                | 44 |
| 3.5.2 NODE MCU ESP8266 Wi-Fi Module .....              | 46 |
| 3.5.3 4-Channel Relay Module (5v) .....                | 48 |
| 3.5.4 IR Sensor .....                                  | 53 |
| 3.5.5 16*2 LCD Display.....                            | 56 |
| 3.5.6 Servo Motor.....                                 | 59 |
| 3.5.7 RC-522 RFID .....                                | 61 |
| 3.5.8 Flame Sensor .....                               | 63 |
| 3.5.9 Transformer .....                                | 64 |
| 3.5.10 LM7805 IC – Power Supply.....                   | 66 |
| 3.5.11 Breadboard.....                                 | 68 |
| 3.5.12 Breadboard Wires .....                          | 68 |
| 3.5.13 AC-DC 12V 3A Arduino Power Adapter.....         | 69 |
| 3.5.14 NTE960 5V Positive Voltage Regulator .....      | 70 |
| Chapter 4: System Design & Implementation .....        | 72 |
| 4.1 Circuit Diagram.....                               | 73 |
| 4.2 Block Diagram .....                                | 73 |
| 4.3 Flow Chart.....                                    | 74 |
| 4.3.1 Flow Chart for Car Parking .....                 | 74 |
| 4.4 Implementation.....                                | 76 |
| 4.4.1 Circuit board .....                              | 76 |
| 4.4.2 Parking System .....                             | 77 |
| 4.4.3 RFID Door Control.....                           | 80 |
| 4.4.4 Home Automation System.....                      | 83 |

|                                             |    |
|---------------------------------------------|----|
| Chapter 5: Result.....                      | 87 |
| 5.1 Result of the Project .....             | 88 |
| Chapter 6: Conclusion and Future Work ..... | 90 |
| 6.1 Conclusion.....                         | 91 |
| 6.2 Future Work .....                       | 91 |
| 6.3 Closing Remarks.....                    | 93 |
| References .....                            | 94 |

# List of Figures

|                                                                                        |    |
|----------------------------------------------------------------------------------------|----|
| Figure 1: Smart car parking System.....                                                | 3  |
| Figure 2: Opening door using RFID technology .....                                     | 4  |
| Figure 3: Wireless home automation system .....                                        | 5  |
| Figure 4: Gantt-chart.....                                                             | 8  |
| Figure 5: Low Cost Smart Parking System for Smart Cities .....                         | 12 |
| Figure 6: IoT Based Smart Parking System for Large Parking Lot.....                    | 13 |
| Figure 7: Automatic Access Control prototype using Arduino and RFID .....              | 14 |
| Figure 8: A Low-Cost Entry Door Using Database Based On RFID And Microcontroller ..... | 15 |
| Figure 9: An IoT based Home Automation Using Android Application.....                  | 16 |
| Figure 10: Wireless Home Automation System Using IoT and PaaS .....                    | 17 |
| Figure 11: Project Flow .....                                                          | 20 |
| Figure 12: Searching Arduino IoT Cloud .....                                           | 21 |
| Figure 13: Login Page.....                                                             | 22 |
| Figure 14: Create a new account (adding date of birth).....                            | 23 |
| Figure 15: Create a new account (adding other information) .....                       | 24 |
| Figure 16: New account is created.....                                                 | 25 |
| Figure 17: Select the IoT cloud for marking things.....                                | 25 |
| Figure 18: Things creating web page.....                                               | 26 |
| Figure 19: Set up for the new thing .....                                              | 26 |
| Figure 20: Add variable form .....                                                     | 27 |
| Figure 21: Variable name and type.....                                                 | 27 |
| Figure 22: Variable permission.....                                                    | 28 |
| Figure 23: Variable Update policy.....                                                 | 28 |
| Figure 24: Final overview of the Things.....                                           | 29 |
| Figure 25: Dashboard Design .....                                                      | 30 |
| Figure 26: Blank Dashboard .....                                                       | 30 |
| Figure 27: Blank Dashboard .....                                                       | 31 |
| Figure 28: Give a new name for the switch .....                                        | 31 |
| Figure 29: Linked variable with the Things (1) .....                                   | 32 |
| Figure 30: Linked variable with the Things (2).....                                    | 32 |
| Figure 31: Switch is Created.....                                                      | 33 |
| Figure 32: Final overview of the dashboard .....                                       | 33 |
| Figure 33: Sketchbook of our Home Automation.....                                      | 34 |
| Figure 34: Board and port selection.....                                               | 35 |
| Figure 35: Uploading the code.....                                                     | 36 |
| Figure 36: ReadMe File .....                                                           | 36 |
| Figure 37: Code generated by Arduino IoT Cloud .....                                   | 37 |
| Figure 38: Secret SSID and Password .....                                              | 37 |
| Figure 39: Installation of Arduino IDE.....                                            | 38 |
| Figure 40: Arduino Code Example .....                                                  | 39 |
| Figure 41: How to Add Library in Arduino IDE (1) .....                                 | 39 |

|                                                           |    |
|-----------------------------------------------------------|----|
| Figure 42: How to Add Library in Arduino IDE (2) .....    | 40 |
| Figure 43: Adding Liquid crystal I2C Library.....         | 40 |
| Figure 44: Adding mfrc522 Library .....                   | 41 |
| Figure 45: Adding one wire Library .....                  | 41 |
| Figure 46: Selecting Board for compile.....               | 42 |
| Figure 47: Selecting port for compile .....               | 42 |
| Figure 48: Verify the code .....                          | 43 |
| Figure 49: Upload the code.....                           | 43 |
| Figure 50: Done compiling .....                           | 44 |
| Figure 51: Arduino UNO .....                              | 45 |
| Figure 52: Node MCU .....                                 | 47 |
| Figure 53: 4 Channel Relay Module (5v) .....              | 48 |
| Figure 54: 4 Channel Relay Module Pin Configuration ..... | 50 |
| Figure 55: 4 Channel Relay Schematic.....                 | 52 |
| Figure 56: How to use 4 Channel Relay .....               | 53 |
| Figure 57: IR Sensor Module.....                          | 54 |
| Figure 58: IR Transmitter or IR LED .....                 | 54 |
| Figure 59: IR Receiver or Photodiode .....                | 55 |
| Figure 60: IR Sensor Detection Technique.....             | 55 |
| Figure 61: 16*2 LCD Display.....                          | 56 |
| Figure 62: I2C driver Pinout. ....                        | 58 |
| Figure 63: Servo Motor.....                               | 59 |
| Figure 64: Pulse Diagram .....                            | 60 |
| Figure 65: Servo Motor pinout. ....                       | 61 |
| Figure 66: RC-522 RFID .....                              | 61 |
| Figure 67: RC-522 RFID pinout.....                        | 62 |
| Figure 68: Flame Sensor .....                             | 64 |
| Figure 69: Transformer .....                              | 65 |
| Figure 70: Resultant Output Waveform.....                 | 65 |
| Figure 71: LM7805 IC .....                                | 66 |
| Figure 72: Circuit Diagram of Amplifier.....              | 67 |
| Figure 73: Breadboard .....                               | 68 |
| Figure 74: Jumper Wire. ....                              | 68 |
| Figure 75: 12v Arduino Power Adapter.....                 | 69 |
| Figure 76: 5v Voltage Regulator.....                      | 70 |
| Figure 77: 5v Voltage Regulator pinout. ....              | 70 |
| Figure 78: Circuit Diagram .....                          | 73 |
| Figure 79: Block Diagram .....                            | 74 |
| Figure 80: Flow Chart for Car Parking .....               | 75 |
| Figure 81: Flow Chart for RFID card .....                 | 76 |
| Figure 82: Circuit Board .....                            | 77 |
| Figure 83: Circuit Diagram for Parking System.....        | 78 |
| Figure 84: Car Parking System. ....                       | 80 |

|                                                |    |
|------------------------------------------------|----|
| Figure 85: Circuit Diagram for RFID .....      | 80 |
| Figure 86: RFID card Working.....              | 82 |
| Figure 87: Home Automation Diagram .....       | 83 |
| Figure 88: Home automation .....               | 85 |
| Figure 89: Temperature Details .....           | 85 |
| Figure 90: Flame Sensor and Buzzer .....       | 86 |
| Figure 91: Full System Set up Picture (1)..... | 88 |
| Figure 92: Full System Set up Picture (2)..... | 89 |

## List of Tables

|                                                     |    |
|-----------------------------------------------------|----|
| Table 1: Distributed Tasks .....                    | 7  |
| Table 2: Financial Report.....                      | 7  |
| Table 3: Arduino UNO Specifications.....            | 44 |
| Table 4: Arduino Uno Pinout.....                    | 46 |
| Table 5: 4 Channel Relay Module Pinout.....         | 51 |
| Table 6: Pin Configuration and Description .....    | 54 |
| Table 7: Technical Specifications .....             | 62 |
| Table 8: Pinout Details for Voltage Regulator ..... | 71 |

# **Chapter 1: Introduction**

## 1.1 Motivation

Several problems are increasing day by day in our country, parking problems and security problems are one of them. Our main reason for doing this project is to figure out how to get out of these problems and work to make people's lives easier and more comfortable. We have done this project in several parts. For example, we can create a neat and beautiful parking system anywhere in our home or office. And we can easily open the parking gate using some sensor and also monitor the empty slots for parking our car. For security, we use an RFID card to open the main gate of the use we can also use it in every single room. Also, we used home automation in this project. Home automation is a system through which we can control many parts of our daily activities through the Internet. For example, we can control all the electrical connections in our house through a mobile app which is a wireless system. And this we can control anywhere in the house in any condition. With this, we can get messages about room temperature. If there is a fire in any part of the house, the alarm will sound. Our main aim in doing this project is to solve some problems and introduce people to the Internet of Things. So that people are properly aware of the Internet of Things and willing to use it.

## 1.2 Objective

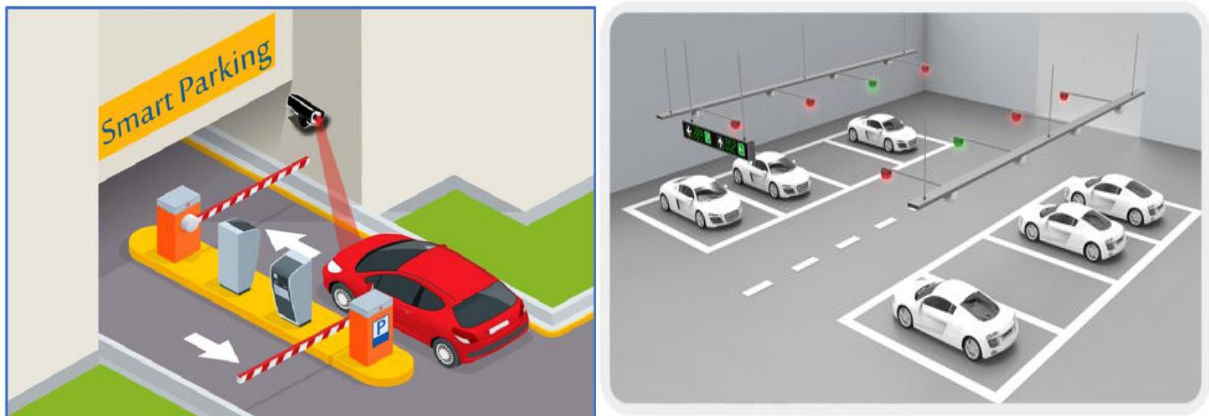
Every action has a purpose. Our main objective of this project is to solve some problems of society and give people some idea about the Internet of Things. Presently the parking system is at a very deplorable level. For example, a car owner parks his car on the side of the road, which often leads to traffic jams. The main goal of our project is to solve this problem, in this project we will be able to monitor car parking. If we want, we can pay for the parking. Another main goal of ours is that using the Internet of Things we will be able to control the light fan AC or any 220-volt suite through apps. This will reduce the wastage of both our time and electricity. We have used the Internet of Things to solve this problem many times we forget to turn off the light fan after leaving the house. It is a completely wireless connection and works as a wifi module. Besides, we have worked on how to understand room temperature very easily. If there is a fire somewhere in the house, we will know about it at the right time. Finally, it can be said that the main goal or objective of our project is to work on how to make people's life beautiful and interesting and to give people an idea about technology.

## 1.3 Feature

- Save time and money while providing prompt service.
- To construct a wireless home automation system controlled by any smart device.
- To design and implement a cost-effective home automation system yet an efficient one.
- To construct a wireless parking gate opening or closing via mobile application.
- In terms of car parking, see which slot is empty or full through the display.
- Main door opening using an RFID card with the servo control system.
- Switching on and off room light fan using the internet of things through the mobile application.
- Room temperature is monitored through the display.
- Alarm system if there is any fire found in any room.

## 1.4 Problem Statement

In this project, we are going to solve some problems which play a very important role in our daily life. For example, when we come home after a busy day, we have to wait in front of the gate until the doorman opens the gate to enter the house, so that we don't have to wait, we have used an ir sensor which we can turn the gate on and off automatically. And at the same time, I can see which parking slot is vacant through the display in front of the gate. And accordingly, the car can be parked at a specific place.



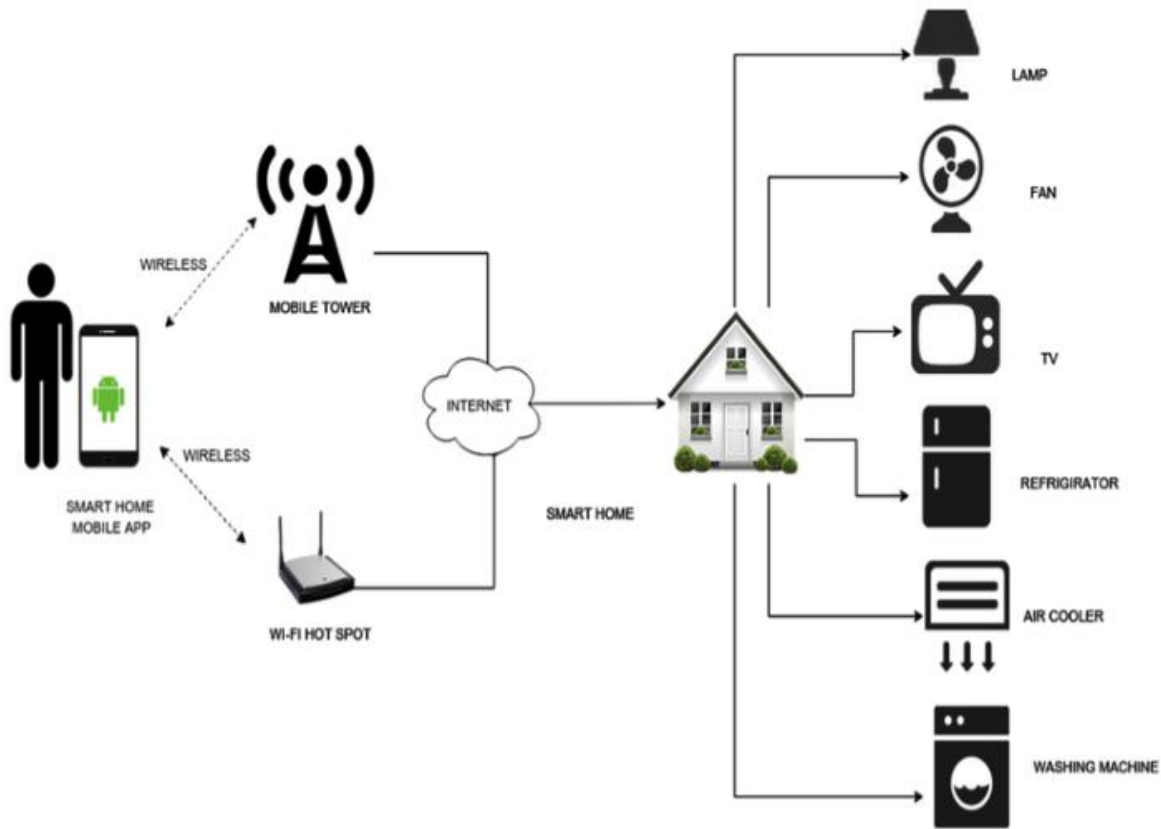
**Figure 1: Smart car parking System**

For security purposes we use an RFID card which is mainly used to open the main door of the house. Only authorized cards can enter the house. So, we don't need to carry the key. That's why we can get special security.



**Figure 2: Opening door using RFID technology**

We attempted to stand home automation accessible via mobile apps and motion detection, which controls the complete automation system in our work. In general, our designed system receives commands via mobile apps and in addition, detects motions using a sensor and transmits signals to the circuit. The ESP8266 Micro-controller processes the dual signals and regulates the appliances. The appliance is turned on if both incoming signals are high. Codes, on the other hand, are designed in such a way that if any of the signals is low, the device is turned off. We can also be notified about the fire.



**Figure 3: Wireless home automation system**

## 1.5 Social Impact

Home automation has made our lives easier day by day. This technology is now being used everywhere in the world so quickly. But it hasn't reached every home in our country yet. It will, though, in the future. Home automation has a big effect on how people interact with each other. We know that there are three big groups of people. All three of these generations are affected by this automation. Home automation devices are made to help the people who use them. We have social impact on this project. Through this project, we will be able to solve many problems in society among which we can eliminate the traffic congestion caused by parking in our society. Now we can monitor everything about parking. As a result of this, no car will be left unnecessarily on the road for parking causing the traffic jam. By doing this, the problem of society will be solved. Also through this project, we will be able to improve the security system as well as wirelessly control all the electrical loads in our house. If there is a sick person in the house, who cannot get out of bed, then he can control everything with the help of it. Everything has advantages and disadvantages it will make people's life easier and on the other hand, its use will make people lazy.

## 1.6 Ethics Behind This Project

There is a massive ethical concern in terms of home automation devices. This automation system should not harm anyone or should be helpful to the users. Every type of unethical issue is being deducted from this project. This project is built to help the users rather than harm them. Our project will help the users in the time of car parking. The gate of the parking can be controlled when the user wants. Also, the user can see the free parking slot from the display. This doesn't harm the users rather than it helps them. It helps the users a lot, it saves time. The user also can unlock the door using an RFID card. It will give the best security to the user. The user also can control the switching of any electric line via mobile application even if the user is not at home. It will also help the user all the time. The user can monitor the temperature and humidity from the display and get a fire alarm in case of fire from anywhere. Security and safety are the two most concerning ethical issues in IoT. These projects ensure these two sides are fairly dealt with. This project gives ultimate security and safety.

## 1.7 Life-Long Learning

When we start our project, we face a lot of problems, because we are new in this section. We only have a little knowledge about electric parts. Then we started to study this project. We know about a lot of electric device working modules. Implementing this project has been a challenge and we have overcome most of these challenges. Because of this, there is a lot of learning that has been done and all the members of this project are able to increase their knowledge.

The following new skills were learned through execution so far.

- Deep knowledge about new hardware devices.
- 220v Power supply and management system.
- Arduino IDE setup and how it works.
- Different Arduino IDE library setup.
- How to connect ESP8266 to Arduino IOT cloud.
- New software tools (Arduino IOT cloud and Arduino Editor).
- Project management skills.
- Time management skills.
- Teamwork.
- Crisis analysis.
- Taking risk and solving problem.

## 1.8 Team-member Contribution to This Project

**Table 1: Distributed Tasks**

| Task                                   | Emon | Ridoy | Durjoy | Any | Total |
|----------------------------------------|------|-------|--------|-----|-------|
| <b>Search &amp; acquire components</b> | 25%  | 25%   | 25%    | 25% | 100%  |
| <b>Design Subsystems</b>               | 30%  | 25%   | 25%    | 20% | 100%  |
| <b>Test Subsystems</b>                 | 30%  | 30%   | 25%    | 20% | 100%  |
| <b>Report writing</b>                  | 25%  | 25%   | 25%    | 25% | 100%  |
| <b>Presentation</b>                    | 30%  | 25%   | 25%    | 20% | 100%  |

## 1.9 Project Management

### 1.9.1 Financial Report

**Table 2: Financial Report**

| No | Description                             | Quantity | Unit Cost (taka) | Total Cost (taka) |
|----|-----------------------------------------|----------|------------------|-------------------|
| 1  | Arduino UNO                             | 3        | 1180             | 3540              |
| 2  | Arduino cable                           | 1        | 150              | 150               |
| 3  | ESP8266 Wi-Fi module                    | 1        | 500              | 500               |
| 4  | Servo motor mini                        | 4        | 195              | 780               |
| 5  | IR sensor                               | 4        | 100              | 400               |
| 6  | 16*2 LCD display with I2C driver        | 2        | 430              | 860               |
| 7  | MFRC 522                                | 1        | 230              | 230               |
| 8  | 4-way relay 5V                          | 1        | 315              | 315               |
| 9  | Flame sensor                            | 1        | 90               | 90                |
| 10 | DHT 11                                  | 1        | 180              | 180               |
| 11 | 5v Fan                                  | 2        | 40               | 80                |
| 12 | 12v LED light                           | 2        | 50               | 100               |
| 13 | Holder + Bulb                           | 1        | 140              | 140               |
| 14 | Jumper wire set + PCB board + Regulator | 1        | 190              | 190               |
| 15 | Glue guns + Glue stick                  | 1        | 479              | 479               |
| 16 | Soldering iron                          | 1        | 550              | 550               |

|    |                          |   |              |               |
|----|--------------------------|---|--------------|---------------|
| 17 | 12v power adapter 2A     | 1 | 370          | 370           |
| 18 | Power transformer 12v 3A | 1 | 345          | 550           |
| 19 | Transport cost + other   |   |              | 1000          |
|    |                          |   | <b>Total</b> | <b>10,504</b> |

## 1.9.2 Gantt-chart

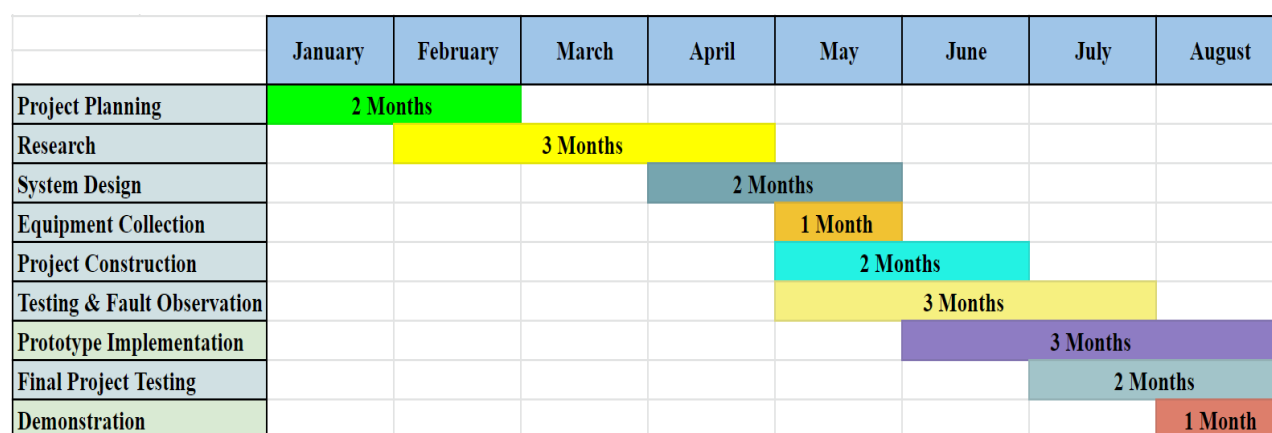


Figure 4: Gantt-chart

## 1.10 Report Layout

This dissertation is structured as follows:

### Chapter 1: Introduction

In this chapter, we have discussed the motivation, feature, problem statement, objective and social impact, Gantt Chart, financial report etc.

### Chapter 2: Literature Review and Background Study

Here we also discuss the background study and previous work with a short description.

### Chapter 3: Methodology and Requirement Analysis

Here we discuss the functional requirement and non-functional requirement which was used in our project.

### Chapter 4: System Design & Implementation

Here we discuss all types of diagrams, Flow-chart and project pictures.

### Chapter 5: Result

Here we discuss all types of analysis of output.

**Chapter 6:** Conclusion and Future work

Future development is discussed in this section which is more important.

## **Chapter 2: Literature Review and Background Study**

## 2.1 Introduction

To reach the goals, there is number of steps that must be taken. All the steps are very important. Both making software and putting it on hardware are parts of these processes.

Here's how to do these things given below:

- Using a microcontroller to set up a wireless network connection between the Android and the home automation system.
- Use ESP8266 and UNO as micro-controller to make a simple but reliable home automation system. This system will connect the Android to the home appliances.
- To find a good app that works well with the ESP8266 board so that the home appliances can be controlled.
- Write code for the UNO so that it can work properly. Also, write code for the ESP8266 board and create Arduino IOT could so that it can talk to the Android app.

## 2.2 Background Study

The average person might think that home automation is a relatively new idea. That is true if you want smart home solutions that are easy for consumers to use and don't cost a lot of money. But the technological progress that brought us here has been going on for a long time. Many technology historians say that in 1898 when Nikola Tesla made a remote control for a toy, this was the real start of easy-to-use consumer automation. As promising as this was, it would be several decades before electrical appliances were common in homes, and even longer before technology could really deliver on the promise of a futuristic home with these appliances that could be controlled remotely. The Home of the Future could be seen at the 1933 A Century of Progress International Exposition, also known as the "Chicago World's Fair." It was made to look like something from science fiction, but the inside didn't live up to the promise because the technology wasn't there yet. After the electric digital computer was made in 1940, computer technology really took off in the 1940s and 1960s. Jim Sutherland, an engineer at Westinghouse, made the ECHO IV in 1966. It was the first real home automation device because it could control temperature and appliances and store and later retrieve shopping lists, recipes, and other family notes. When ARPAnet, the forerunner of the Internet we know today, came out in 1969, the world became truly connected. The X10 Home Automation Project started in 1975. We're finally getting to the point where we're talking about useful things for real homes. The X10 devices worked with the AC wiring already in a building and were used to control small appliances and lighting. The 1980s changed the way people bought things. Motion-sensing lights, automatic garage door openers, programmable thermostats, and security systems were now common and affordable. The American Association of Home Builders came up with the term "smart house" in 1984.

Then, in 1990, Dan Lynch, President of the Interop Internet networking show, issued a challenge. In response, John Romkey and Simon Hackett built a toaster that could be connected to the Internet and controlled from it. The Internet of Things (IoT) was born, but Kevin Ashton wouldn't coin the

term for another nine years. In the same year, Microsoft added its own ideas about what a smart home should look like and how it should work. Microsoft predicted a lot of things that smart homeowners today take for granted, like security systems, smart locks, smart thermostats, and smart lighting controls. Throughout the 2000s, smart systems and devices have changed quickly. It is thought that by 2012, 1.5 million automated home systems were already set up. In 2014, Amazon released the Amazon Echo for Prime members. At first, it was marketed as a way to control music with your voice, but when Alexa was added, it became clear that the device could also be used as a smart home hub.

IoT devices are more common than ever today, and the price of smart home systems keeps going down, which makes them a good choice for homeowners. But the home automation industry has had some growing pains because of software and systems that aren't open source. Consumers often have to choose between getting the devices they really want and getting devices that work well together in a single installation. At Zeus Integrated Systems, we design, plan, and install the best smart home automation systems. We can make a smart home solution just for you, no matter if you live in a mansion in Westchester or a brownstone in New York City.

## 2.3 Previous Work

### 2.3.1 Low-Cost Smart Parking System for Smart Cities

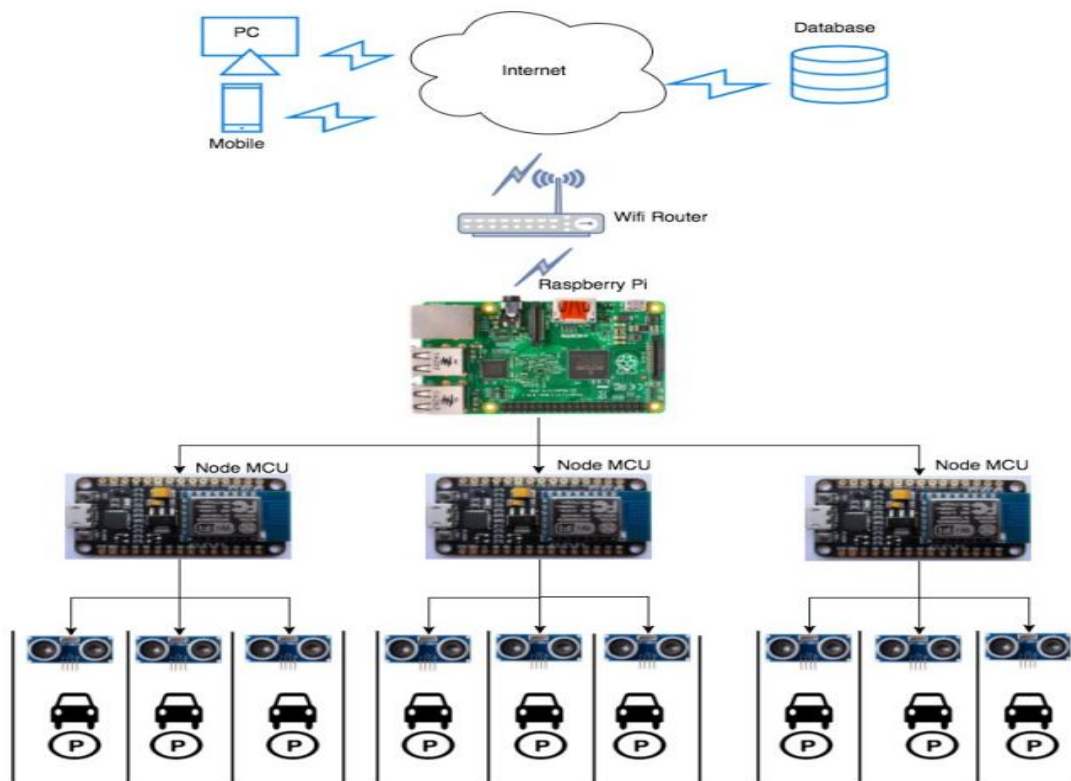
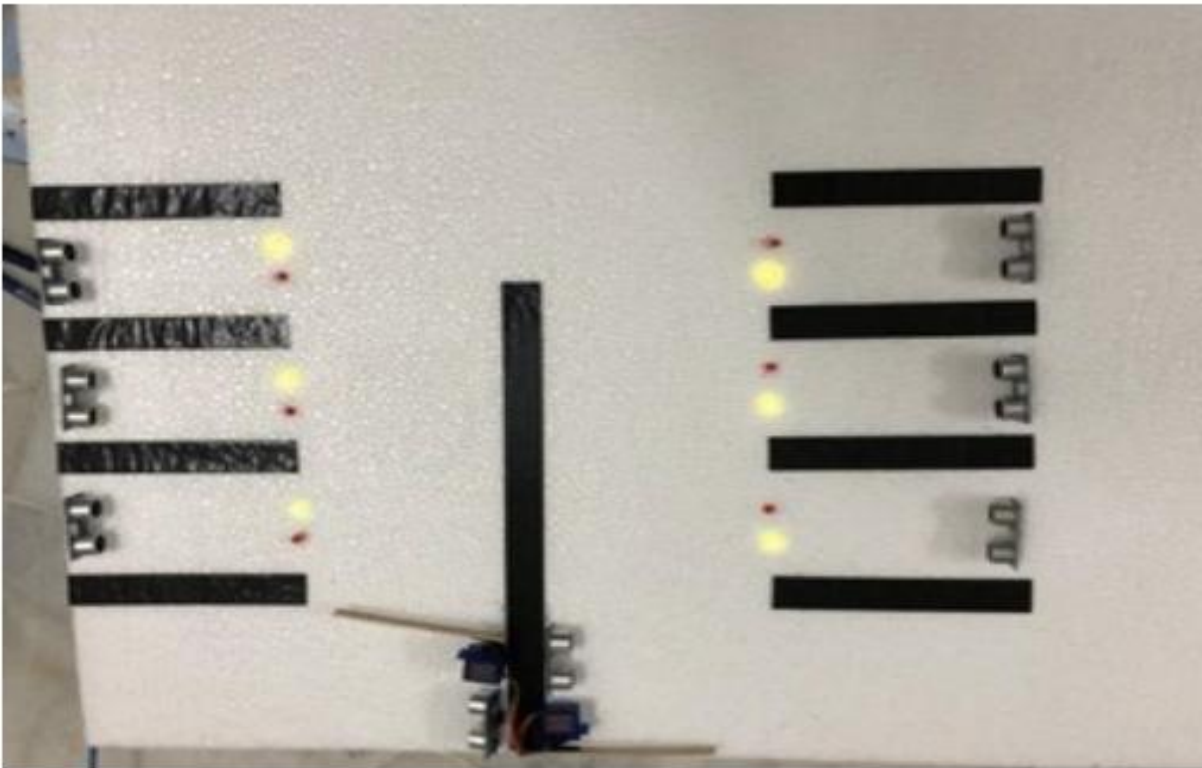


Figure 5: Low Cost Smart Parking System for Smart Cities

(Vakula, 2017) said about low cost parking systems for smart cities. In this paper, the author try to explain the solution for parking system which is budget-friendly. Now, a lot of people are interested in the idea of "smart cities." With the rise of the Internet of Things, smart cities are becoming more and more possible. In the field of the Internet of Things, there is a lot of research going on to improve the quality of services in cities and to make urban infrastructure more productive and reliable. Internet of Things is helping to solve some of the most common problems in cities, like finding a place to park and getting around in a hurry. This paper describes a parking system for smart cities that is based on the Internet of Things. The proposed parking system has an Internet of Things (IoT) module that is set up on-site to manage the parking spaces. A platform in the form of a portal that lets people book parking spots. Here the author used 3 wifi modules and six ultrasonic sensors Each parking spot has an ultrasonic sensor that checks to see if the spot is taken (Occupied or Vacant). This section shows a typical diagram of how three parking spots in a parking lot are connected to each other. The three ultrasonic sensors (HCSR04) are connected to a NODEMCU board that is based on the ESP8266 chip. The Raspberry Pi3 gets information from all the NODEMCU and adds the number of free slots to the database. Using data from the database, a mobile app and an online portal are made to show how many free parking spots are in a certain area. Users are also given a way to book a parking spot online through a user interface.

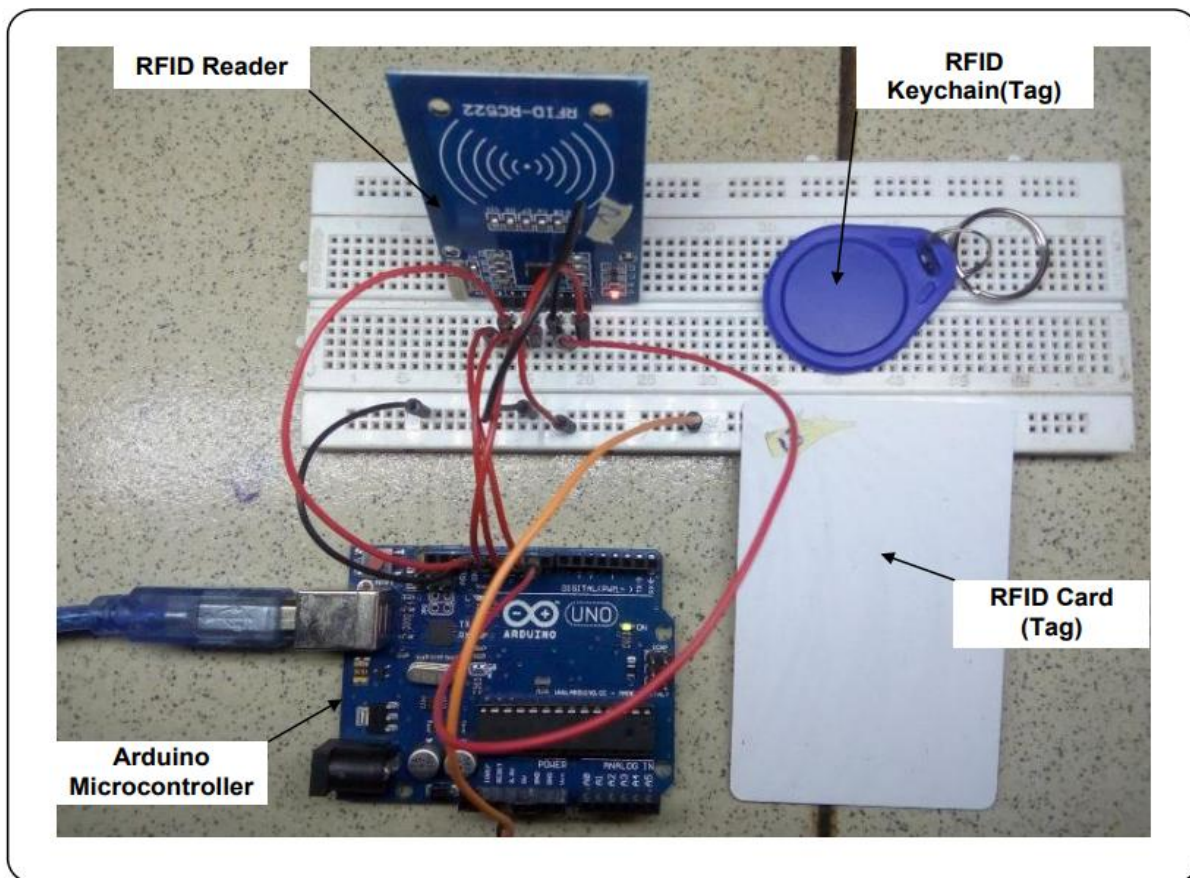
### 2.3.2 IoT Based Smart Parking System for Large Parking Lot



**Figure 6: IoT Based Smart Parking System for Large Parking Lot**

(Ramasamy, 2018) said about IoT based smart car parking which is mainly used for large parking. This paper deals with the car parking system. For example, when the car is taken to a big shopping complex or a big office court, there is a lot of trouble with car parking. And a traffic jam is created. In order to reduce this traffic jam, IOT based parking system has been developed to manage the parking efficiently. A successful cloud-based smart parking system has been developed with the aim of reducing searcher congestion. This project uses a microcontroller Uno and a WiFi modulator along with six ultrasonic sensors and a servo motor. The red light will be on to indicate when there is a car in the parking slot and the green light will be on when there is no car. These are discussed in detail in this paper. The hardware design uses an Arduino Mega 2560 as a microcontroller, an ESP-8266 WiFi Module to send information about available parking spaces to the cloud, an ultrasonic sensor HC-SR04 to tell if a parking space is free or not, and LEDs to show if a space is taken or not. The Arduino is linked to all of the hardware. All the driver will be able to use a mobile phone to access the system and see information about the availability of parking spots. An app that was made with BLYNK mobile app.

### 2.3.3 Automatic Access Control prototype using Arduino and RFID



**Figure 7: Automatic Access Control prototype using Arduino and RFID**

(Orji, 2018) said as technology has gotten much better, the number of wireless security-based applications has grown quickly. Different people made or put in place a lot of access control systems based on different kinds of wireless communication technologies. Radio Frequency Identification (RFID) is a contactless technology that is used in many industries to do things like control access, track books in libraries, manage the supply chain, and so on. The author takes one Arduino UNO and one RFID card. The author tries to explain that an RFID-based automatic access control system was made with Arduino. RFID technology and Arduino are used together in the system to do what needs to be done. When the RFID reader at the entrance sees an RFID tag, it records the user's unique identifier (UID) and checks to see if it matches the UID that has already been stored. If the captured user UID matches one of the stored UIDs, access is granted. If not, access is denied. The results show that the system is cheap, works well, and is a reliable way to let people in or keep them out of a secure area.

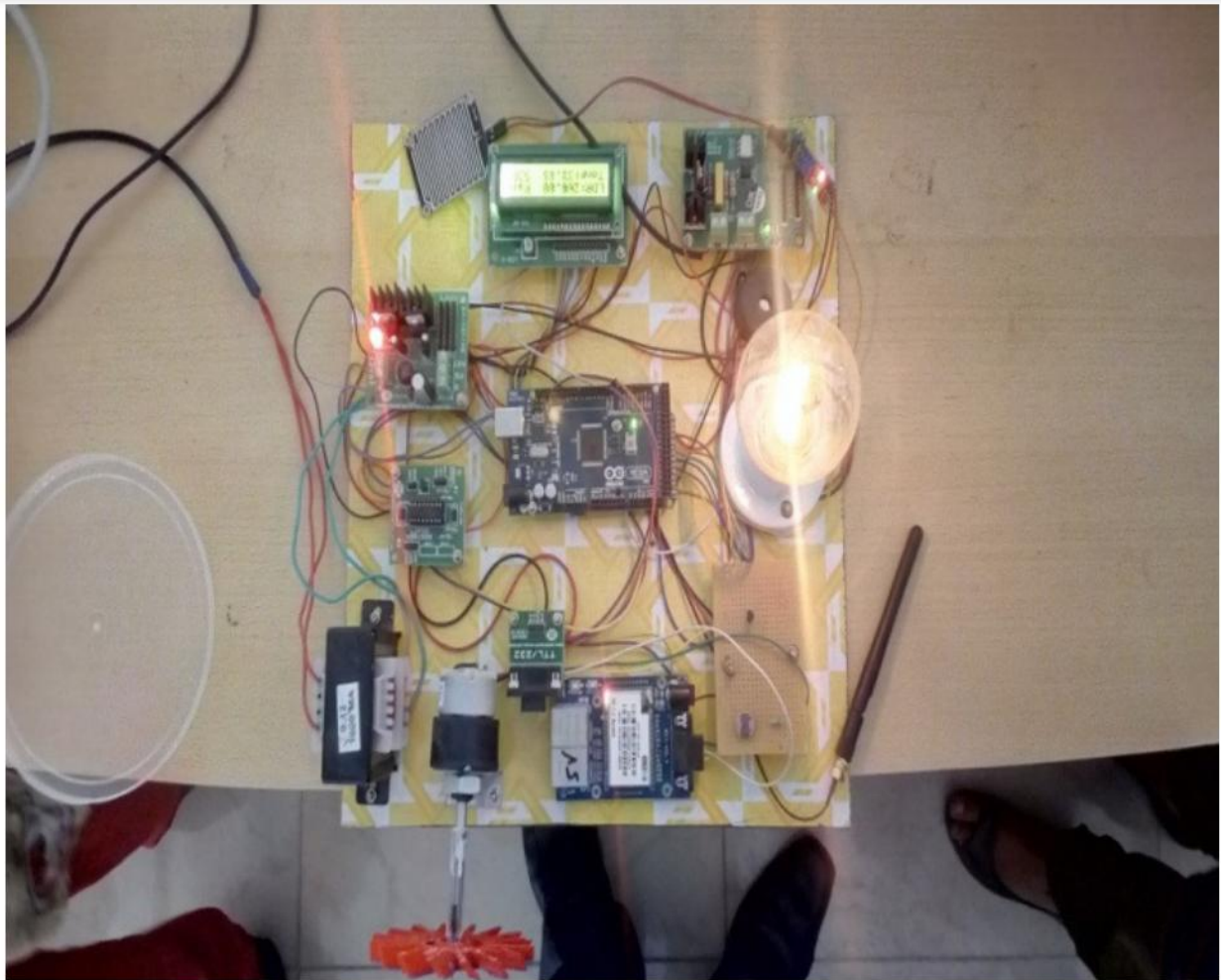
#### **2.3.4 A Low-Cost Entry Door Using Database Based On RFID And Microcontroller.**



**Figure 8: A Low-Cost Entry Door Using Database Based On RFID And Microcontroller**

(Wibowo, 2017) said the world's technology is getting better and better very quickly. As these technologies get better, more and more homes and apartments are putting in smart home systems. Putting in a smart home system is good for a home's security, especially when it comes to accessing appliances and other things that should be smart. Radiofrequency identification is used in one of these systems (RFID). This paper used a method of identification that uses radio waves on a certain frequency along with an Arduino microcontroller as the core to process input and output (I/O) components that are connected to the database through the interface application. The goal of making this device is to add an extra layer of security so that a door can only be opened by people who have the right to do so. This is done by using an RFID tag that is read by an RFID reader to verify who has access.

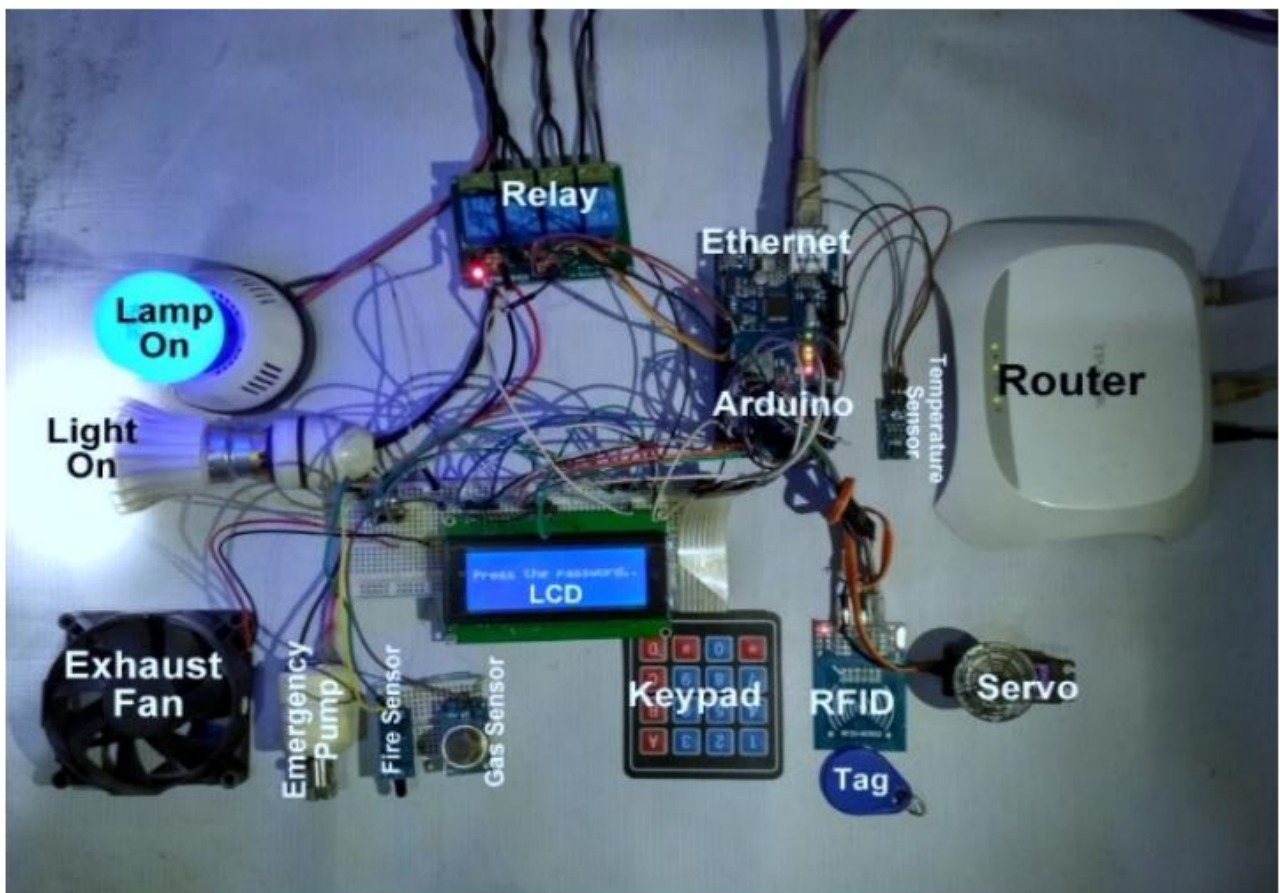
### **2.3.5 An IoT based Home Automation Using Android Application**



**Figure 9: An IoT based Home Automation Using Android Application**

(Reddy, 2016) said as technology gets more and more intrusive, the design challenges of home automation are becoming more and more obvious. Home monitoring, control, and programming by the end user have not yet become commonplace. This could be a real challenge when it comes to making a home system that is fully independent and scalable and can support devices and technologies with different functions and protocols. The author explains how to use an Android app to control and monitor home appliances over the internet. There are a number of commercial home automation systems you can buy. But these are made to be used only a few times. So, each appliance can be controlled from inside the house or from outside the house. This is very helpful for people who have physical problems. The practical goal of this paper was to make a virtual Android home automation system that can be used in real life. All the home appliances are controlled by sending commands to the Arduino from an Android phone. The main function of this system is to control the voltage levels of home appliances, such as the speed of a fan based on the temperature or the brightness of light based on the amount of light in the room, etc. Another function is that we can check the status of our home appliances from our Android phones. In this system, we use different sensors for different tasks, such as a temperature sensor, a rain sensor, and an LDR.

### 2.3.6 Wireless Home Automation System Using IoT and PaaS



**Figure 10: Wireless Home Automation System Using IoT and PaaS**

(Nitu, 2019) said how a flexible, secure, low-cost, advanced home control and monitoring system was made and put into place. Home security services include protection against gas leaks, fires, and trespassers. If someone who shouldn't be there tries to get in, the system sends a text message to the authorized phone number. Sensors are part of the system. When the gas sensor detects a problem, the system sounds an alarm through the buzzer and automatically turns on the kitchen exhaust fan. Again, when the fire sensor finds something strange, the system sounds an alarm with a buzzer and turns on an emergency pump by itself. Both times, the system sends an SMS to the user. The system has a Teleduino server, a micro-web server, and an Altair Smart Core server so that home appliances can be controlled and monitored from anywhere, at any time. When the state of an appliance changes, the information is saved right away, and the system sends an email to the user. If you log in to the Altair SmartCore web server, you can check this stored data from anywhere. So, the proposed system lets authorized homeowners to use the "Home Automation" android or web app and a server real IP to check on and control their home appliances from afar.

## **Chapter 3: Methodology and Requirement Analysis**

## 3.1 Introduction

In this section, we'll discuss all types of diagrams like circuit diagram, block diagram, flow chart etc. Also, describe briefly the Arduino IOT cloud and pictorial Explanation of all requirements with the working principle.

## 3.2 Project Flow

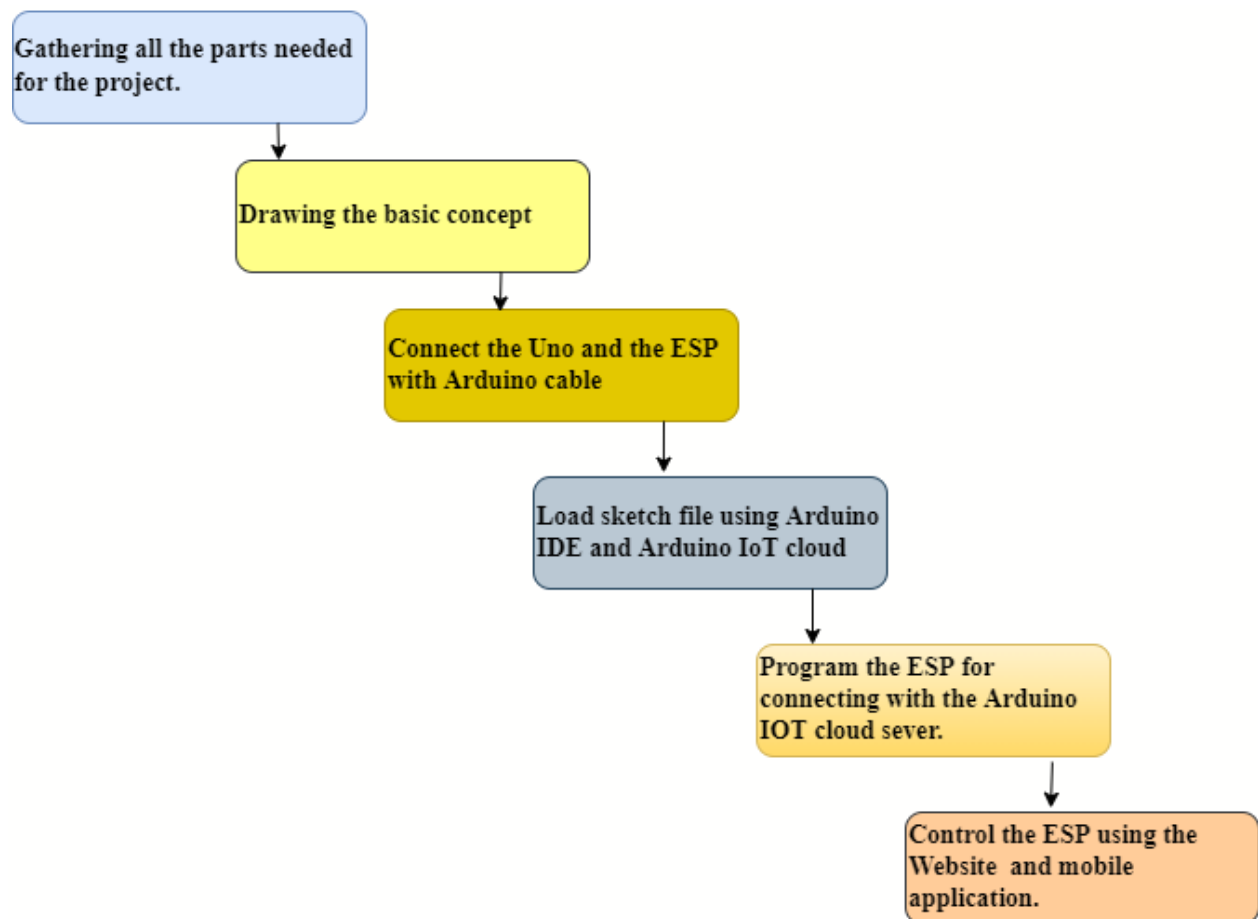


Figure 11: Project Flow

## 3.3 Arduino IoT Could

### 3.3.1 What is Arduino IoT Cloud?

An IoT cloud is a huge network that supports IoT devices and applications. This includes the infrastructure, servers, and storage needed for operations and processing to happen in real-time.

This Arduino IoT is used by us in Home Automation System in our project. It is very easy to use.

### 3.3.2 How does IoT connect to Arduino?

There are three steps to connect a new device to Arduino IoT Cloud: Put the thing on IoT Cloud. The device can now be used with IoT Cloud. How to set things up

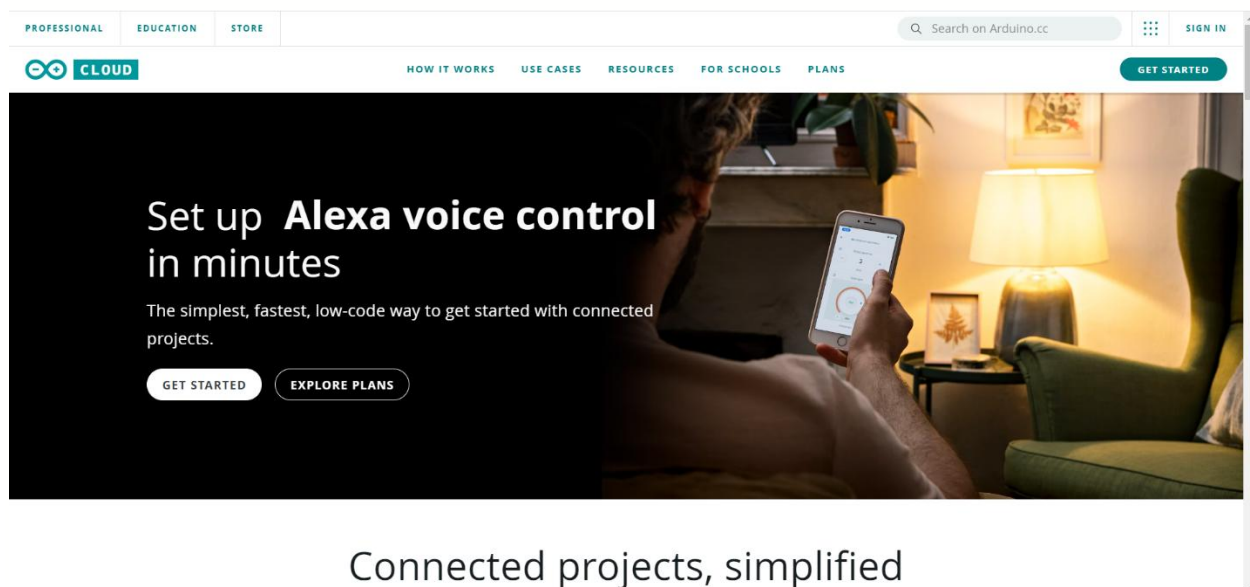
Step 1: Put the thing in. Open the Devices tab on Arduino IoT Cloud.

Step 2: Make a Thing and set it up.

Step 3: Send the sketch of the Thing.

### 3.3.3 How to find Arduino IOT cloud?

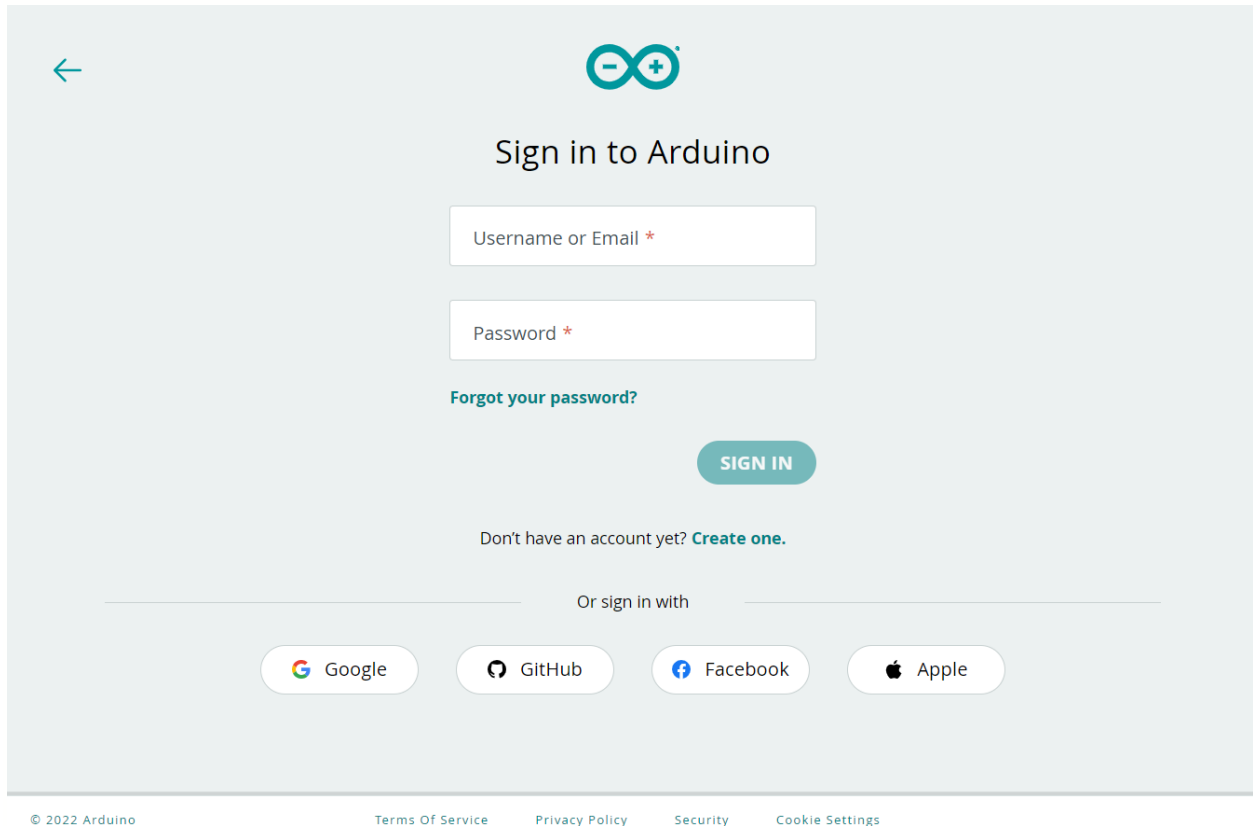
First, we need to open any browser from a laptop or pc and search Arduino IOT cloud. If we search, we will find a web page like this.



**Figure 12: Searching Arduino IoT Cloud**

### 3.3.4 How to create an account in the IoT cloud?

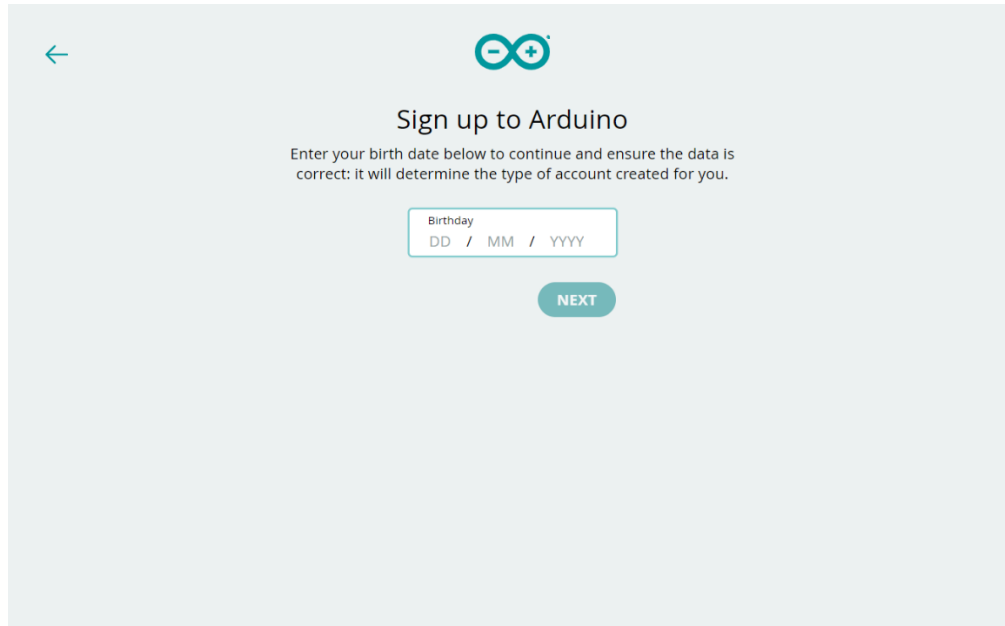
To create an account, we need to click on the **SIGN IN** button from Arduino IoT cloud page. Then we can see a Login page.



The screenshot shows the login page for the Arduino IoT cloud. At the top left is a back arrow icon. In the center is the Arduino logo (an infinity symbol with a minus and plus sign). Below the logo is the heading "Sign in to Arduino". There are two input fields: "Username or Email \*" and "Password \*". Below the password field is a link "Forgot your password?". A teal "SIGN IN" button is positioned below the input fields. Below the button is the text "Don't have an account yet? [Create one.](#)". A horizontal line separates this section from the "Or sign in with" section, which contains four buttons: "Google", "GitHub", "Facebook", and "Apple". At the bottom of the page is a footer with the text "© 2022 Arduino" on the left and links for "Terms Of Service", "Privacy Policy", "Security", and "Cookie Settings" on the right.

**Figure 13: Login Page**

Then we click on '**Create One**' to create a new account for IoT cloud for our project.



The screenshot shows a light gray background with a teal back arrow in the top left corner. At the top center is the Arduino logo, which consists of two interlocking circles with minus and plus signs inside. Below the logo, the text "Sign up to Arduino" is centered. Underneath that, a smaller line of text reads: "Enter your birth date below to continue and ensure the data is correct: it will determine the type of account created for you." In the center of the page is a white rectangular input field with a thin teal border. Inside the field, the word "Birthday" is at the top, and below it are the labels "DD / MM / YYYY". At the bottom center of the page is a teal rounded rectangular button with the word "NEXT" in white capital letters.

**Figure 14: Create a new account (adding date of birth)**

Then we need to add a valid email account and user name and password for privacy. Accept all the terms of service. After selecting Education click on the SIGN-UP button.

## Sign up to Arduino

Enter your email \*

.....@gmail.com

Choose a username \*

.....

Choose a password \*

.....

Minimum 8 characters long

☒ I have read the [Privacy Policy](#) and accept the [Terms of Service](#) \*

☒ I would like to receive the Arduino newsletter.

☒ I agree to see personalised commercial offers from Arduino based on my browsing and purchasing behaviour.

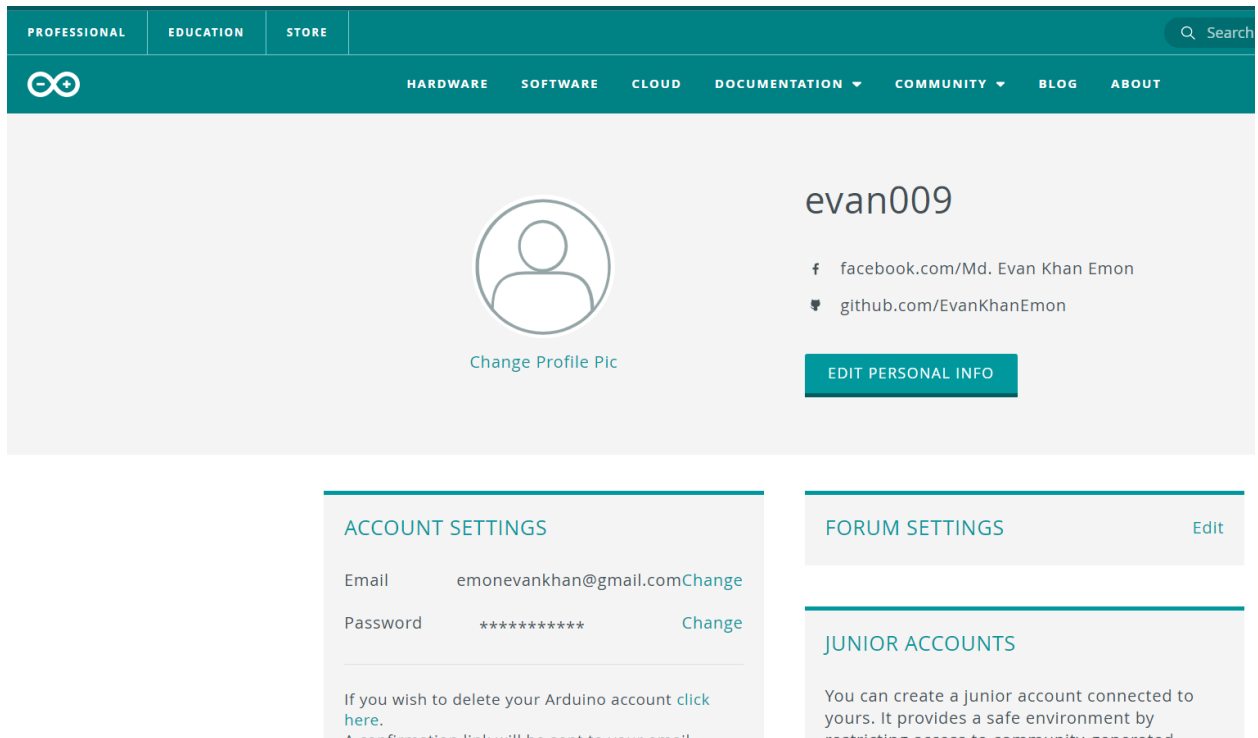
☒ I would like to get email updates about special deals and commercial offers from Arduino.

☐ Everything ☐ Maker ☒ Education ☐ Professional

**SIGN UP**

**Figure 15: Create a new account (adding other information)**

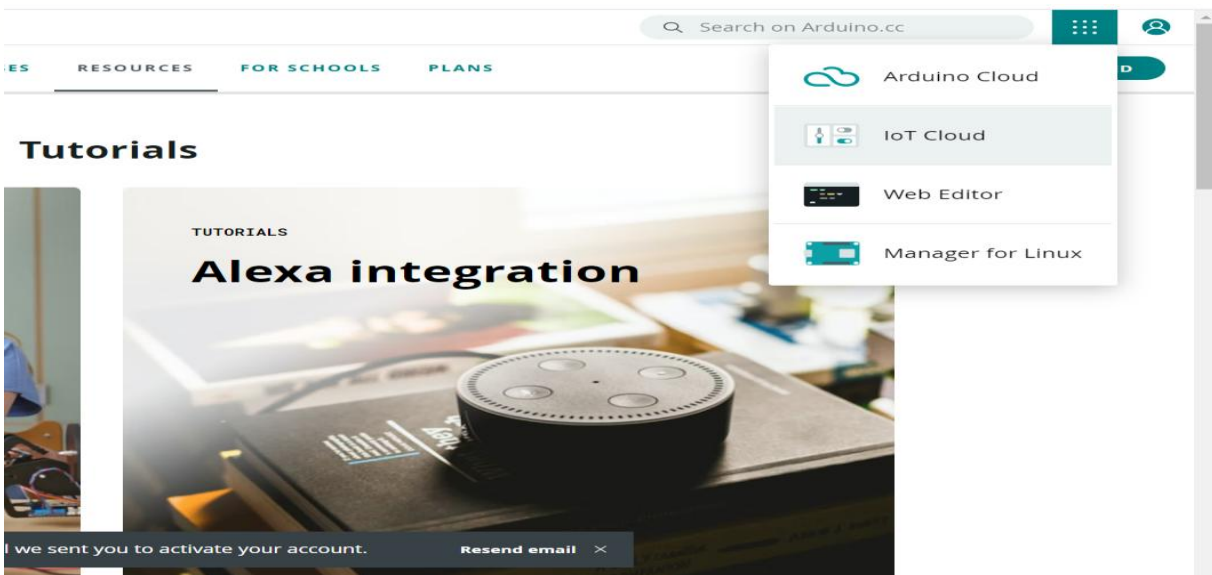
Finally, we create an account for our project. Now it will help to create things and create a dashboard for the Home automation system.



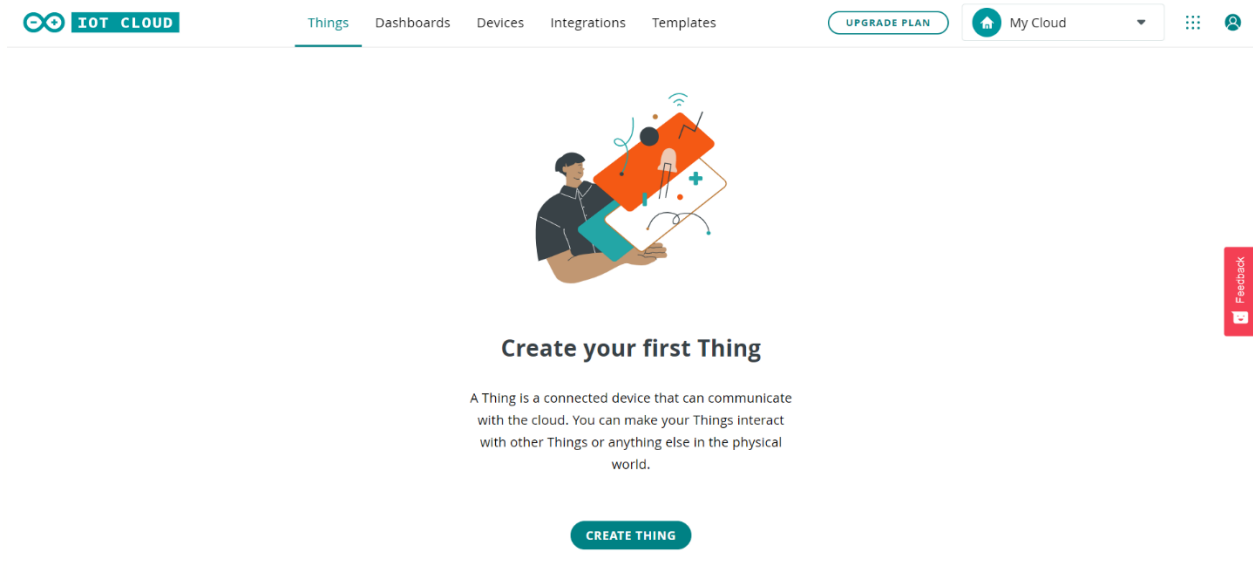
**Figure 16: New account is created**

### 3.3.5 How to create things in the IoT cloud

After creating an account click on three dots and select the option IoT cloud. This will open a new tab where we can create Things.

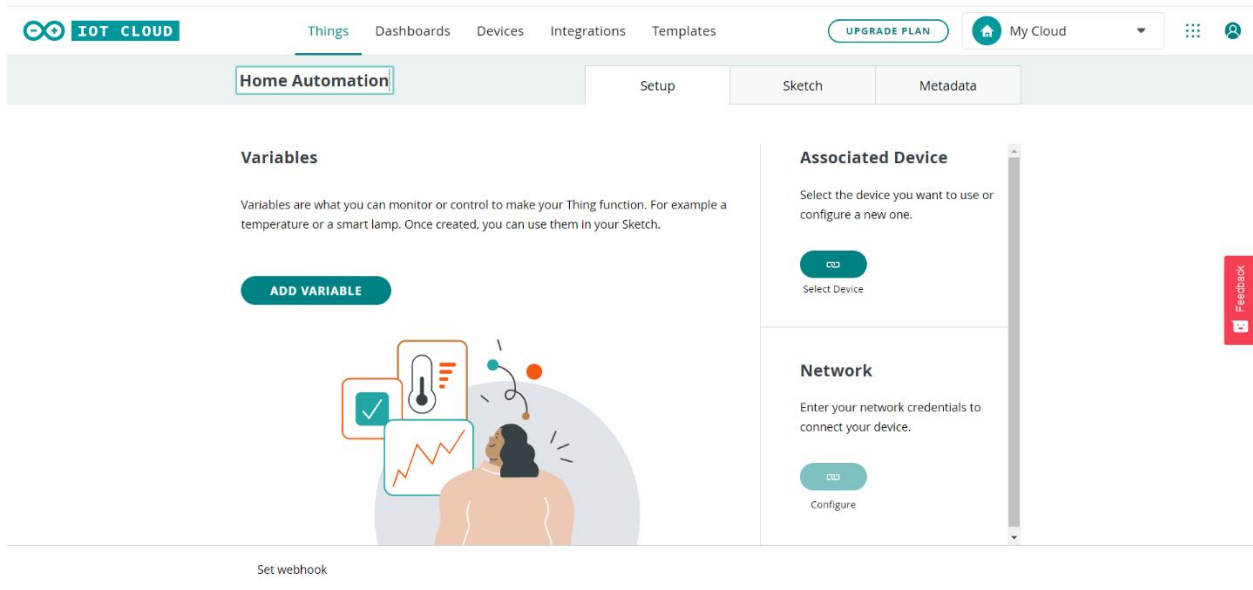


**Figure 17: Select the IoT cloud for marking things**



**Figure 18: Things creating web page**

Then click on a button named **CREATE THING** to create a new thing. After clicking, we can see a page like this. Here we give a name for the Things and click on **ADD VARIABLE**



**Figure 19: Set up for the new thing**

Then we can see a form to add a variable.

Add variable ✕

Name

↻ Sync with other Things ⓘ

Select variable type ▼

Declaration ⓘ

---

**Variable Permission** ⓘ

☒ Read & Write
 ☐ Read Only

**Variable Update Policy** ⓘ

**Figure 20: Add variable form**

We are using a 4-way relay shield to control the load. That's why we give a variable name **relay1**, select the variable type **Boolean**, select variable permission **Read & Write** and select the variable **On Change**. Then click on **ADD VARIABLE**.

Add variable ✕

Name

relay1

↻ Sync with other Things ⓘ

Boolean ▼

Alexa compatible
✓ Basic types
Energy

Light and color
Size and motion
Time

Boolean eg. true

**Variable Update Policy** ⓘ

**Figure 21: Variable name and type.**

Add variable
X

Name
relay1

Sync with other Things

Boolean eg. true

Declaration
bool relay1;

Variable Permission
Read & Write
Read Only

Variable Update Policy

Figure 22: Variable permission

Add variable
X

Boolean eg. true

Declaration
bool relay1;

Variable Permission
Read & Write
Read Only

Variable Update Policy
On change
Periodically

ADD VARIABLE
CANCEL

Figure 23: Variable Update policy

This process is the same for all 4 variables because we are using a 4-way relay shield in our project. After adding all the variables things will look like this.

The screenshot shows the 'Things' interface with a top navigation bar containing 'Things', 'Dashboards', 'Devices', 'Integrations', and 'Templates'. A 'UPGRADE PLAN' button and a user profile icon are on the right. Below the navigation bar is a sub-header with 'Home Automation', 'Setup', 'Sketch', and 'Metadata'. The main content area is divided into two panels. The left panel, titled 'Variables', has an 'ADD' button and a table with the following data:

|                          | Name ↓                        | Last Value | Last Update          |   |
|--------------------------|-------------------------------|------------|----------------------|---|
| <input type="checkbox"/> | <b>relay1</b><br>bool relay1; | true       | 24 Aug 2022 17:17:01 | ⋮ |
| <input type="checkbox"/> | <b>relay2</b><br>bool relay2; | true       | 24 Aug 2022 17:17:06 | ⋮ |
| <input type="checkbox"/> | <b>relay3</b><br>bool relay3; | true       | 24 Aug 2022 17:17:06 | ⋮ |
| <input type="checkbox"/> | <b>relay4</b><br>bool relay4; | true       | 24 Aug 2022 17:17:06 | ⋮ |

The right panel, titled 'Associated Device', shows an 'Arduino\_IOT' device with the following details:

- ID: 36d75330-93fb-42c5-9776-...
- Type: ESP32 Dev Module
- Status: Offline

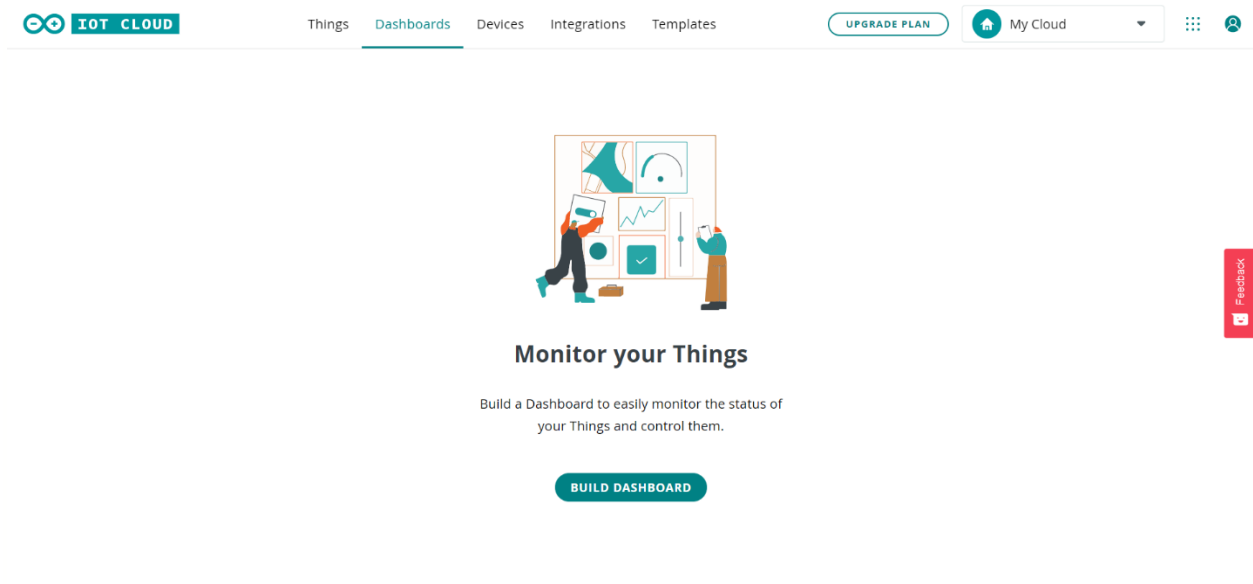
Below the device details are 'Change' and 'Detach' buttons. At the bottom of the right panel is a 'Network' section with the following information:

- Wi-Fi Name: Emon
- Password: .....
- Secret Key: .....

Figure 24: Final overview of the Things

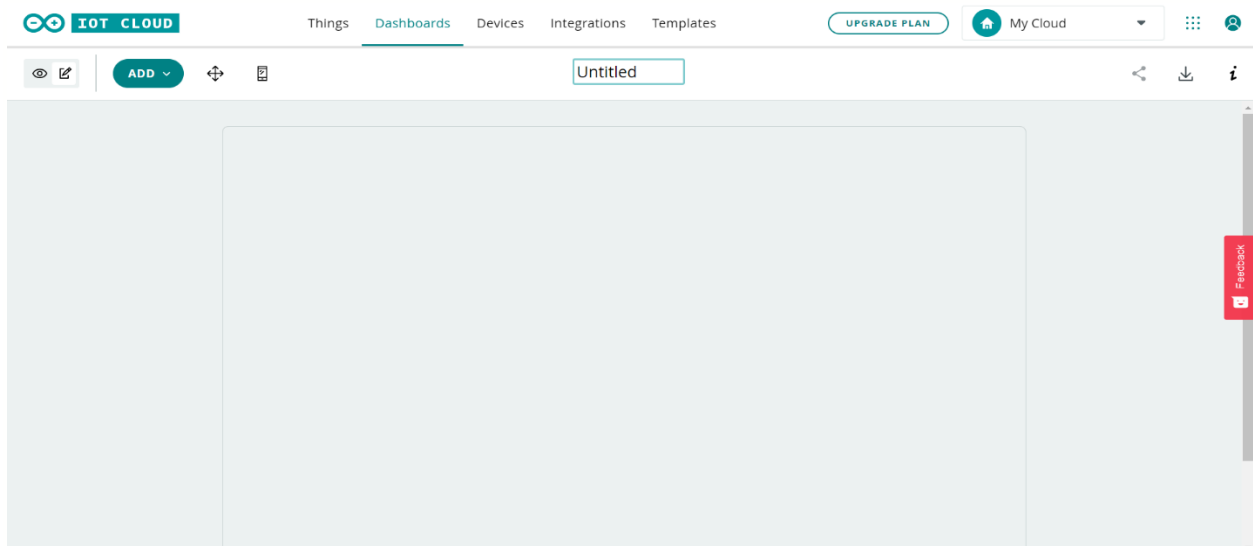
### 3.3.6 How to create a dashboard in the IoT cloud.

After creating **Things**, on the same page, we can see the **Dashboard** option. Click on the dashboard option which can monitor the Things.



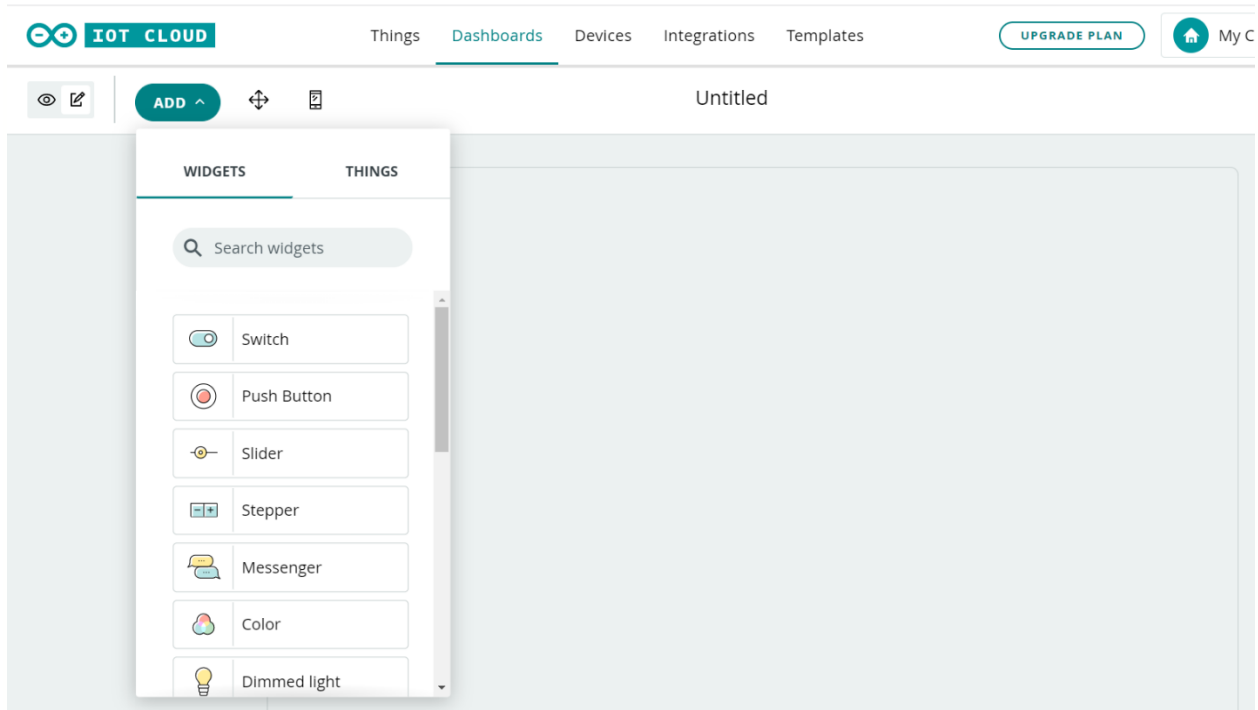
**Figure 25: Dashboard Design**

Then click on **BUILD DASHBOARD** to make a beautiful dashboard which can help to control the load.



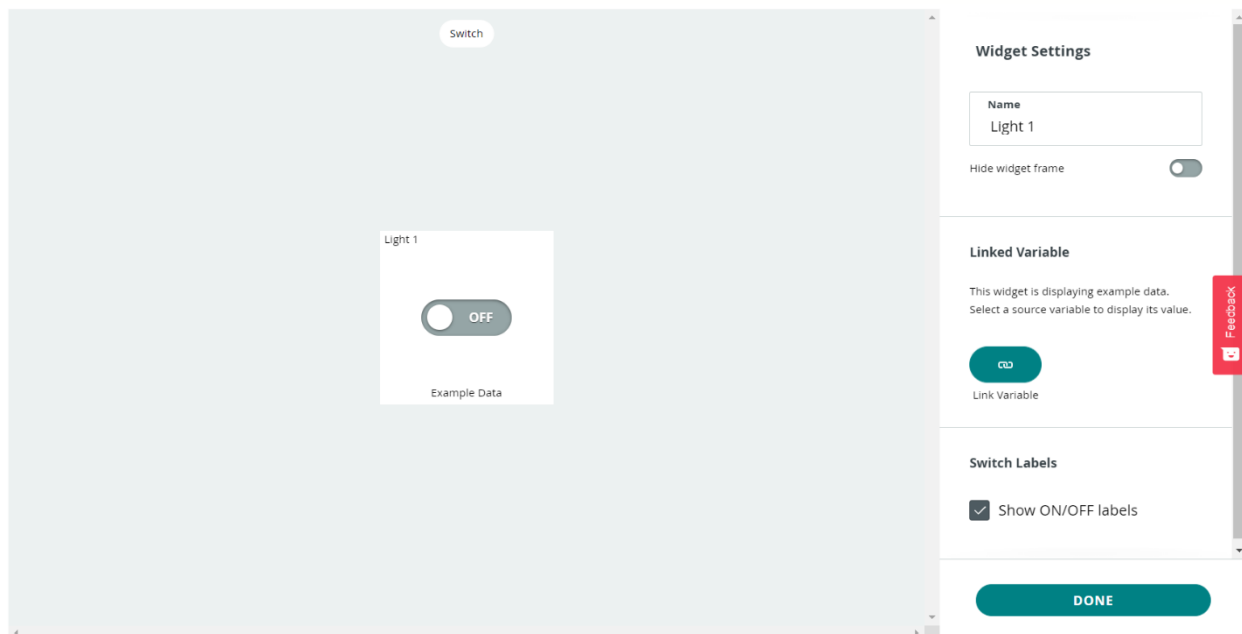
**Figure 26: Blank Dashboard**

Now click on **ADD** button to add Switch. By using this we can on off the load easily.



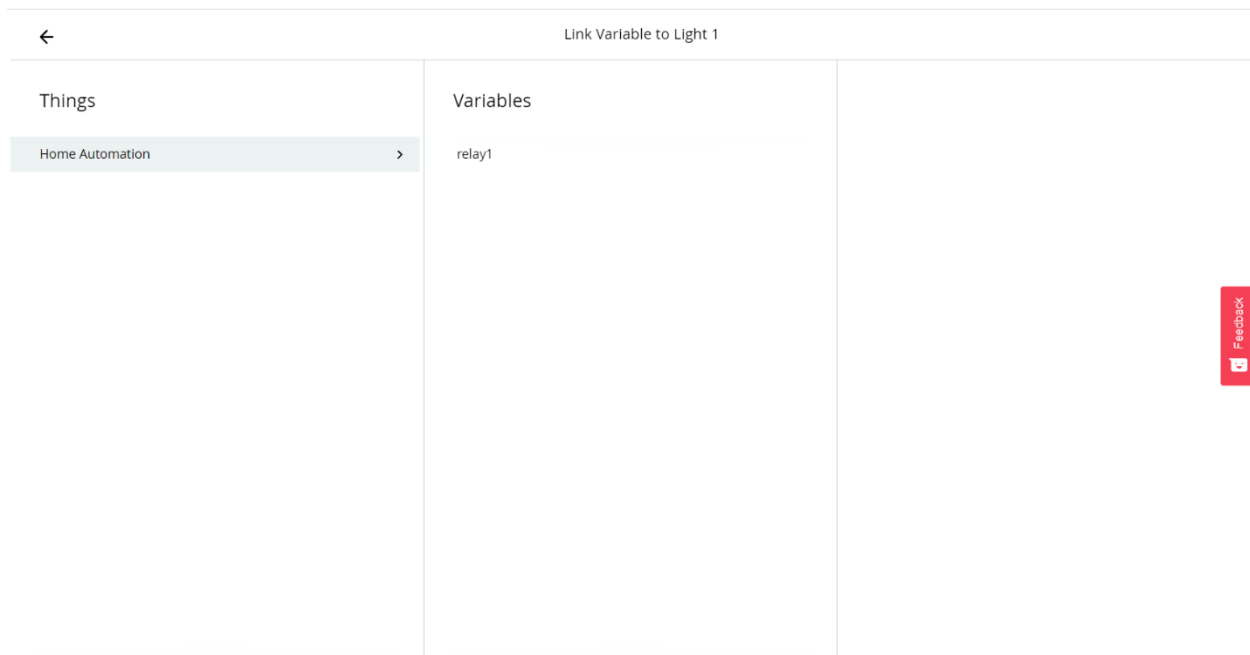
**Figure 27: Blank Dashboard**

After selecting the switch, we can change the name of the switch to Light 1.



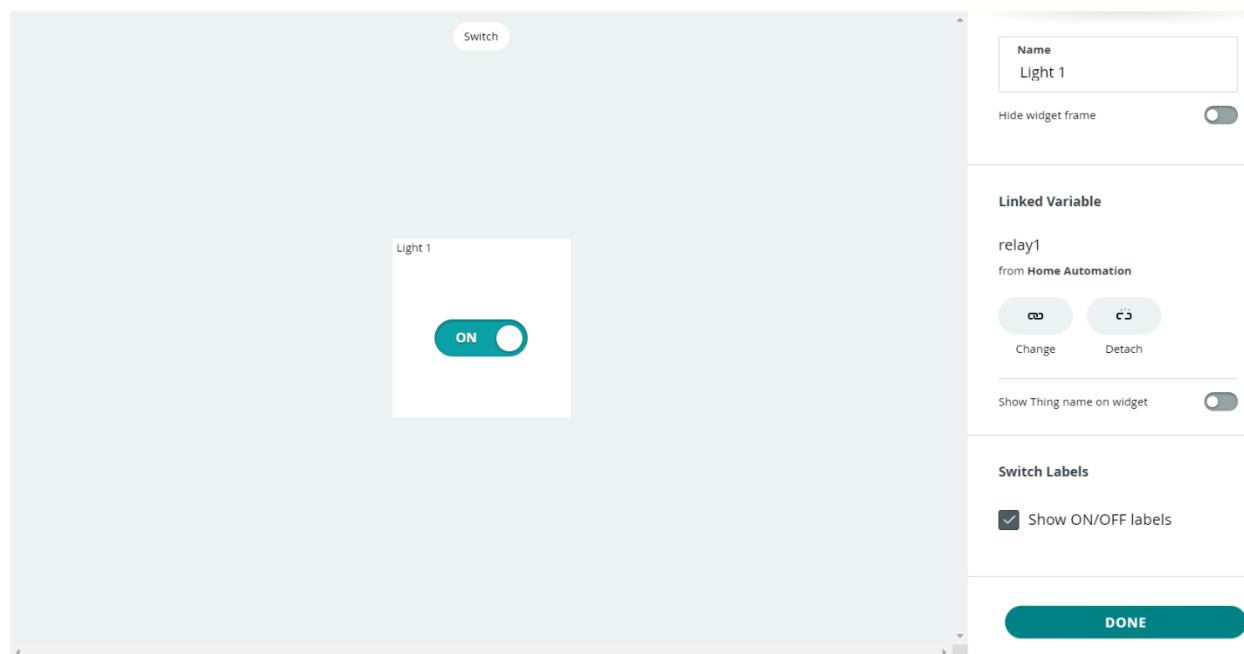
**Figure 28: Give a new name for the switch**

After giving a new name to the switch click on the **Linked variable** button to connect the switch with the variables that we created before in the Things.



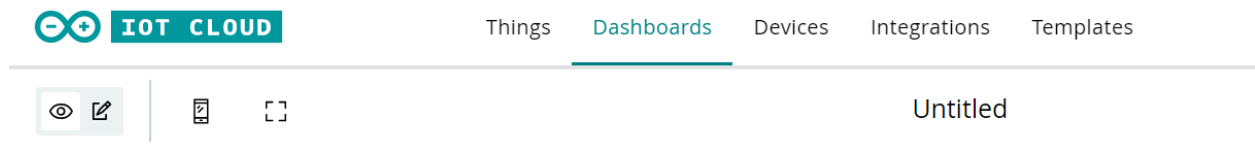
**Figure 29: Linked variable with the Things (1)**

Then click on relay1 and also click on the **LINK VARIABLE** button.



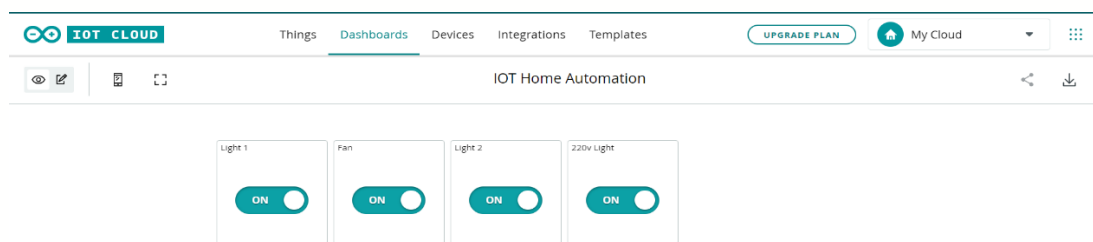
**Figure 30: Linked variable with the Things (2)**

Then we click on the **DONE** button. Finally, the switch is ready to perform.



**Figure 31: Switch is Created**

This process is the same for all the switches because we are using a 4-way relay shield in our project. After adding all the switch **Dashboard** will look like this.



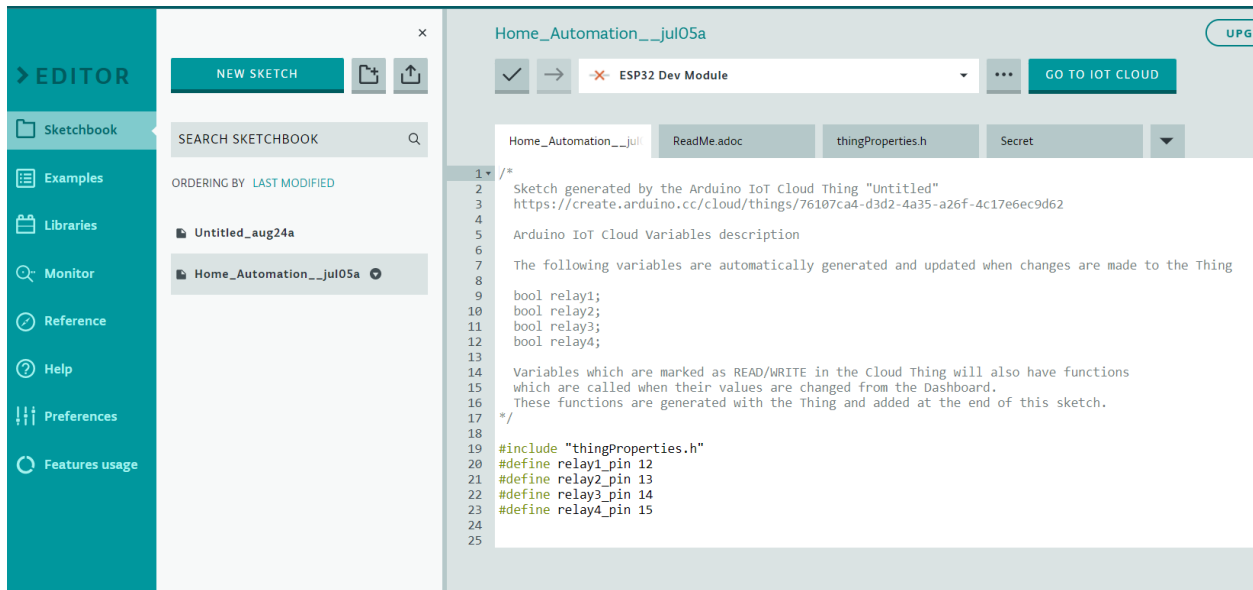
**Figure 32: Final overview of the dashboard**

We can also edit this switch from the dashboard at any time in case of changes. We can also delete

the switch easily.

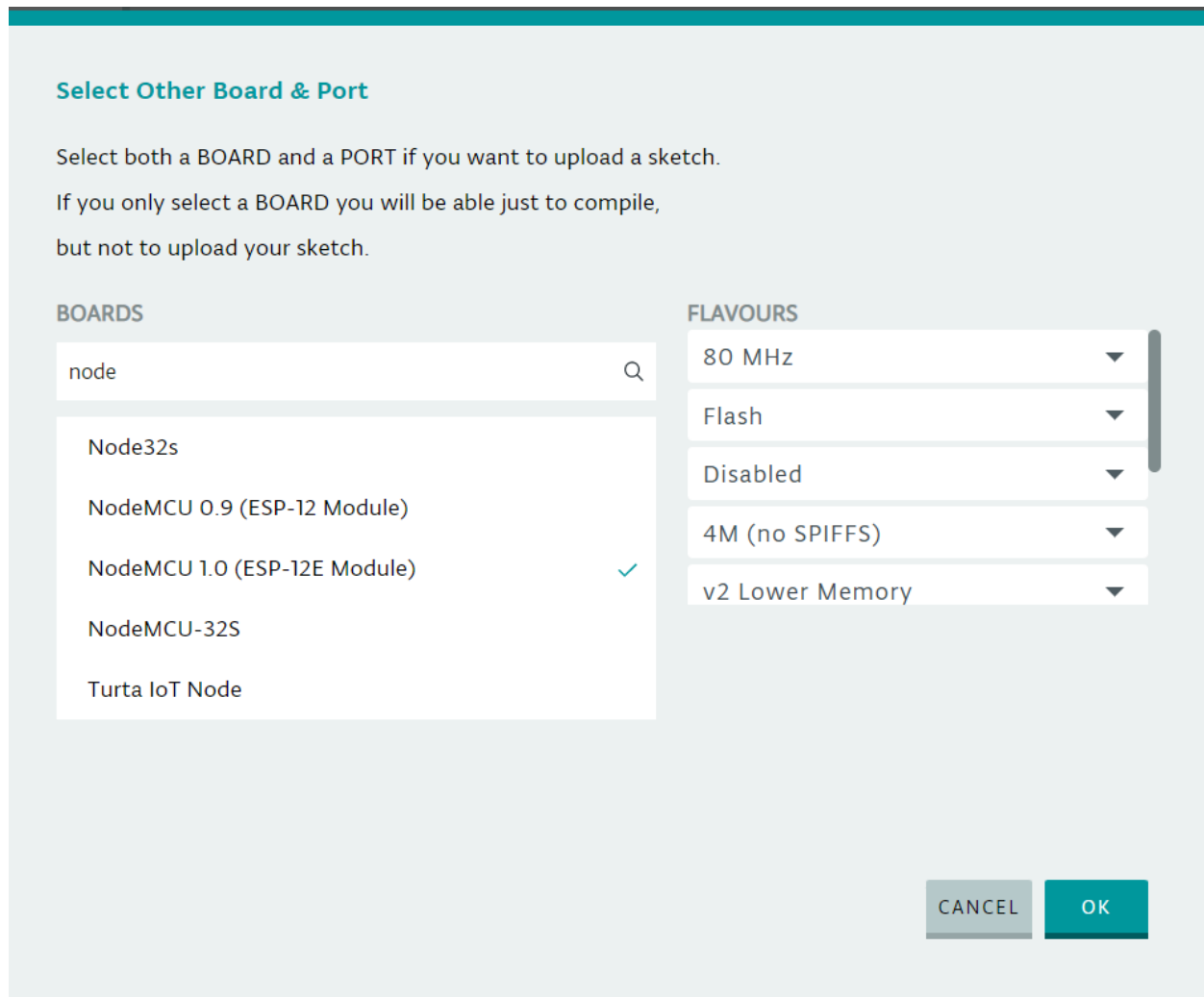
### 3.3.7 Arduino Editor.

Using the Arduino Web Editor, we can write code and upload sketches to any Arduino or Genuino board. After installing a simple plug-in, Sketchbook will be stored in the cloud and available from any device. Even a zip file can be used to import Sketchbook. When we create the thing some codes are auto-generated in the sketchbook. We add just a few conditions.



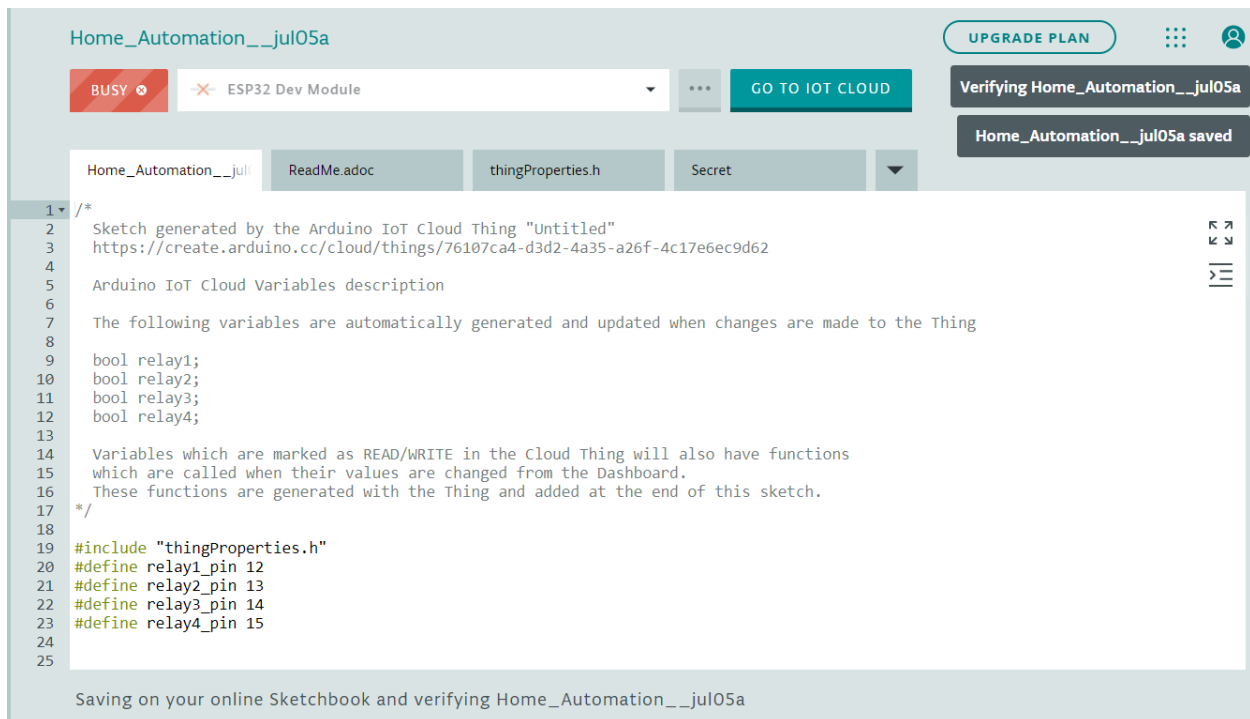
**Figure 33: Sketchbook of our Home Automation**

To upload the code to any device we need to select the board and the port first. Then click on the OK button.



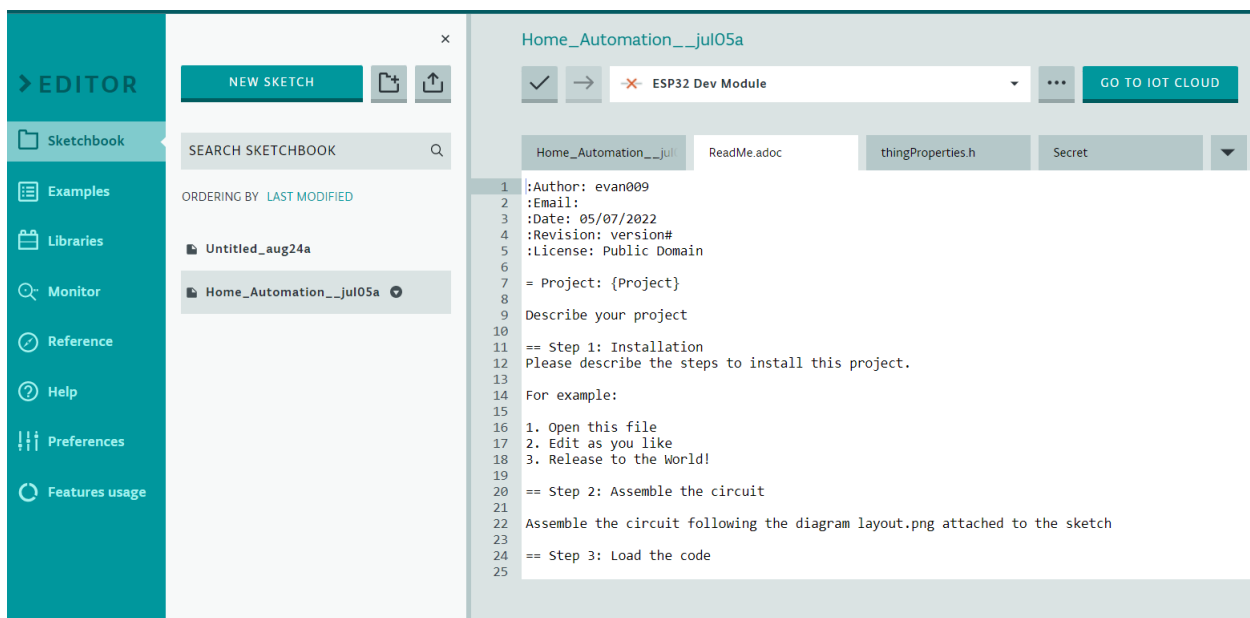
**Figure 34: Board and port selection**

After selecting the board and port the code is ready to upload to a device. We need to connect the Device with a USB cable from the computer. Then click on the arrow button.

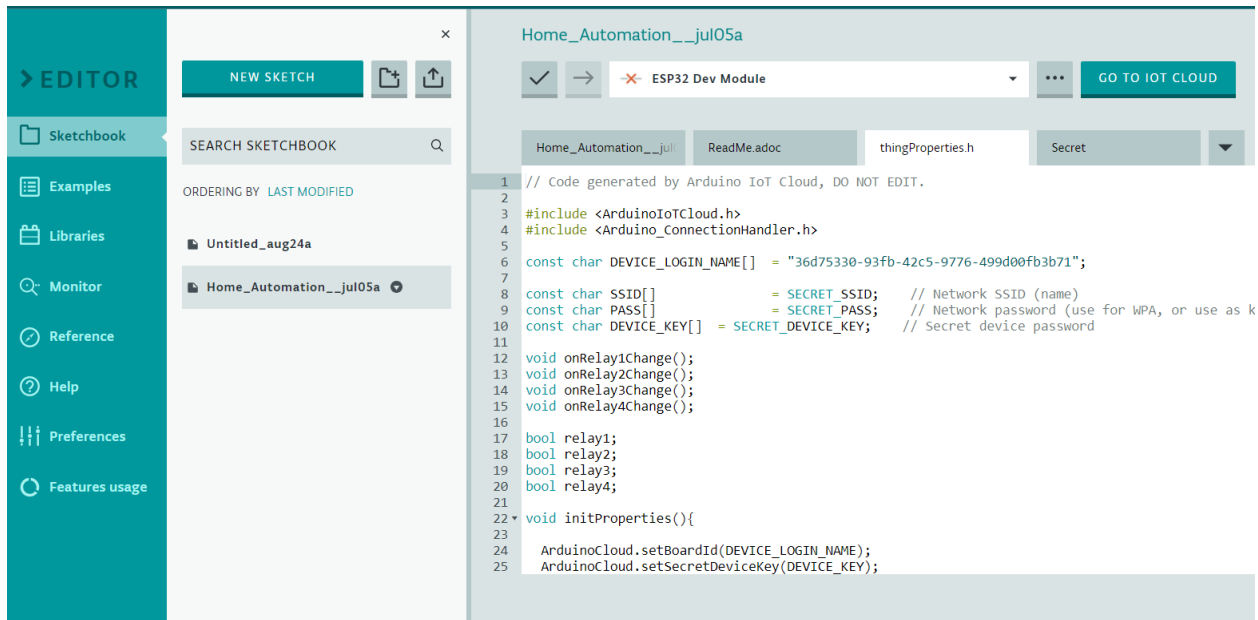


**Figure 35: Uploading the code**

Other part of the Arduino editor is given below.

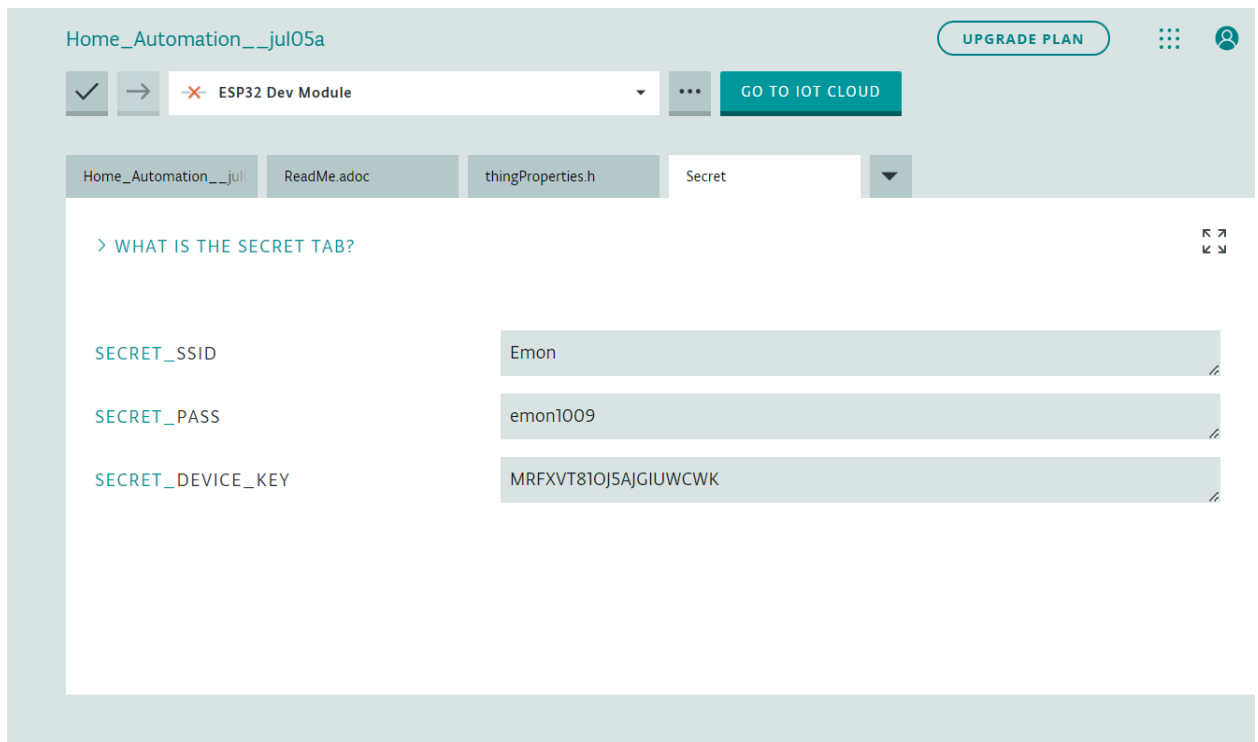


**Figure 36: ReadMe File**



**Figure 37: Code generated by Arduino IoT Cloud**

This is the Id and password for the device connection.

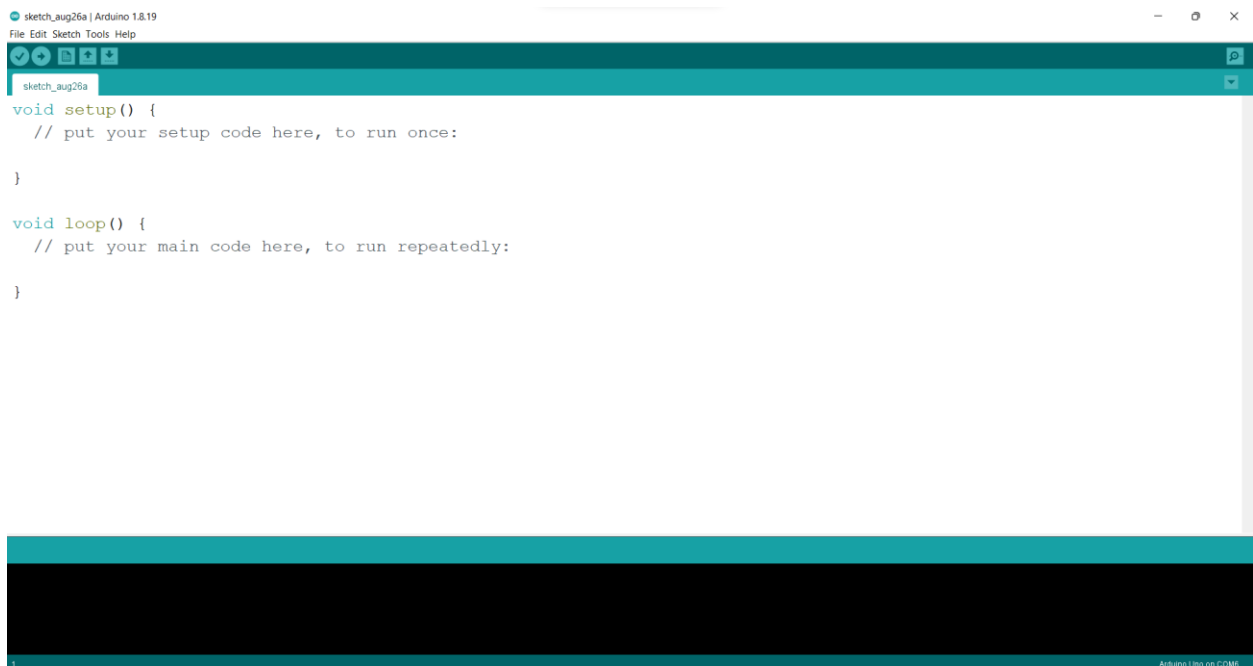


**Figure 38: Secret SSID and Password**

## 3.4 Arduino Setup

### 3.4.1 Arduino IDE

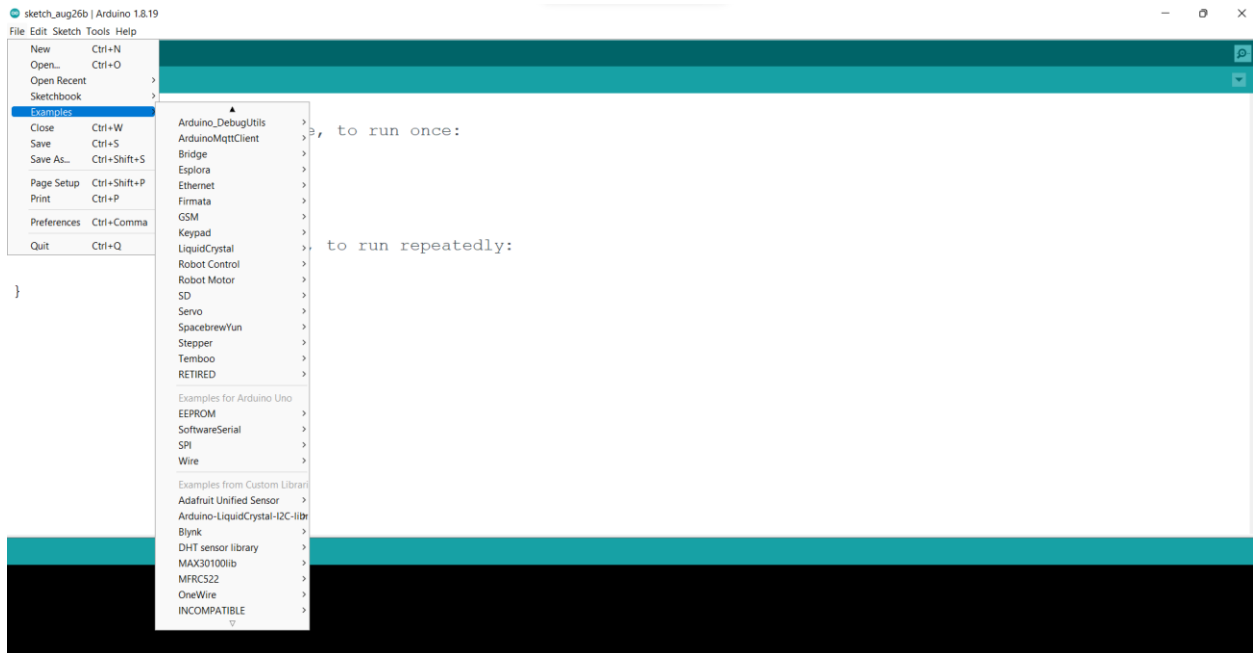
First, we need to download Arduino IDE from the browser. The Arduino Integrated Development Environment, or Arduino Software (IDE), has a text editor for writing code, a message area, a text console, a toolbar with buttons for standard functions, and several menus. It connects to the Arduino hardware, so we can upload programs and communicate. After downloading the Arduino IDE install it.



**Figure 39: Installation of Arduino IDE**

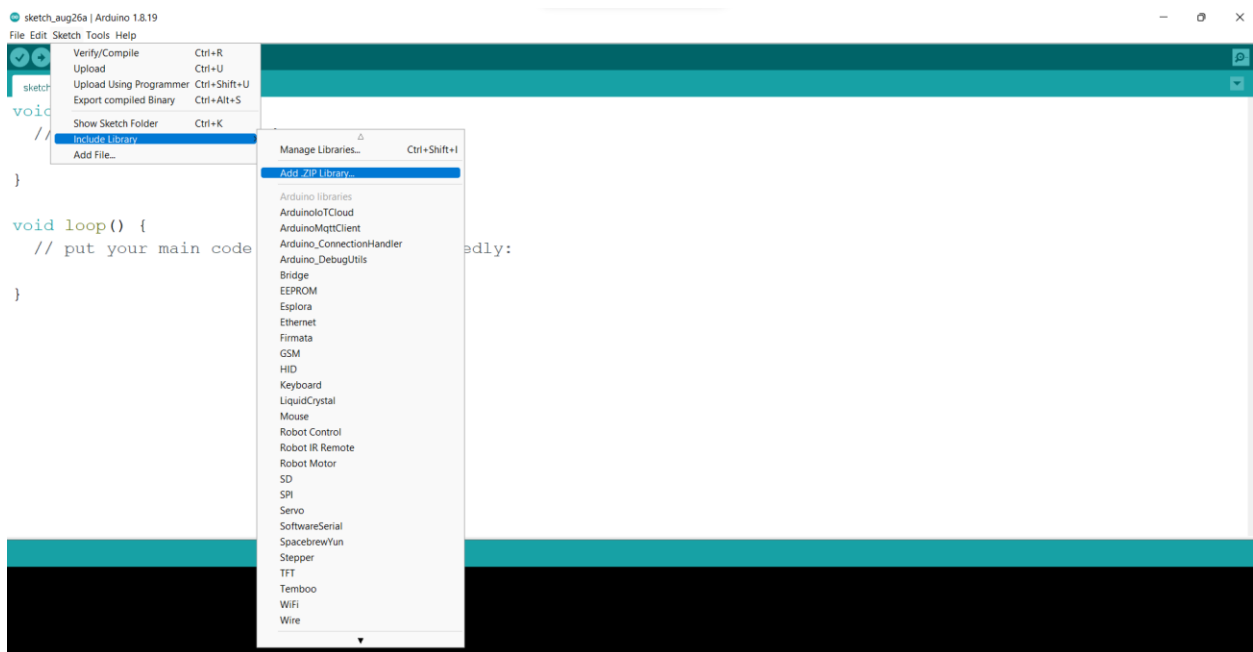
### 3.4.2 Arduino Code Example

If we click on File then Example we will see a lot of options. These are the example code for different equipment. So, we can easily test any equipment.



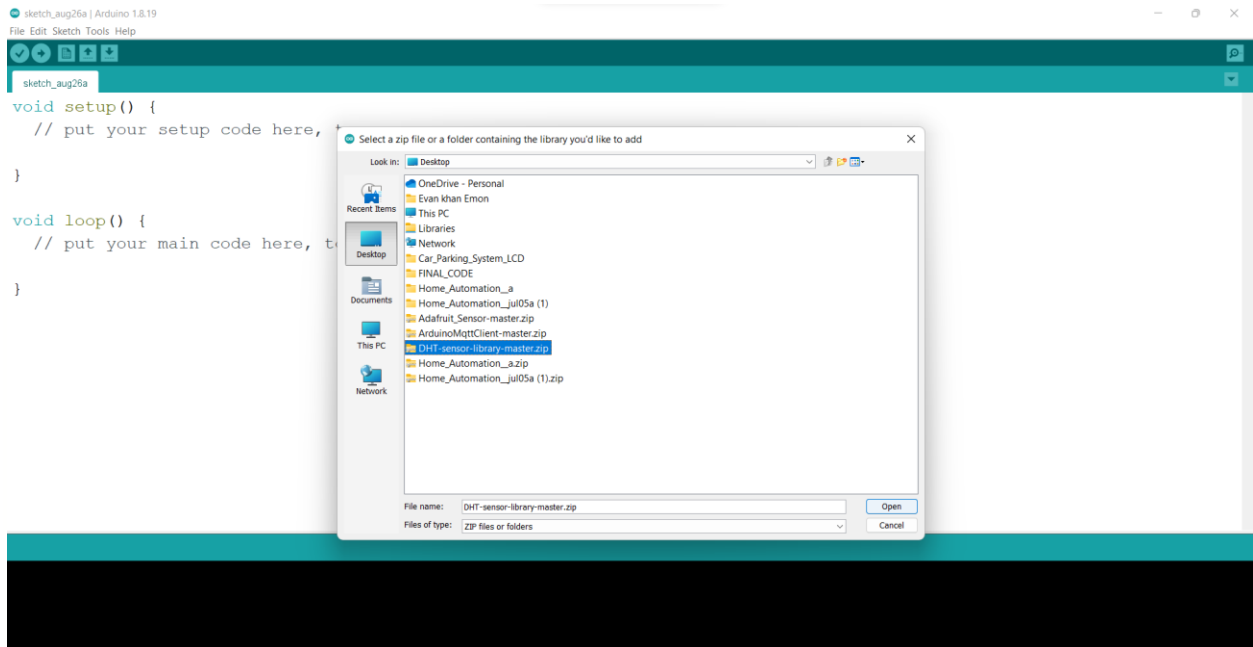
**Figure 40: Arduino Code Example**

### 3.4.3 How to Add Library in Arduino IDE



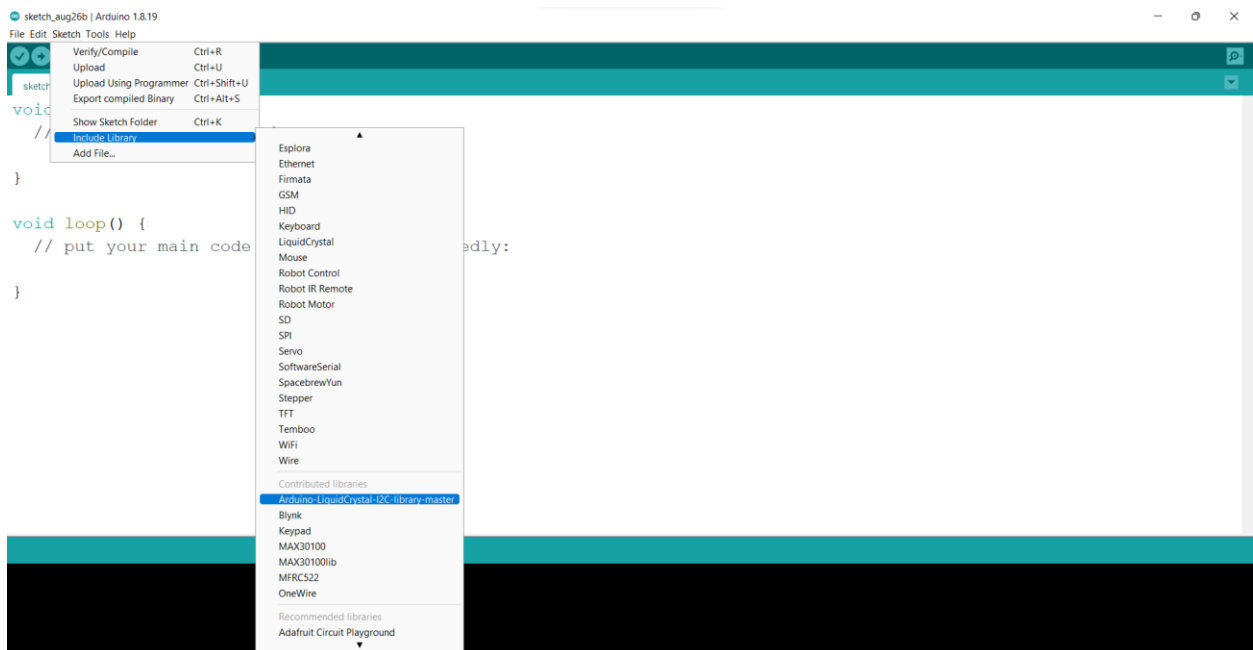
**Figure 41: How to Add Library in Arduino IDE (1)**

This is the way to add an Arduino library file in Arduino IDE. Note that, you must download the zip file from the internet that you wanted to add.

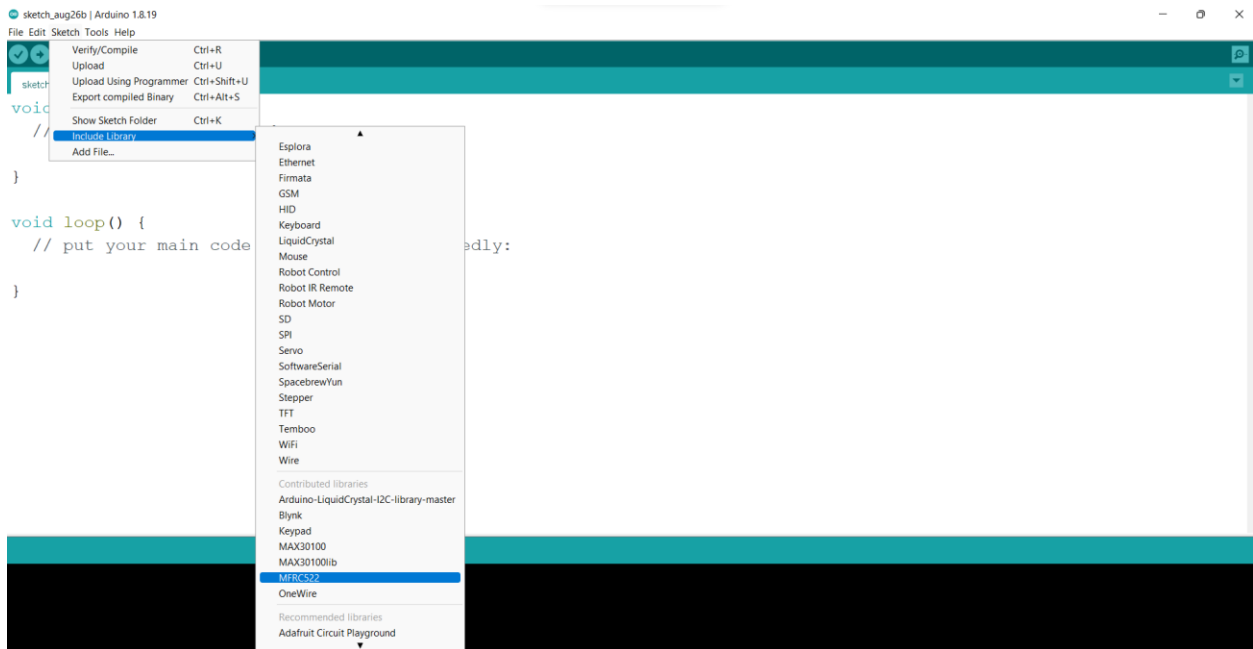


**Figure 42: How to Add Library in Arduino IDE (2)**

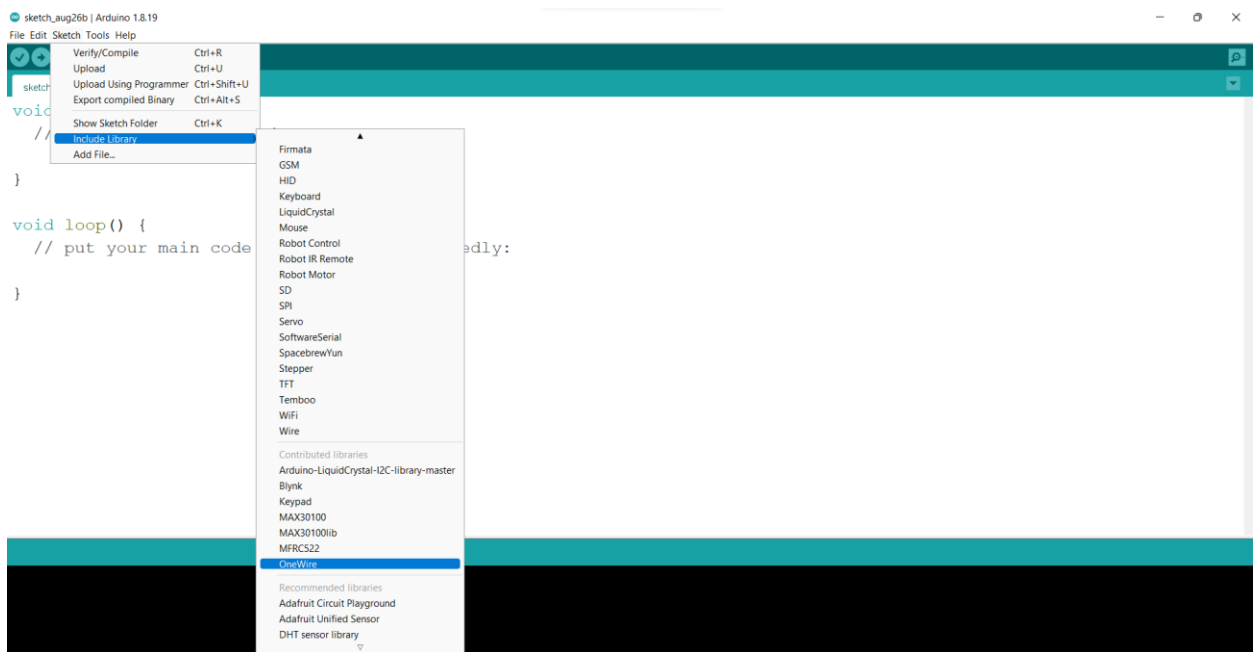
We are using different components in our project. So, we need to add few libraries for the code. Like Liquid Crystal I2C library, mfrc522 library, wire library etc.



**Figure 43: Adding Liquid crystal I2C Library**



**Figure 44: Adding mfrc522 Library**



**Figure 45: Adding one wire Library**

### 3.4.4 How to Compile Sketch

To compile the sketch, write full code for the project and select the board and port. After selecting the board and port, if there is no error the sketch will upload to the selected board. And ready to perform.

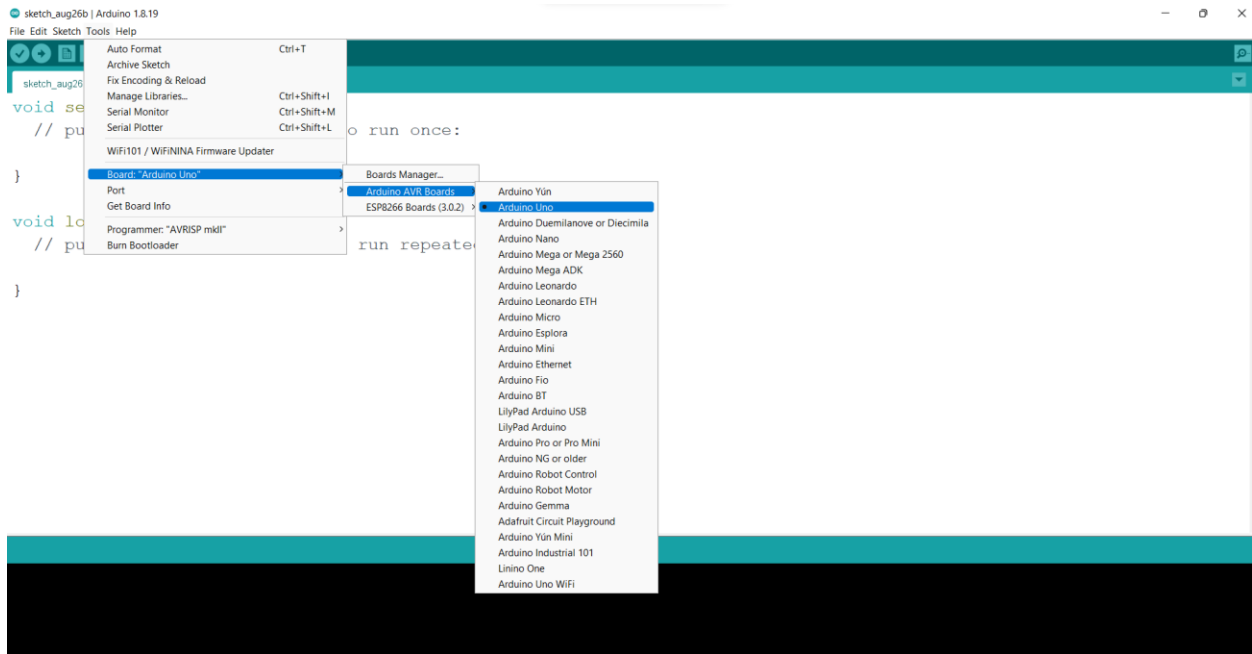


Figure 46: Selecting Board for compile

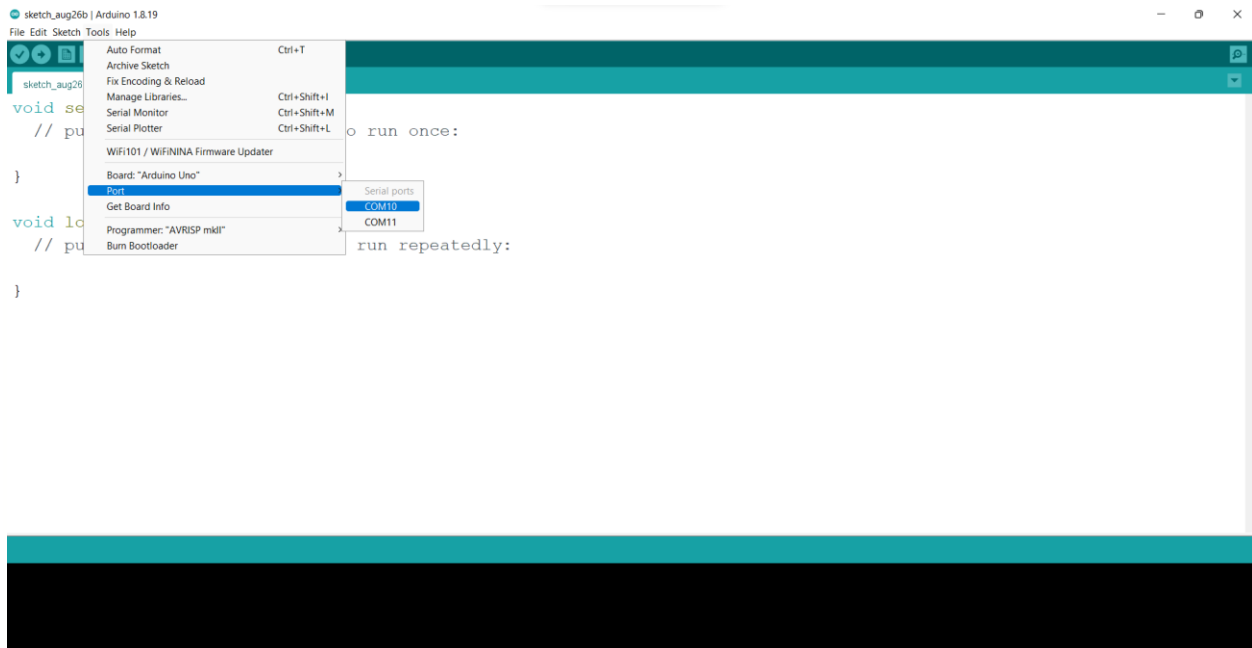
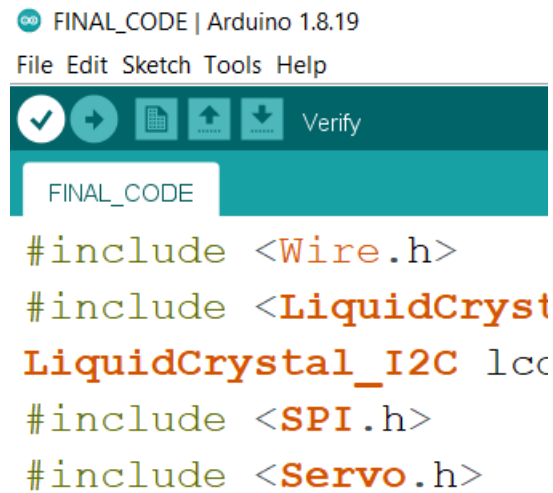


Figure 47: Selecting port for compile



**Figure 48: Verify the code**



**Figure 49: Upload the code**

If there is no error in the code, they will successfully upload to the selected Board.

```

FINAL_CODE | Arduino 1.8.19
File Edit Sketch Tools Help

FINAL_CODE
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD address to 0x27 for a 16 chars and 2 line display
#include <SPI.h>
#include <Servo.h>
#define buzzer A2
#define SS_PIN 10
#define RST_PIN 9
Servo myservo;
#include "DHT.h"
#define flame A1
#define DHTPIN A0 // Digital pin connected to the DHT sensor
#define DHTTYPE DHT11 // DHT 11
DHT dht(DHTPIN, DHTTYPE);
#define ir_enter 6
#define ir_back 2

#define ir_car1 3
#define ir_car2 8

Done compiling

Sketch uses 10486 bytes (32%) of program storage space. Maximum is 32256 bytes.
Global variables use 597 bytes (29%) of dynamic memory, leaving 1451 bytes for local variables. Maximum is 2048 bytes.

```

**Figure 50: Done compiling**

## 3.5 Pictorial Explanation of all Requirements

### 3.5.1 ARDUINO UNO

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It holds a USB port, a power jack, an ICSP header, a reset button, and 14 digital I/O pins (6 of which can be utilized as PWM outputs). It has everything you need to get started with the microcontroller; you just plug it into a computer with a USB wire or power it with an AC-to-DC adapter or battery. The Uno is a microcontroller board form on the ATmega328P. It has 14 digital input/output pins, 6 analogue inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button Arduino Software (IDE) makes code writing and uploading easy. The system is built in Java and Processing. This software is Arduino-compatible. Arduino boards use many microprocessors and controllers. Microcontrollers are typically programmed using C and C++ features.

### Arduino UNO Specifications

**Table 3: Arduino UNO Specifications**

|                             |            |
|-----------------------------|------------|
| Microcontroller             | ATmega328P |
| Operating Voltage           | 5V         |
| Input Voltage (recommended) | 7-12V      |
| Input Voltage (limit)       | 6-20V      |

|                         |                                                               |
|-------------------------|---------------------------------------------------------------|
| Digital I/O Pins        | 14 (of which 6 provide PWM output)                            |
| PWM Digital I/O Pins    | 6                                                             |
| Analog Input Pins       | 6                                                             |
| DC Current per I/O Pin  | 20 mA                                                         |
| DC Current for 3.3V Pin | 50 mA                                                         |
| Flash Memory            | 32 KB (ATmega328P) of which 0.5 KB is used by the bootloader. |
| SRAM                    | 2 KB (ATmega328P)                                             |
| EEPROM                  | 1 KB                                                          |
| Clock Speed             | 16 MHz                                                        |
| Length                  | 68.6 mm                                                       |
| Width                   | 53.4                                                          |
| Weight                  | 25g                                                           |

## Arduino Uno Pinout

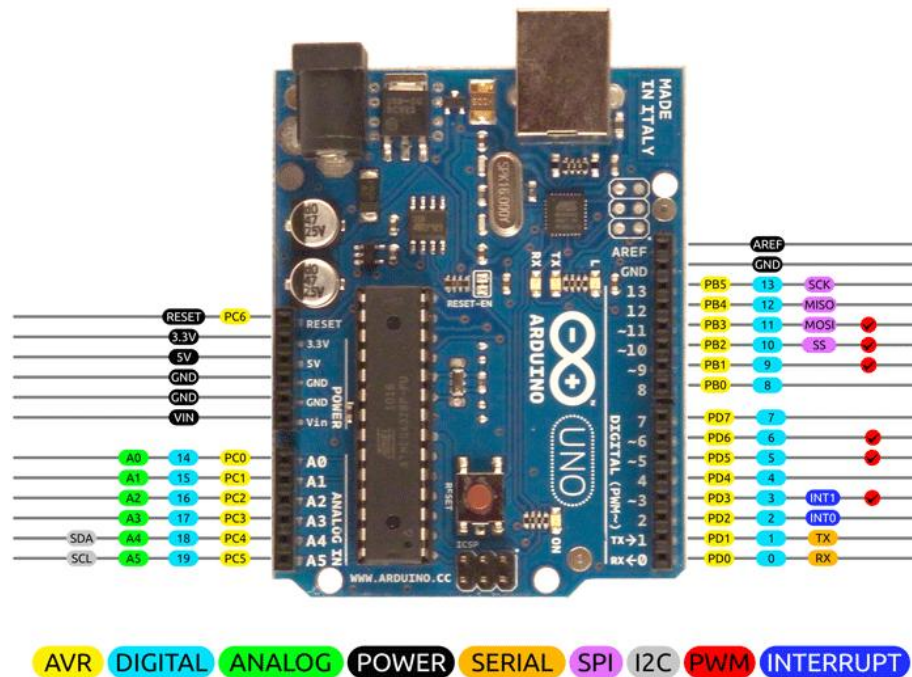


Figure 51: Arduino UNO

**Table 4: Arduino Uno Pinout**

| <b>Classification of Pins</b> | <b>Pin Name</b>                            | <b>Description</b>                                                                                                                                                                        |
|-------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power                         | Vin, 3.3V, 5V, GND                         | Vin: Arduino's external voltage input.<br>5V: Regulated power source for microcontroller and board components.<br>3.3V: On-board regulator generates 3.3V. 50mA max.<br>GND: ground pins. |
| Reset                         | Reset                                      | Performs a factory reset on the microcontroller.                                                                                                                                          |
| Analog Pins                   | A0 – A5                                    | Used to give analog input between 0V and 5V.                                                                                                                                              |
| Input/Output Pins             | Digital Pins 0 - 13                        | Pins can function either as inputs or outputs.                                                                                                                                            |
| Serial                        | 0(Rx), 1(Tx)                               | Used for both receiving and sending serial data via the TTL protocol.                                                                                                                     |
| External Interrupts           | 2,3                                        | To trigger an interrupt.                                                                                                                                                                  |
| PWM                           | 3, 5, 6, 9, 11                             | Enables 8-bit PWM output.                                                                                                                                                                 |
| SPI                           | 10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK) | Used for communication within the SPI.                                                                                                                                                    |
| Inbuilt LED                   | 13                                         | To activate the built-in light indicator.                                                                                                                                                 |
| TWI                           | A4 (SDA), A5 (SCA)                         | Used for communication with the TWI.                                                                                                                                                      |
| AREF                          | AREF                                       | To give reference voltage for voltage source.                                                                                                                                             |

### 3.5.2 NODE MCU ESP8266 Wi-Fi Module

Node MCU is a network that can link objects and allow data transfer via the Wi-Fi protocol. It is based on the ESP8266 microcontroller. The ESP8266 Wi-Fi Module is a self-contained SOC with an integrated TCP/IP protocol stack that lets any microcontroller connect to your Wi-Fi network. The ESP8266 can either run an application on its own or take over all Wi-Fi networking tasks from another application processor. This means that you can just connect it to your Arduino device and get approximately the same Wi-Fi capabilities as a Wi-Fi Shield. The ESP8266 module is a board that is very cheap and has a very large and growing community.

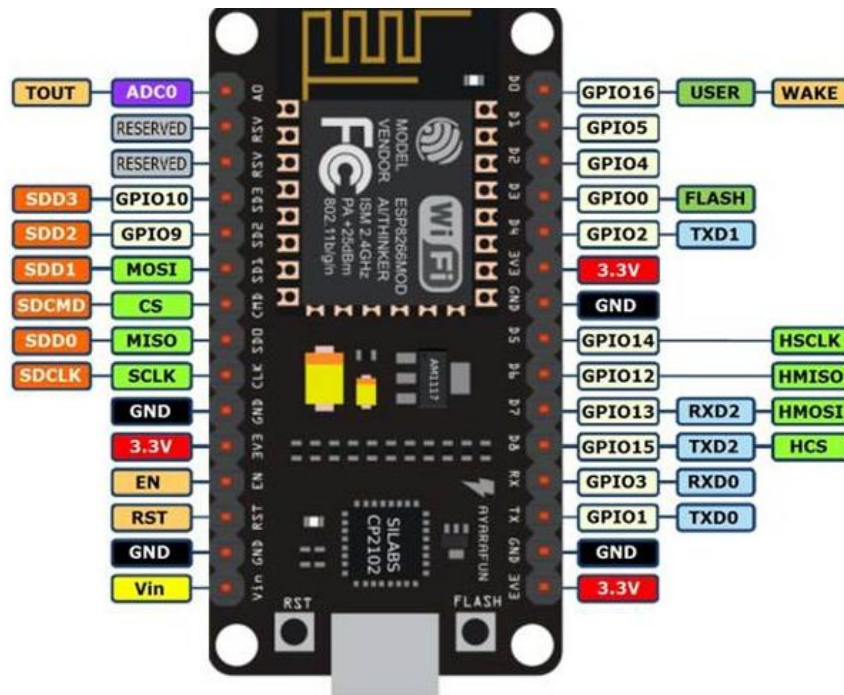


Figure 52: Node MCU

## ESP8266 Node MCU Pinout

The ESP8266 Node MCU has a total of 30 pins, which are all responsible for its connectivity to the external world. Here are the relationships between them:

GND is a ground pin of the ESP8266 Node MCU development board. There are four power pins. VIN pin and three 3.3V pins. I2C Pins are used to connecting sensors and other I2C devices to your project. I2C Master and I2C Slave can both be used. GPIO Pins ESP8266 Node MCU has 17 GPIO pins which can be assigned to various functions such as I2C, I2S, UART, PWM, IR Remote Control, LED Light and Button programmatically. ADC Channel the Node MCU is embedded with a 10-bit precision SAR ADC. UART Pins ESP8266 Node MCU has 2 UART interfaces, i.e. UART0 and UART1, which provide asynchronous communication (RS232 and RS485) and can communicate at up to 4.5 Mbps. SPI Pins ESP8266 have two SPIs (SPI and HSPI) that can be used in slave and expert modes. ESP8266's SDIO pins interface SD cards directly. PWM Pins The board has 4 channels of Pulse Width Modulation (PWM). Control Pins are used to controlling ESP8266.

## Features:

- Wi-Fi Direct (P2P).
- Built-in TCP/IP protocol stack.
- Built-in TR switch, balun, LNA, power amplifier and matching network.
- Built-in PLLs, regulators, DCXO and power management units.

- +19.5dBm output power in 802.11b mode.
- Power down leakage current of <10uA.
- 1MB Flash Memory.
- Built-in low-power 32-bit CPU could be used as an application processor.
- SDIO 1.1 / 2.0, SPI, UART.
- STBC, 1×1 MIMO, 2×1 MIMO.
- A-MPDU & A-MSDU aggregation & 0.4ms guard interval.
- Wake up and transmit packets in < 2ms.
- Power consumption of < 1.0mW (DTIM3).

### 3.5.3 4-Channel Relay Module (5v)



**Figure 53: 4 Channel Relay Module (5v)**

A relay is an electrical device that is typically employed to manage high voltages with low voltage. The circuit is completed with a coil looped around a pole and two little metal flaps. One node is stationary, while the other is mobile. When electricity flows into the coil, a magnetic field is generated that pulls the moving node toward the stationary node, therefore completing the circuit. So, we can complete the circuit for the high voltage to go by providing a modest amount of electricity to the coil.

This is a USB-controlled relay board with four channels. The USB relay board has 4 SPDT 10A relays. One relay unit may control 4 220V/120V devices.

This is a relay board with 4 SPDT Relays operated through the USB port of your computer. This USB relay module's main goal is to help you build projects for home automation.

### Relay Module Specifications

- 3.75V up to 6V is the supply voltage.
- Trigger current – 5mA.
- When the relay is engaged, the current is between 70 and 300 milliamperes.

- Relay higher contact voltage – 250VAC, 30VDC.
- Current limit for the relay is 10A.

The four-channel relay module has four 5V relays with the related switching and isolating components, making connecting with a microcontroller or sensor simple with few components and connections. There are two terminal blocks with six terminals each, and every block is shared by two relays. The terminals are screw-type, making connections to the main supply wiring simple and adaptable. The four relays just on the module are rated for 5V, which indicates the relay is engaged when there is roughly 5V across the coil. Each relay's contacts are rated for 250VAC, 30VDC, and 10A, as indicated on the relay's body.

Switching transistors buffer high-current relay coils and low-current inputs. They magnify input signals to operate relay coils. The isolation jumper is optional. The input jumper has V-CC, GND, and input pins for female jumper wire connections

## **Relay Module Inputs**

The module receives power through the pin labelled VCC and ground through the pin labelled GND. The relays are activated by low inputs to IN1, IN2, IN3, and IN4.

## **Relay Module Outputs**

There are four relays with dry contact outputs each. That is, each relay has a common (COM) terminal, a normally open (NO) port, and a normally closed (NC) port.

## **Relay Load (Contact Capacity of Relay)**

- 7 A @ 230-250 VAC
- 10 A @ 120 VAC
- 10 A @ 24 VDC
- CN1 – CN4 Connector: Relay 1 to 4 (S1 to S4) Output (Normally Open/Normally Close)
- CN5 Connector: Control Signal Input, Trigger 2 to 5 VDC and Supply Input 12 VDC
- D2, D4, D6, D8: Relay On/Off LED Indication
- CN6 Connector: Axillary Supply 12 VDC

## **Relay Advantages**

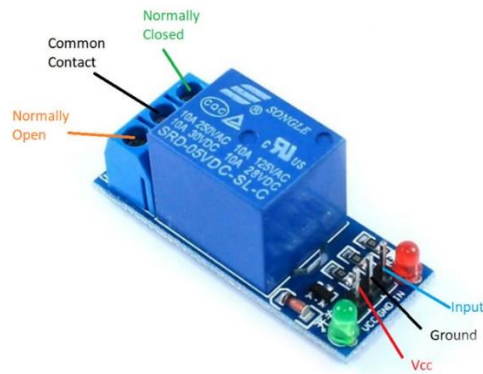
- A remote device's control is straightforward and simple.
- It requires less electricity to activate, yet it is capable of activating high-power devices as well.

- It is simple to modify one's contacts.
- Using just one signal, several connections may be activated and deactivated at the same time.
- It is possible to separate the activating portion.
- It is able to switch between AC and DC.
- It performs exceptionally well when subjected to extreme temperatures.

## Relay Disadvantages

- When the connections of relay modules are utilized repeatedly, this might cause them to deteriorate over time.
- When the contacts are opened and closed, there is a potential for noise to be produced.
- The amount of time needed to transition is considerable.

## Relay Module Pin Configuration



**Figure 54: 4 Channel Relay Module Pin Configuration**

**Normally Open (NO):** This pin is open except we supply a command to the relay module's signal pin. This pin is considered to be normally open. Therefore, in order to form a communication through the NO pin, the common contact pin must first break its link through the NC pin.

**Common Contact (CC):** This pin is utilized to link through the load that we would like to switch utilizing the module. It is referred to as "Common Contact."

**Normally Closed (NC):** A normally closed circuit is formed when the COM pin and this NC pin are linked to one another. Therefore, once the relay is switched by supplying an operational high/low signal towards the signal pin from a microcontroller, this NC connection will become broken.

**Signal Pin:** It controls relays. This pin is active low or high. In active low conditions, the relay triggers when we supply a low signal to the signal pin, but in an active high instance, we can provide the high signal.

**5V VCC:** This pin requires 5V DC.

**Ground:** Connects the power supply's GND terminal.

These modules function on an active high signal to improve the relay coil's contact with the usually open terminal.

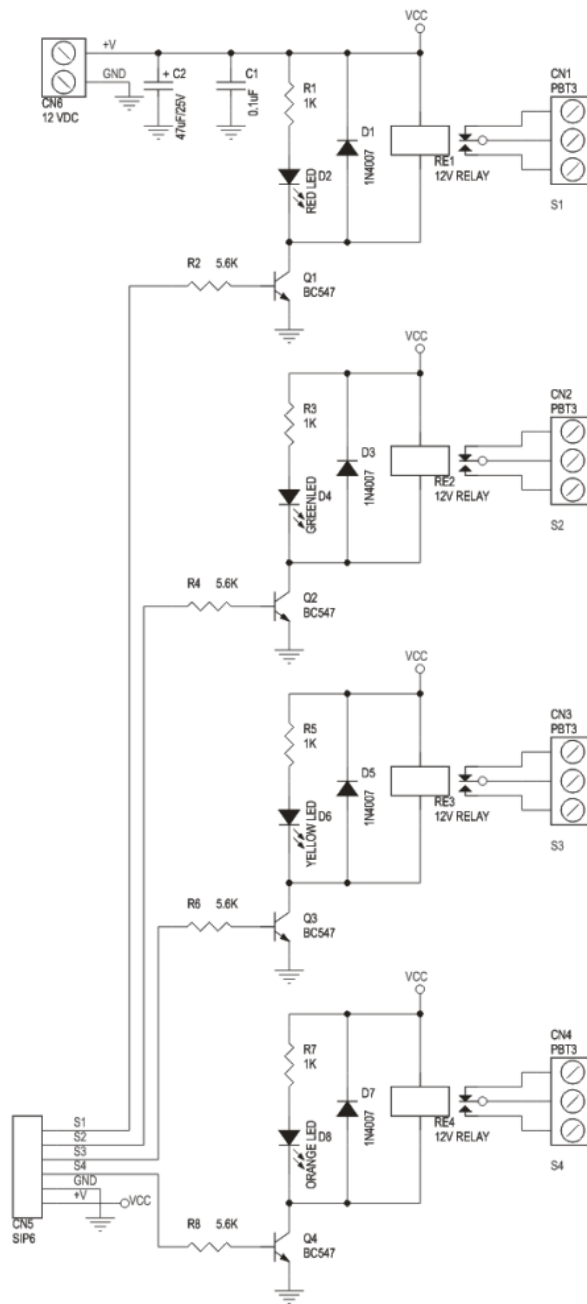
## Components Present on A Four-Channel Relay Module

5V relay, terminal blocks, male headers, transistors, optocouplers, diodes, and LEDs.

**Table 5: 4 Channel Relay Module Pinout**

| Pin No | Pin Name           | Details                                                |
|--------|--------------------|--------------------------------------------------------|
| 1      | GND                | Ground PIN                                             |
| 2      | IN1                | Input to trigger the operation of relay 1              |
| 3      | IN2                | Input to trigger the operation of relay 2              |
| 4      | IN3                | Input to trigger the operation of relay 3              |
| 5      | IN4                | Input to trigger the operation of relay 4              |
| 6      | V <sub>CC</sub>    | Relay module power                                     |
| 7      | V <sub>CC</sub>    | Power supply selection jumper                          |
| 8      | JD-V <sub>CC</sub> | Pin for supplementary power supply on the relay module |

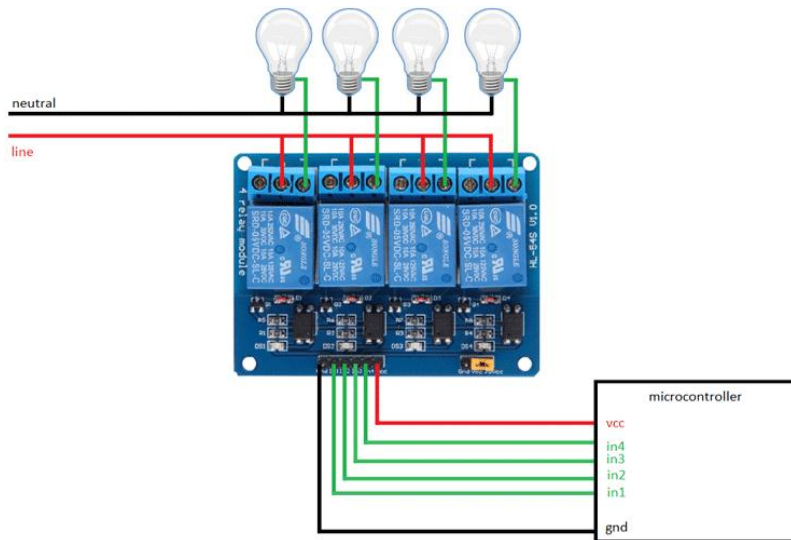
## 4 Channel Relay SCHEMATIC



**Figure 55: 4 Channel Relay Schematic**

## How To Use The Four-Channel Relay Module.

Since there are four relays within the same module, the four-channel can be utilized to switch more than one load at the same time. This is beneficial when creating a central hub that can power multiple loads far away. It may be used for applications such as home automation, where the module can be put in the main switchboard, attached to loads in other parts of the house, and managed by a microcontroller from a central location.



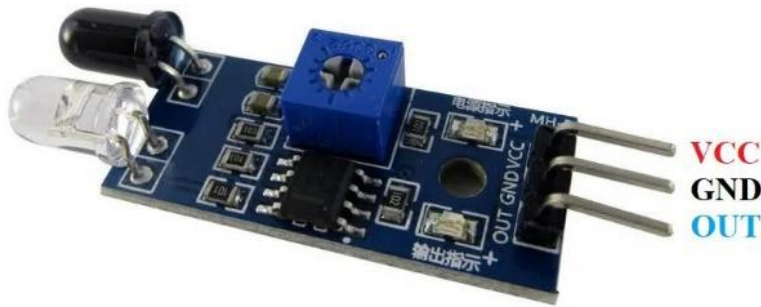
**Figure 56: How to use 4 Channel Relay**

In this diagram, the NO terminals of the relay have been attached to four different loads, which are shown by the light bulbs. Each relay's live wire is linked to the common terminal. Whenever the relays are turned on, the live wire is attached to the load and powers it. This configuration can be turned around by integrating the load to the NC terminal, which keeps it on until the relay is turned on.

### 3.5.4 IR Sensor

An infrared sensor is a type of electrical sensor that produces light in order to detect certain features of its environment. An IR sensor can detect motion as well as monitor the heat of an item. These sensors simply measure infrared radiation rather than emitting it, which is known as a passive IR sensor. Typically, all things in the infrared range emit some type of heat radiation. An infrared sensor is required in order to detect these types of radiations because they are not visible to the human eye. The detector is only an infrared photodiode that is sensitive to infrared light with the same wavelength as the emitter, which is an infrared light-emitting diode (Light Emitting Diode). The emitter is literally an infrared LED (Light Emitting Diode). In a photodiode, the amount of infrared light that causes a change in the resistances and output voltages of the diode is proportional to the intensity of the IR light.

### IR Sensor Module Pinout Configuration



**Figure 57: IR Sensor Module.**

**Table 6: Pin Configuration and Description**

| Pin Name | Description        |
|----------|--------------------|
| VCC      | Power Input        |
| GND      | Ground             |
| OUT      | Active High Output |

### **IR Sensor Working Method**

Infrared transmitters come in a variety of designs based on the wavelengths, output power, and reaction time they have. The two components of an IR sensor, an IR LED and an IR Photodiode, usually known as a PhotoCoupler or OptoCoupler, are used.

### **IR Transmitter or IR LED**

An infrared transmitter, usually known as an IR LED, is a light-emitting diode (LED) that produces infrared radiation. Although an IR LED seems to be a regular LED, the radiation it emits cannot be seen by the human eye.



**Figure 58: IR Transmitter or IR LED**

## IR Receiver or Photodiode

Infrared radiation from an IR transmitter is collected by infrared receivers or infrared sensors. Photodiodes and phototransistors are used as IR receivers. As opposed to regular photo diodes, infrared photodiodes can pick up on infrared light.



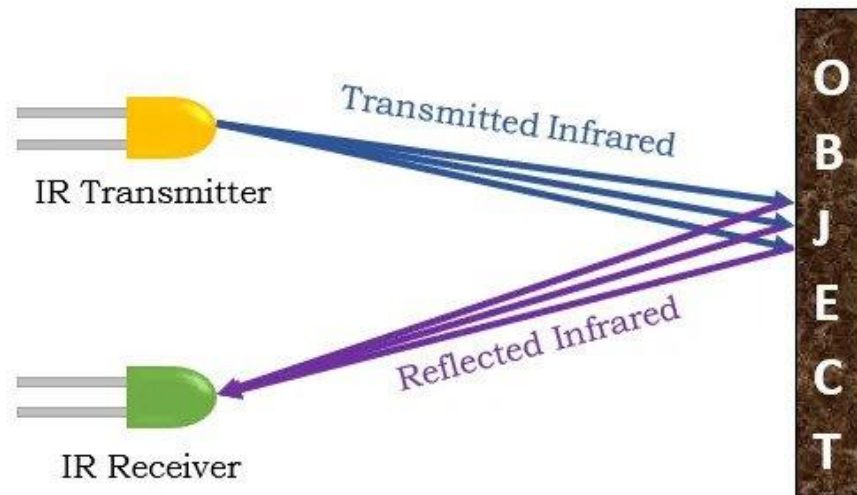
**Figure 59: IR Receiver or Photodiode**

An infrared sensor operates in the following circumstance:

To emit the desired wavelength, an IR source (transmitter) is utilized. This radiation is absorbed and reflected by the target. The radiation that is reflected is recognized by the IR receiver.

The radiation detected by the IR Receiver is then analyzed depending on its intensity. In most cases, IR Receiver output is minimal, and amplifiers are utilized to increase the detected signal.

The typical operation of an IR sensor detection technique is displayed in Figure below.



**Figure 60: IR Sensor Detection Technique.**

## Advantages of IR Sensor

- lower energy consumption

An important benefit of an IR sensor is that it uses less energy.

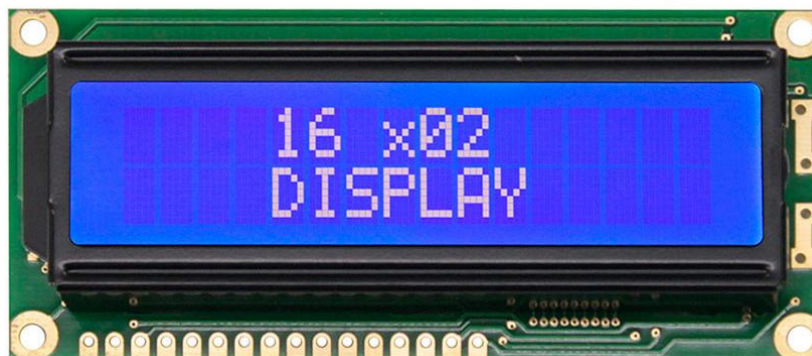
- Contact-less monitoring  
An IR sensor can detect an item without physical touch.
- No data leaking  
There is no data loss due to beam direction with IR sensors.
- Immune to noise  
IR sensors are extremely resistant to noise.
- Antioxidant and anticorrosion  
These sensors are impervious to oxidation and corrosion.
- Low sensitivity to light  
IR sensors may reliably detect motion regardless of the presence or absence of light.

### **Limitations of IR Sensor**

- Visibility is important  
A direct line of sight is required for the optimal operation of an infrared (IR) sensor.
- limited scope  
IR sensors have a restricted range and hence cannot be utilized in all circumstances.
- Sensitive to climate and environmental circumstances  
IR sensors are impacted by climate, temperature, and humidity fluctuations. Thus, fog, rain, and dust can conflict with its operation.
- Less data transfer rate  
IR sensors also lack a high transmission rate, which is another restriction.

### **3.5.5 16\*2 LCD Display**

An I2C LCD display is often made up of an HD44780-based character LCD display and an I2C LCD adapter. Let's get to recognize them individually at a time.



**Figure 61: 16\*2 LCD Display.**

An I2C LCD display is often made up of an HD44780-based character LCD display and an I2C LCD adapter. Let's get to recognize them individually at a time. That make up a standard I2C LCD display. Let's take it one at a time and get to know each of them. The advantage of an I2C LCD is the ease of wiring. To control the LCD, you simply need two data pins.

### **Technical Specification**

- Compatible with Arduino/Genuino UNO, Leonardo, Mega, 101 (Intel Curie), Micro, Nano, Mini.
- I2C Address: 0x20-0x27 (0x20 default).
- Backlit.
- Input voltage: 5V.
- Interface: I2C/TWI x1, Gadgeteer interface x2.
- Adjustable contrast.
- Size: 80x36x20mmz(3.1x1.4x0.7in).

### **Character LCD Display**

These LCDs are great for displaying solely text or characters, which is consistent with their name. For instance, a 16-by-2-character liquid crystal display (LCD) features an LED backlight and is capable of displaying 32 ASCII characters over two rows of 16 characters each.

If you look closely you can see tiny rectangles for each character on the display and the pixels that make up a character. Each of these rectangles is a grid of 5×8 pixels.

### **I2C LCD Adapter**

PCF8574 is an 8-bit I/O expander chip, and it serves as the adapter's central processing unit. This chip is responsible for converting the data coming from an Arduino's I2C bus into the parallel data that is necessary for an LCD display.

The board also includes a trimpot, which allows for more precise adjustments to be made to the contrast of the display.

In addition, it has a jumper just on board that can be used to control the amount of power that is sent to the backlight. You may adjust the brightness of the backlight by removing the jumper and applying an additional voltage to the header pin that is labelled with an LED. This will allow you to do so.

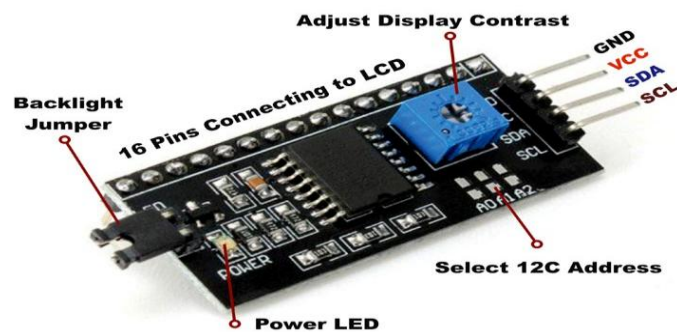
### **I2C Address of LCD**

It is possible that you will need to establish a distinct I2C address for the LCD adapter if you are utilizing numerous devices on the same I2C bus. This will ensure that the LCD adapter does not cause a conflict with another I2C device.

This is accomplished by the adapter having three solder jumpers or solder pads labeled A0, A1, and A2.

Each of these is utilized in order to firmly establish the address. The address can be changed by temporarily shorting out a jumper with a glob of solder.

It is crucial to note that a number of different businesses, including Texas Instruments and NXP Semiconductors, among others, are responsible for the production of the PCF8574 chip. And the manufacturer of the chip in your LCD determines the I2C address it uses.



**Figure 62: I2C driver Pinout.**

**GND:** A ground pin is denoted by GND. Establish a connection between it and the ground on the Arduino.

**VCC:** Both the module and the LCD receive their power from VCC. You can attach it to the 5V output on the Arduino or to an external source of 5V power.

**SDA:** I2C communicates via the SDA data pin. You will need to attach it to the I2C data pin on the Arduino.

**SCL:** I2C uses the SCL pin as its clocking input. Establish a connection between it and the I2C clock pin on the Arduino.

## **Hooking up an Arduino Uno to an I2C LCD display**

I2C LCDs are significantly easier to connect than ordinary LCDs. Instead of 12 pins, you just need to connect four. To begin, connect the Arduino's VCC and GND pins to the board's 5V output.

All that's left are the I2C communication pins. It is important to remember that each Arduino board has a unique set of I2C pins that must be linked. The SDA (data line) and SCL (clock line) are located on the pin headers near the AREF pin on Arduino boards using the R3 configuration. Both A5 (SCL) and A4 (SCL) are commonly referred to (as SDA).

## Adjusting LCD Contrast

After you have completed the initial wiring of the LCD, you will need to make adjustments to the contrast of the panel. You will discover a potentiometer on the I2C module, and you can turn it using a little screwdriver.

To supply electricity to the LCD, plug in the USB port found on the Arduino. You will notice that the backlight is illuminated. The first row of rectangles will become visible as soon as you begin to adjust the knob just on the potentiometer. If that does occur, you should be very proud of yourself! Your LCD display is operating normally.

After this is finished, we will be able to begin programming the LCD.

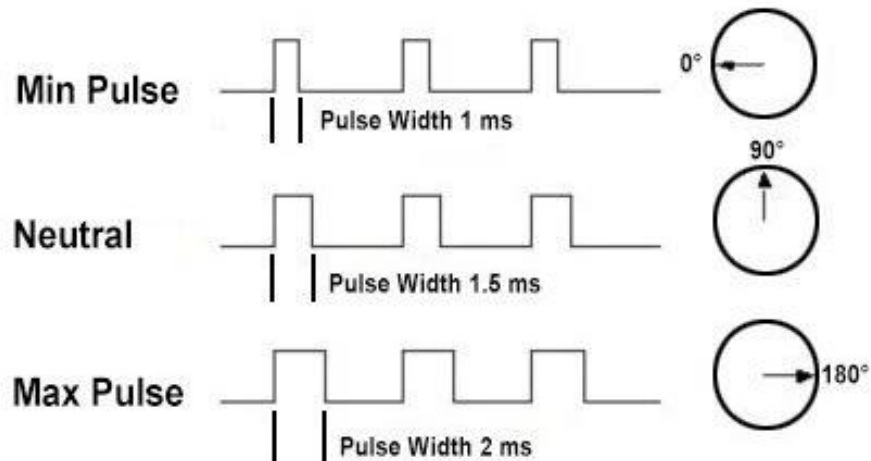
### 3.5.6 Servo Motor

A servo motor is a type of motor that can rotate with pinpoint accuracy. This type of motor typically includes a control circuit that offers a response to the current position of the motor shaft; this feedback allows servo motors to rotate with absolute accuracy. A servo motor is used when you wish to spin an object at a specified angle or distance. It is simply a motor that is controlled by a servo system.



**Figure 63: Servo Motor**

The servo motor can be controlled by delivering a series of pulses to the signal line. A standard analog servo motor anticipates receiving a pulse approximately every 20 milliseconds.



**Figure 64: Pulse Diagram**

The position of the servo motor is set by the length of the pulse.

- In 1ms, the servo angle will be zero.
- If the pulse is high for 1.5ms, then the servo will be at its center position.
- If the pulse is high for 2ms, then the servo will be at 180 degrees.
- 1ms to 2ms pulses rotate the servo shaft 180 degrees.

### **SG-90 Servo Motor Features**

- Operating Voltage is +5V typically
- Torque: 2.5kg/cm
- Operating speed is 0.1s/60°
- Gear Type: Plastic
- Rotation: 0°-180°
- Weight of motor: 9gm

### **SG-90 Servo Motor Pinout**

Servo motors have three connections:



**Figure 65: Servo Motor pinout.**

GND is common in both the motor and the logic. VCC 5V is a positive voltage that powers the servo. The signal is input for the control system.

### 3.5.7 RC-522 RFID

A radio frequency identification system, also known as an RFID system, is made up of two primary components: a tag that is affixed to the item that needs to be recognized and a reader that scans the tag. A radio frequency module and an antenna that emits a high-frequency electromagnetic field comprise a reader. The tag, on the other hand, is typically a passive device (it does not have a battery). It is made up of a microchip that stores and processes data, as well as an antenna for receiving and sending signals.



**Figure 66: RC-522 RFID**

The RC522 RFID module normally comes with an RFID card tag and a 1KB memory key fob tag. The finest aspect is that it can write a tag, which means it can store any message.

The RC522 RFID reader unit is designed to produce an electromagnetic field at 13.56MHz and connect with RFID tags (ISO 14443A standard tags).

The reader can contact a microcontroller via a 4-pin SPI interface at up to 10 Mbps. It also supports I2C and UART protocols for communication. The operational voltage of the module spans from 2.5 to 3.3V, but the logic pins are 5-volt tolerant, so we can easily link it to an Arduino or other 5V logic microcontroller without having a logic level converter.

**Table 7: Technical Specifications**

|                              |                    |
|------------------------------|--------------------|
| Frequency Spectrum           | 13.56 MHz ISM Band |
| Host Compatibility           | SPI / I2C / UART   |
| Operating Power Voltage      | 2.5 V to 3.3 V     |
| Maximum Operating Current    | 13-26mA            |
| Minimum Current (Power down) | 10 $\mu$ A         |
| Logic Inputs                 | 5V Tolerant        |
| Read Range                   | 5 cm               |

### RC522 RFID Module Pinout

The RC522 module can communicate with the outside world through a total of eight pins that are protruding from it. The following are the linkages that have been made:



**Figure 67: RC-522 RFID pinout**

**VCC:** The module receives its power from the VCC supply. This may range from 2.5 volts all the way up to 3.3 volts. It is possible to wire it to the 3.3V output that your Arduino provides. However, keep in mind that able to connect it to the 5V pin would very certainly render your module useless.

**RST:** RST is an input that may be used to reset or shut down the device. The module switches into power-down mode if this pin experiences a low state, a state in which the oscillator is switched off

and the input pins are cut off from their connections to the outside world. On the other hand, the module will be reset if the signal begins to rise.

**GND:** The GND pin serves as the ground, and it must be linked to the corresponding GND port on the Arduino.

**IRQ:** An RFID tag sends an interrupt signal to the microcontroller whenever it is detected nearby, and this signal is sent through the IRQ pin.

**MISO / SCL / Tx:** When the SPI interface is active, the MISO / SCL / Tx pin functions as master-in-slave-out; when the I2C interface is enabled, it functions as serial clock; and when the UART interface is enabled, it functions as serial data output.

**MOSI:** The MOSI input of the RC522 module is the SPI input. MOSI stands for "Master Out Slave In."

**SCK:** SCK, also known as the Serial Clock, is responsible for accepting the clock pulses that are sent either by SPI bus master, also known as Arduino.

**SS / SDA / Rx:** When the SPI interface is active, the SS / SDA / Rx pin functions as a signal input. When the I2C interface is enabled, it functions as a serial data input, while the UART interface is enabled, it functions as a serial data input. It is customary to label this pin by enclosing it inside a square, so that it may serve as a point of reference for determining the identity of other pins.

## How to use RC522 RFID Module

The RC522 can work with voltages between 2.5V and 3.3V, so it is usually powered by 3.3V and must be used with 3.3V communications systems. But this module's connectivity pins can handle 5V, thus it can be used with Arduino and other 5V microcontrollers even without extra hardware. The module can talk over SPI, IIC, and UART, but most people use SPI because it is the fastest and can send data at up to 10Mbps.

Since in use, the reader module will spend the majority of its time waiting for the tag to get close. The Reader has a power down mode that can be used to save power in applications that run on batteries. The IRQ pin on the module can be used to do this. During power down mode, the module will only use 10uA of current at most. Because Miguel Balboa has made the RC522 RFID Arduino library, it is easy to use the module with Arduino.

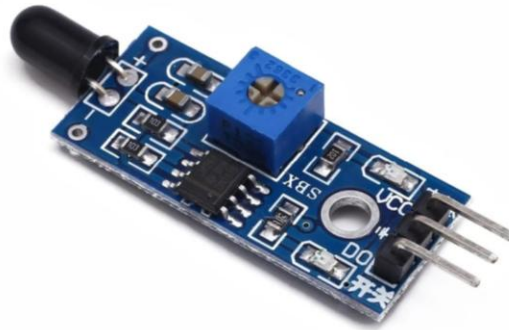
## RFID Working

The RFID module really consists of two separate modules. The first is an RFID tag, while the second is an RFID reader. RFID tags have a coil that, when the tag or card is placed closer enough to a reader, becomes activated. This causes the coil to communicate data to the reader.

## 3.5.8 Flame Sensor

A flame sensor is the most sensitive type of light sensor. This sensor module is specifically designed for use in fire alarms. This detector is sensitive to wavelengths of light between 760 nm and 1100 nm, which are characteristic of a flame. Unfortunately, this sensor is readily harmed by

hot environments. This sensor doesn't have to be right next to the fire, either; it can be set back a fair ways. It's possible to spot a flame from a distance of 100 cm, with a detection angle of 600 degrees. This sensor can be used to produce either an analog or digital signal.



**Figure 68: Flame Sensor**

This sensor/detector uses an electronic circuit and an electromagnetic radiation receiver. This sensor employs infrared flame flash to work through oil, dust, water vapour or ice.

### **Flame Sensor Features & Specifications**

- This sensor has the following characteristics.
- Photosensitivity is really high.
- The response time is quick.
- Simple in its operation
- The level of sensitivity may be adjusted.
- The angle of detection is six hundred degrees.
- It responds to the different flame temperatures.
- The level of precision may be adjusted.
- This sensor's operating voltage ranges from 3.3V to 5V.
- Digital switch o/ps and analog voltage o/ps.
- The PCB is 3cm x 1.6cm.
- Power and digital switch o/p indicators.
- If the flame intensity is less than 0.8m, the flame test can be engaged; if the flame intensity is strong, distance detection will be improved.

### **3.5.9 Transformer**

A transformer is an electrical device that uses electromagnetic induction to transmit energy between two or more circuits.

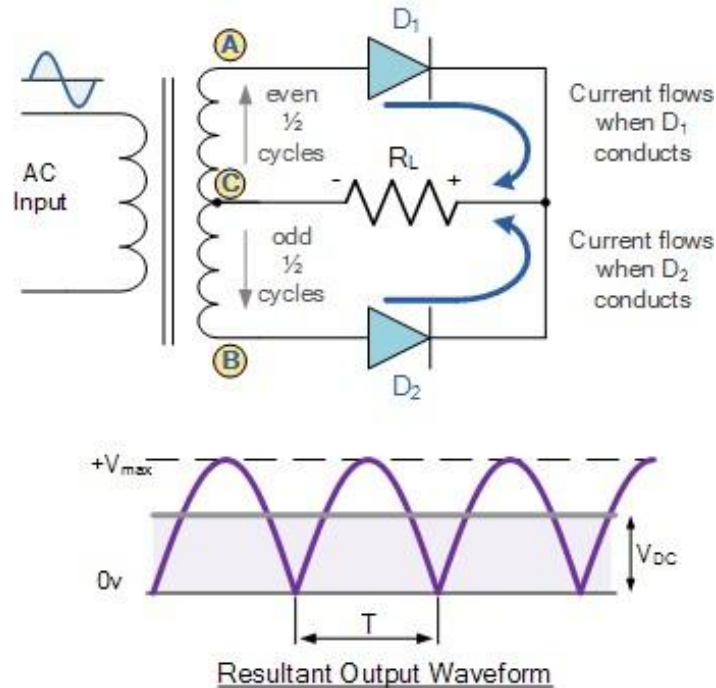
The 12-0-12 3 Amp Step-Down Transformer is a chassis-mounted mains transformer that can be used for a variety of purposes. The primary winding of the transformer is 230V, and the secondary winding is not tapped in the center. The transformer has connecting leads that are isolated and fly in different colours (Approx 100 mm long). The Transformer operates as a step-down transformer,

bringing the AC 230V voltage down to the AC 12V range. The transformer produces 12V and 0V outputs. The Entire Core and Winding information for the transformer is described below.



**Figure 69: Transformer**

The transformer is a piece of static electrical equipment that transmits energy among its winding circuits through the use of an inductive coupling. The transformer's cores are built of silicon steel with high permeability. The magnetizing current is significantly reduced and the flux is constrained to a channel that closely relates to the winding thanks to the steel's permeability, which is many times greater than that of open space.



**Figure 70: Resultant Output Waveform**

### Transformer 12V 3A Specifications

- Input Voltage: 230V AC
- Output Voltage: 12V, 12V or 0V
- Output Current: 3 Amp

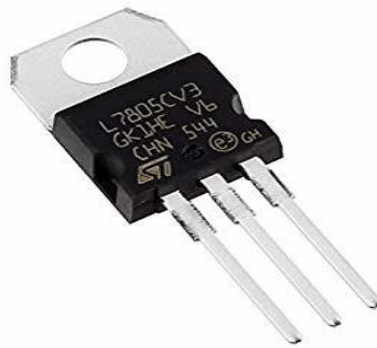
- Mounting: Vertical mount type
- Winding: Copper

### **Transformer 12V 3A Features**

- Soft Iron Core.
- 3 Amp Current Drain.
- 100% Copper Winding

### **3.5.10 LM7805 IC – Power Supply**

The integrated circuit LM7805 is a positive voltage regulator that outputs 5V. It is the integrated circuit of the LM78xx series of positive voltage regulators. This integrated circuit is commonly deployed in commercial equipment and educational projects because to its high reliability.



**Figure 71: LM7805 IC**

Voltage regulators are extremely common in circuits. They maintain a consistent output voltage independent of the input voltage. In our situation, the 7805 IC is an iconic regulator IC used in the vast majority of applications. The designation 7805 has two meanings: "78" indicates that it is a positive voltage regulator, and "05" indicates that it outputs 5V. Therefore, our 7805 will generate a +5V output.

The maximum output current of this IC is 1.5A. However, the IC suffers from significant heat loss, thus a heat sink is required for applications with a higher current consumption. If the input voltage is 12V and the current consumption is 1A, then  $(12 - 5) * 1 = 7W$ . These 7 Watts will be converted into heat.

#### **LM7805 IC Features:**

- L7805CV to give +5V regulated voltage
- Part number: L7805CV
- Numbers: LM7805, 7805Part
- Type: Positive fixed voltage regulator

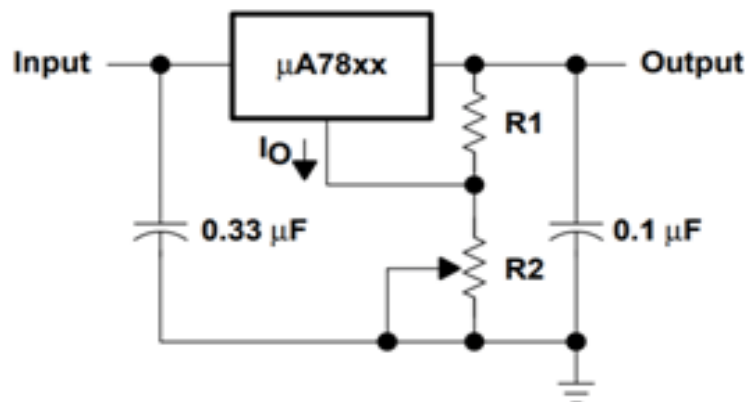
- Output voltage: +5VMax
- Output current: 1A
- Maximum input voltage: 35V

## LM7805 IC Technical Specifications

- TO-220 Package
- 1.5 Amps is the output current.
- Immediately shuts down when a short circuit happens
- Instant shutoff for too much heat
- low price
- Efficient to use in industrial applications
- 5V output that is exact and fixed
- Max input power is 35V DC
- Only 8mA flow when nothing is going on.

## 7805 As Adjustable Output Regulator

This IC may also function as an output current voltage regulator, which means you can change the output voltage using the circuit below.



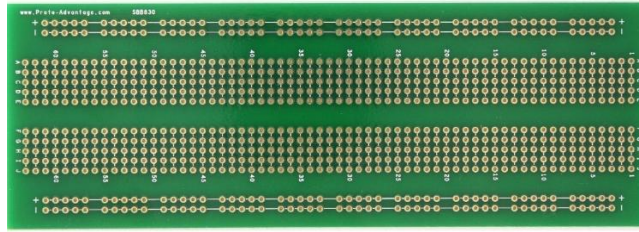
**Figure 72: Circuit Diagram of Amplifier**

In this case, the input voltage may be anywhere between 9V and 25V, and the value of the resistances R1 and R2 can be used to modify the output voltage. The value may be determined by using the formulas that are provided below.

$$V_O = V_{xx} + \left( \frac{V_{xx}}{R1} + I_Q \right) R2$$

Where,  $V_{xx}=5$ ,  $I_Q = 5 \times 10^{-3}$

### 3.5.11 Breadboard



**Figure 73: Breadboard**

These breadboards provide a permanent configuration for your electrical circuits. This kind of breadboard offers a stronger configuration. It has holes for electrical components as well as copper tracing. These components may be attached to the breadboard using a soldering iron, allowing an electrical connection to be made via the copper trace. Jumper wires are soldered between circuit components to provide a current path.

#### Solderable Breadboards Advantages

- These breadboards are extremely strong, and your circuit will be very protected on them.
- This breadboard provides the project with a more specialist appearance.
- When designing a circuit, it saves money and time.

#### Solderable Breadboards Disadvantages

- If something goes wrong in the circuit, de-soldering may destroy the parts.
- We can't use this board again.

### 3.5.12 Breadboard Wires

Breadboards use different kinds of wires depending on what they need to do. So, it's important to choose the right wires for breadboards if you want to build projects. Instead of basic wires, you can use the following types of wires.



**Figure 74: Jumper Wire.**

## Jumper Wires

Jumper wires, which are significantly longer, are used to create smaller circuits. These wires make it more difficult to hunt down faults. Jumper wires include metal pins attached to the end and are flexible, making them easy to put into the board. These cables are freely available yet inexpensive at electronics retailers.

## Jumper Wire Kit

The jumper wire kit is a type of plastic package that contains wires of various diameters, lengths, and colors. Both ends of these wires are wrapped at 90 degrees so they may be easily inserted into the breadboard. These wires are significantly more perfect and clean than long wires. Jumper wires with two or more connections are available in a variety of colours, making it difficult to design the circuit with a consistent colour code.

### 3.5.13 AC-DC 12V 3A Arduino Power Adapter

The Arduino can run on 6 to 20V through the DC Barrel Jack. If the board is supplied with a voltage below 7V, the unit's performance may become inconsistent, and the 5V I/O pin may deliver less than 5V, causing extra circuitry to malfunction. Using an over-voltage power source up to 20V will cause the regulators on the board to overheat. Inefficient and might overheat Arduino! 9V to 12V is advised. This allows the board regulators to readily waste unneeded energy and give the right voltage to the Arduino's I/O pins.



**Figure 75: 12v Arduino Power Adapter.**

## Specification

- Input voltage 110VAC~240VAC
- Output voltage of 12VDC
- Output current of 0~3A

### 3.5.14 NTE960 5V Positive Voltage Regulator

The NTE960 is a voltage regulator integrated circuit that operates at 5 volts and has a maximum output current of up to 1 amp. It features built-in safety against both heat overload and short circuits, and it does not require any additional equipment.

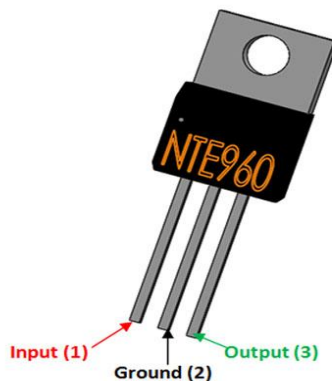


**Figure 76: 5v Voltage Regulator**

#### NTE960 Regulator Features

- 5V Positive Voltage Regulator
- Output current: 1A
- Lowest Input Voltage is 7V
- Voltage drop: 2V
- Load Regulation 100mV
- Protection against overheating overload and short circuits, as well as current limiting, is offered inside.
- Heat Dissipation maximum 125 degree Celsius
- Able to be done in the form of TO-220

#### NTE960 Regulator Pin Configuration



**Figure 77: 5v Voltage Regulator pinout.**

**Table 8: Pinout Details for Voltage Regulator**

| <b>Pin Number</b> | <b>Pin Name</b> | <b>Details</b>                             |
|-------------------|-----------------|--------------------------------------------|
| 1                 | Input (V+)      | Voltage at the Input That Is Not Regulated |
| 2                 | Ground (Gnd)    | It is Connected to Ground                  |
| 3                 | Output (Vo)     | Outputs Regulated +5V                      |

## **Chapter 4: System Design & Implementation**

## 4.1 Circuit Diagram

A circuit diagram is a simplified picture of the parts of an electrical circuit. It can use pictures of the parts or standard symbols to show what each part does. It shows where everything is in relation to everything else and how everything is connected. It is often used to show an electrician what the circuit looks like. We make a circuit diagram of our project. We make our project based on this design.

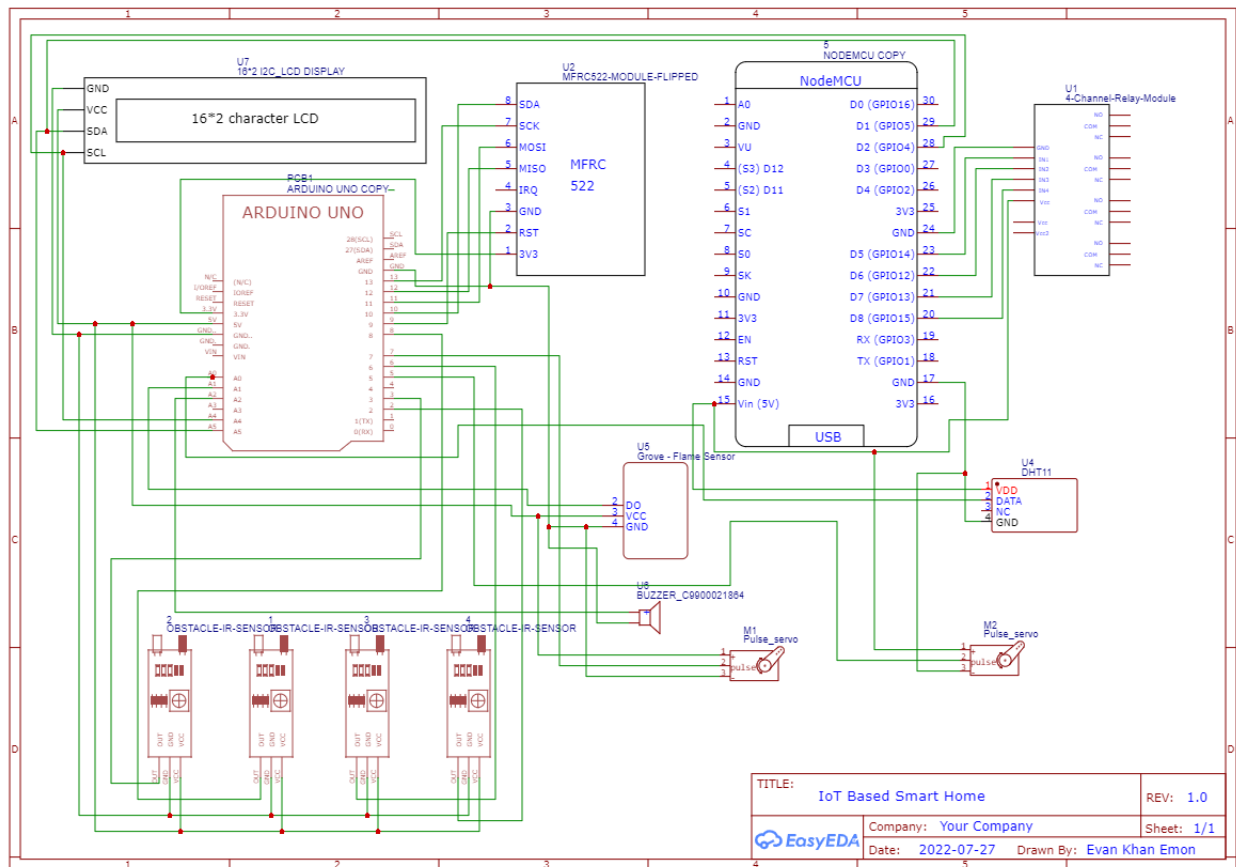
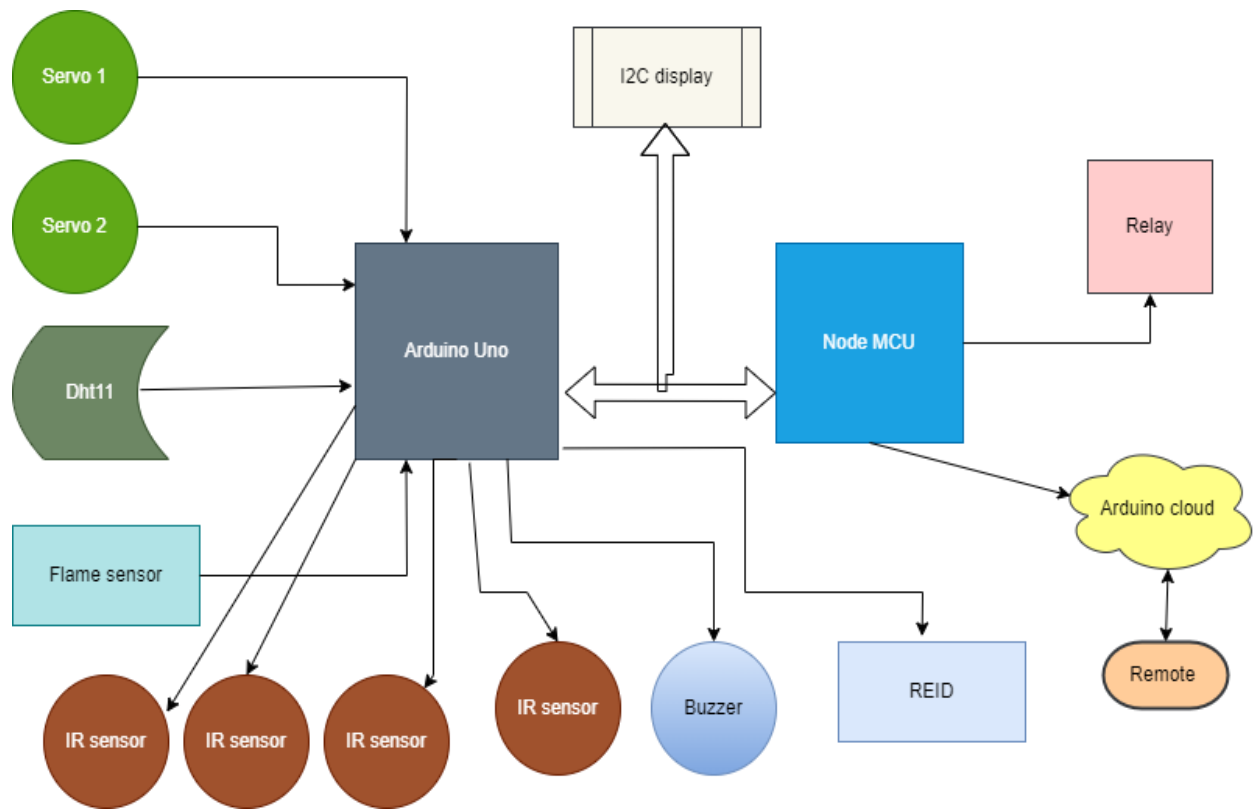


Figure 78: Circuit Diagram

## 4.2 Block Diagram

A block diagram is an advanced flowchart used in engineering. It is used to make plans for new systems or to define and improve ones that are already in place. Its structure gives a high-level look at the main parts of the system, the most influential important people in the process, and the most important working relationships. The Block diagram of our project is given below.

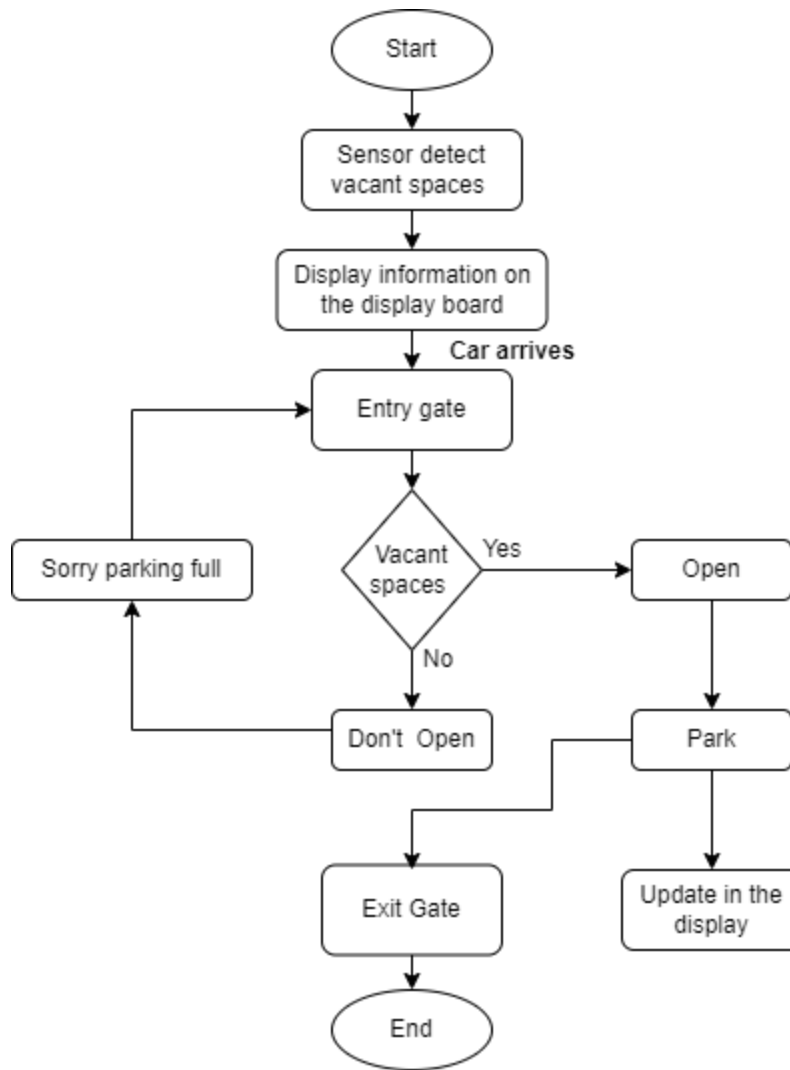


**Figure 79: Block Diagram**

## 4.3 Flow Chart

### 4.3.1 Flow Chart for Car Parking

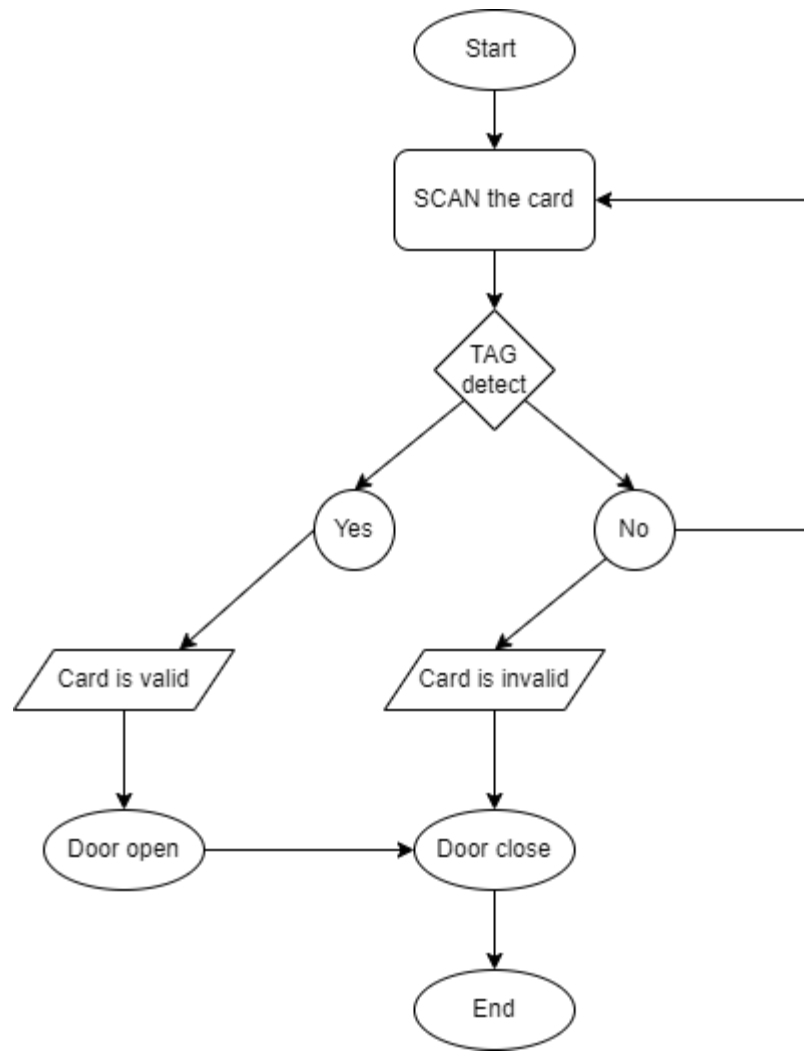
There are three parts in our project. Car parking is the first part of our project. This flow chart is based on the car parking system of the project.



**Figure 80: Flow Chart for Car Parking**

### 4.3.2 Flow Chart for RFID Card System

There are three parts in our project. RFID door opener is the second part of our project. This flow chart is based on the RFID card system of the project.

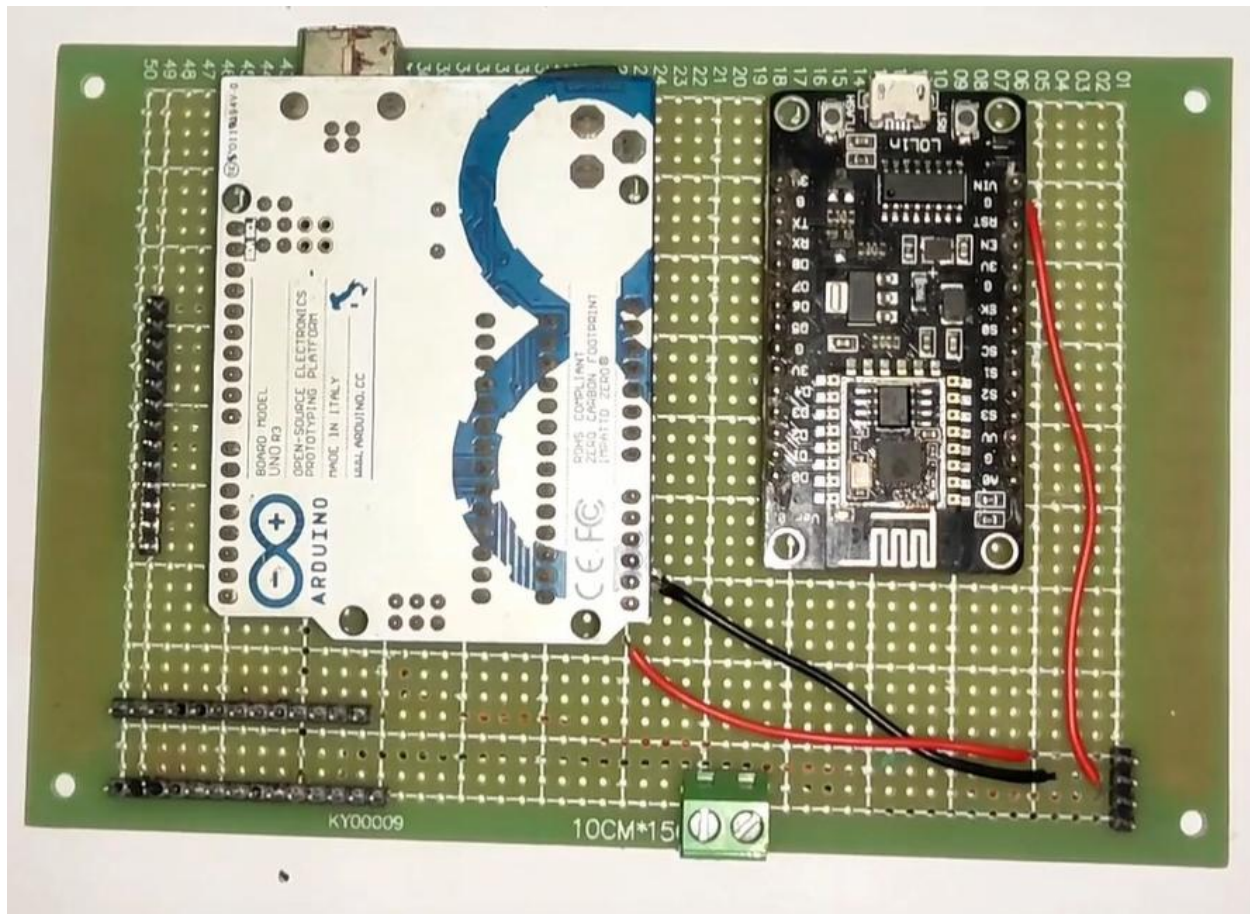


**Figure 81: Flow Chart for RFID card**

## **4.4 Implementation**

### **4.4.1 Circuit board**

After, completing the circuit design, block diagram and flow chart we need to implement this design. We started the implementation by making the circuit board.



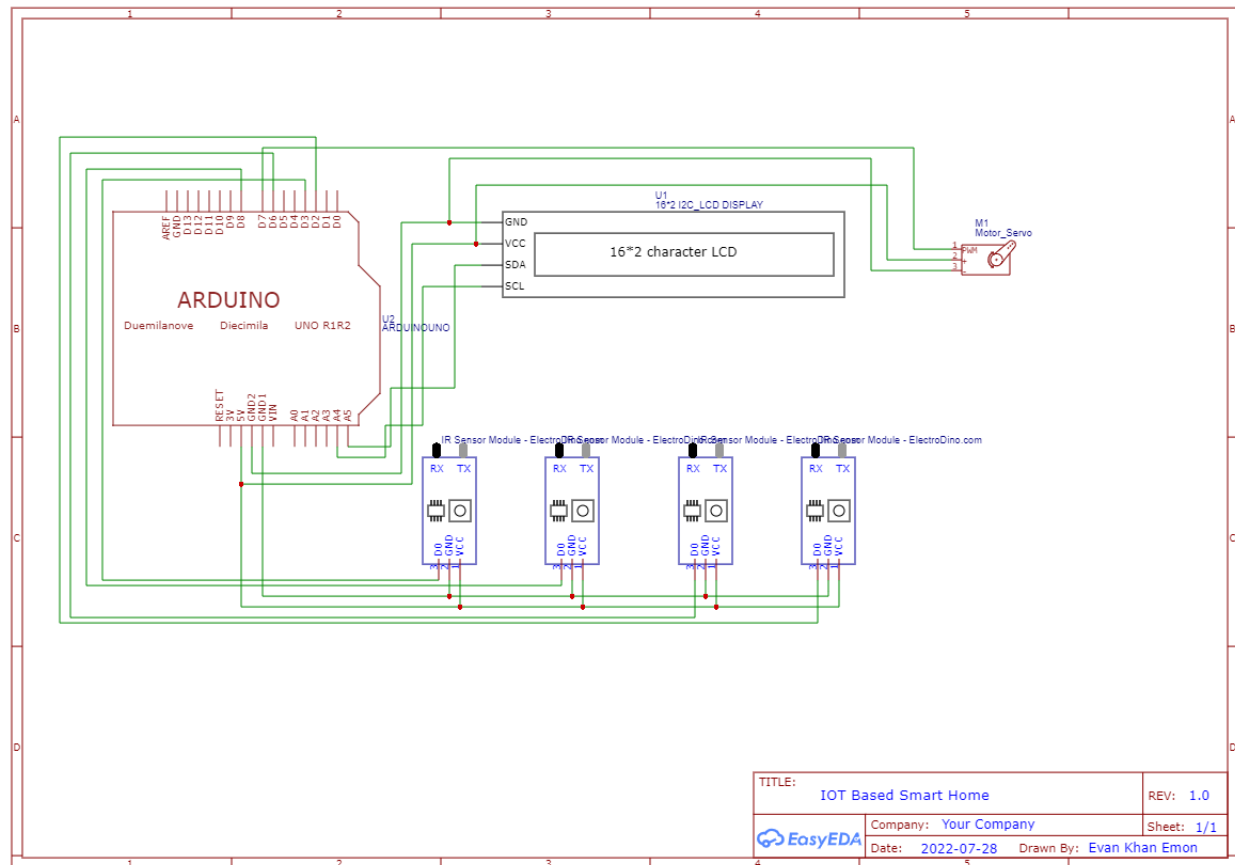
**Figure 82: Circuit Board**

This is the main circuit board of the project. Here we can see the different pins in the circuit board which is used to connect all the component. This is called plug and play circuit. If there is any problem in the component we can easily plug out the components and use a new one.

On this board, there are a lot of pins. In vertical, all the pin is used as Arduino UNO digital pin. In the horizontal, the last row that we can use as a ground for all the components and the second last row we can as a VCC (5v) for all the components. We power this VCC and ground from a 220v transformer. Which transforms 220v AC current to DC current or we also can use a 12v AC-DC adapter.

#### **4.4.2 Parking System**

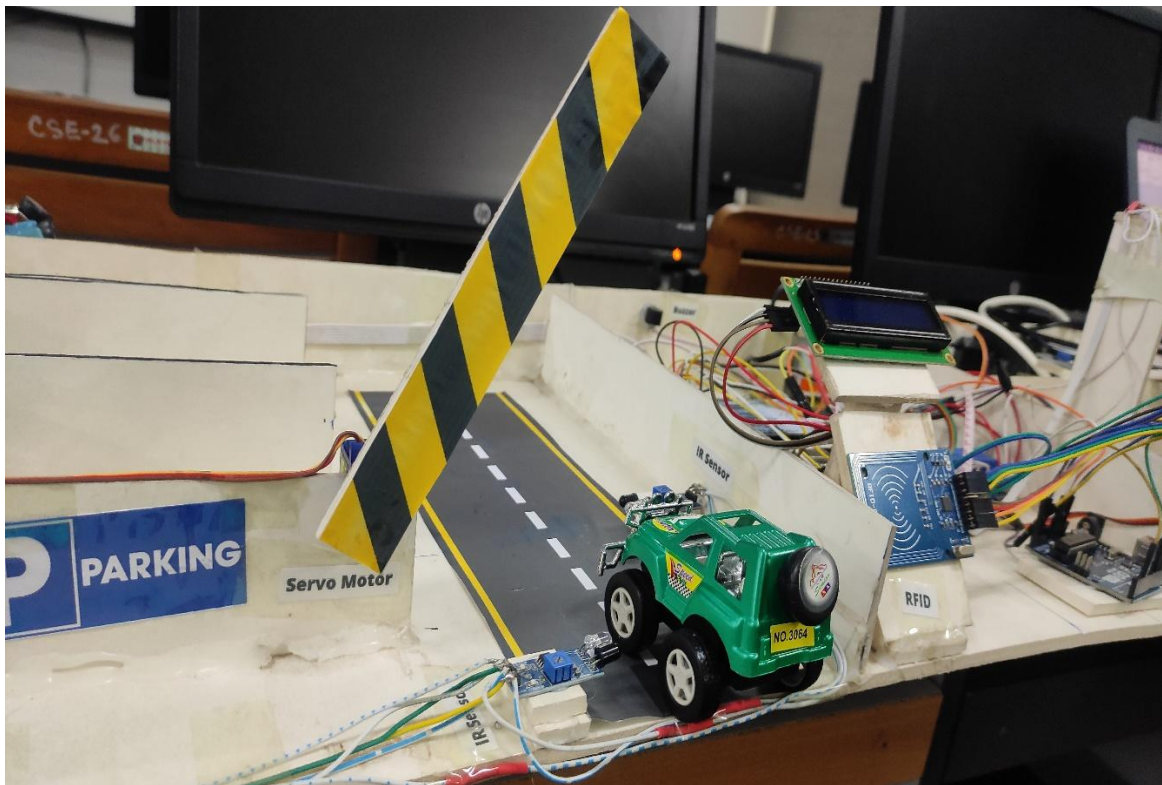
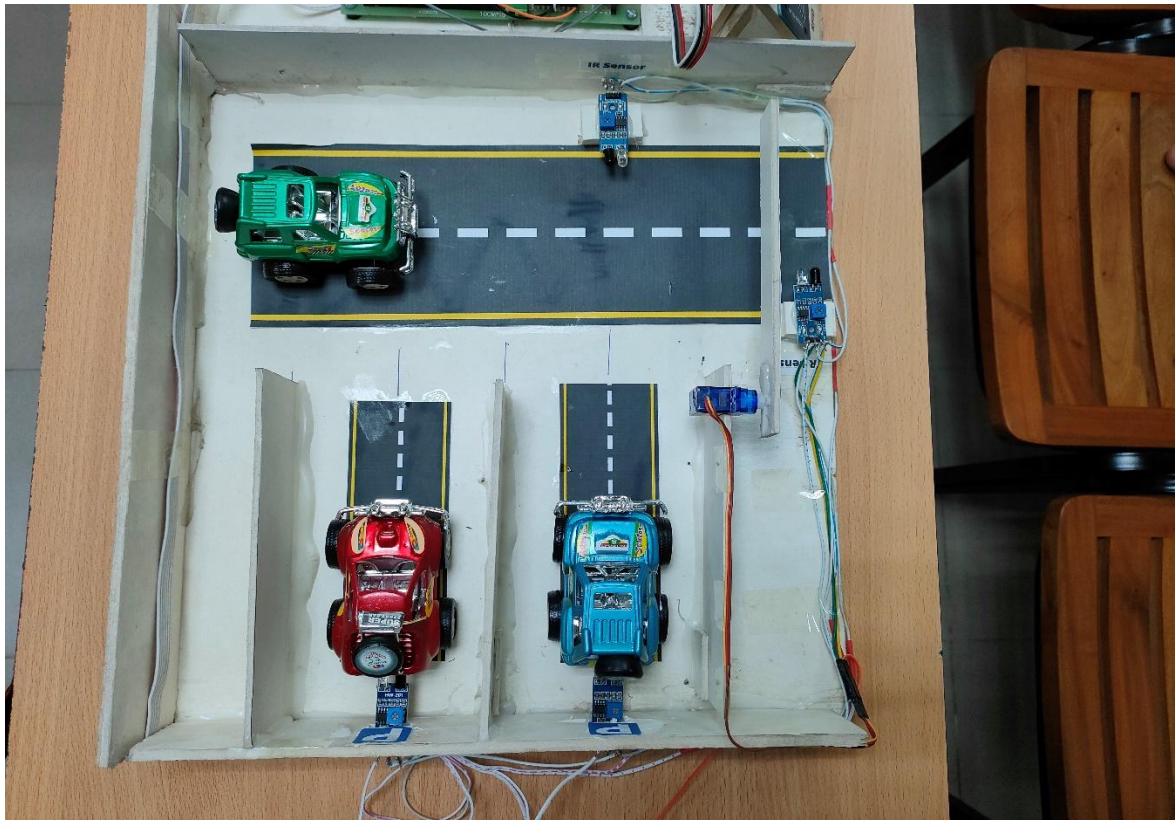
In the parking system, we have used four IR sensors. Two is for parking slots and the other is for opening and closing the parking gate. When a car come in front of the gate the gate will open. After entering the gate will close automatically. Users can park any free slot they want. Free space details will show on the LCD display which is located at the gate. YES for empty NO for full. If all the slots are full the gate will remain closed until the slots are empty.



**Figure 83: Circuit Diagram for Parking System**

This is the circuit diagram. We can see four IR sensors, LCD display, servo motor connected with Arduino UNO.

Some pictures of the car parking System.

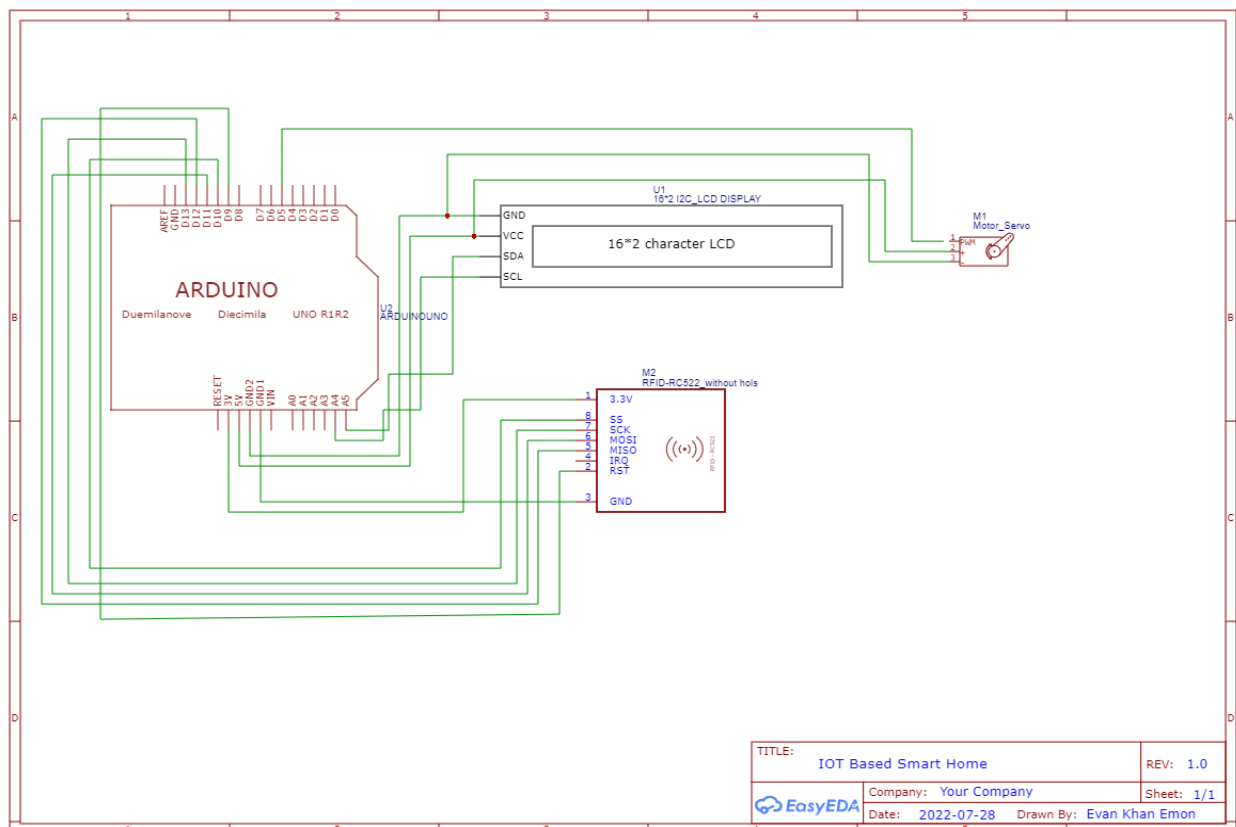




**Figure 84: Car Parking System.**

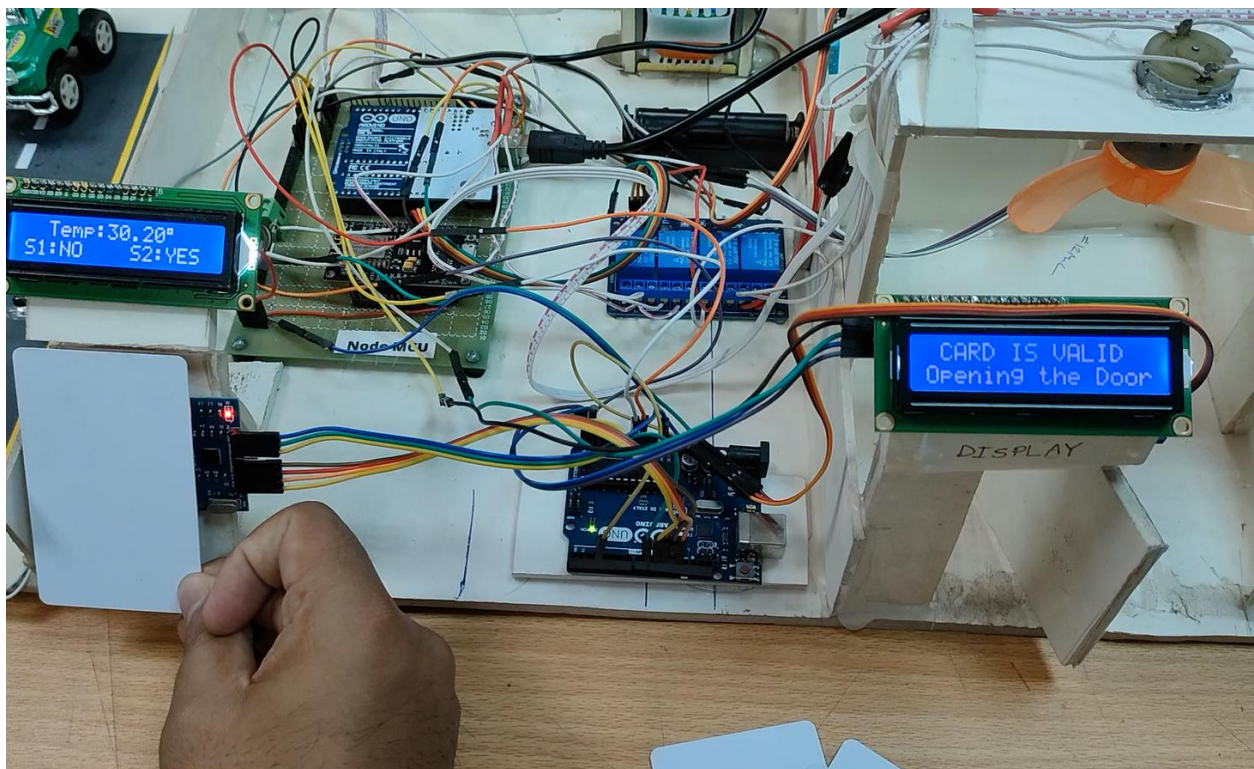
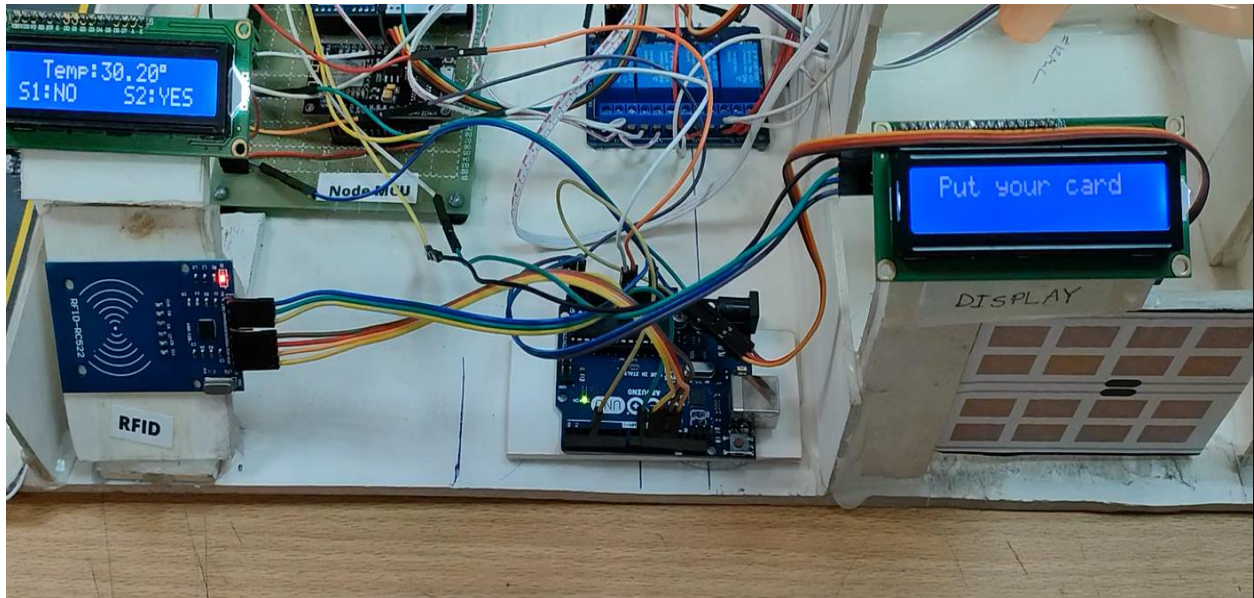
### 4.4.3 RFID Door Control

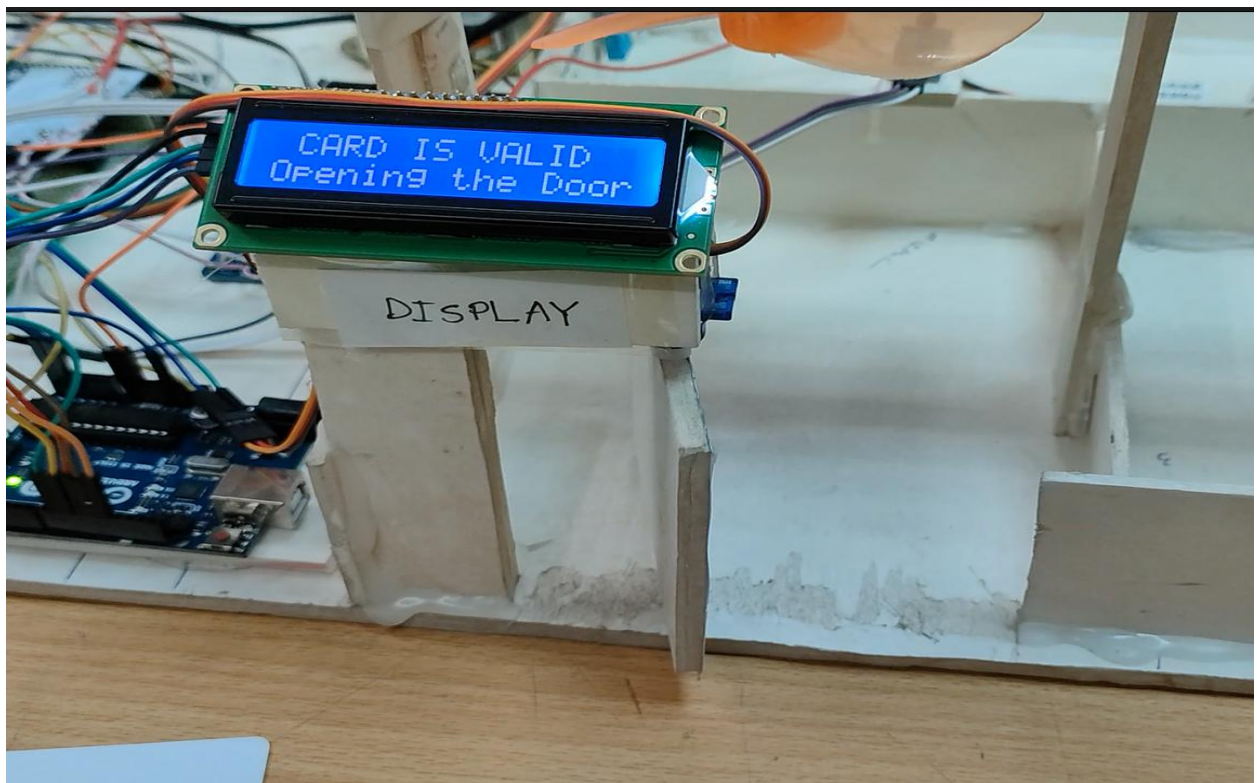
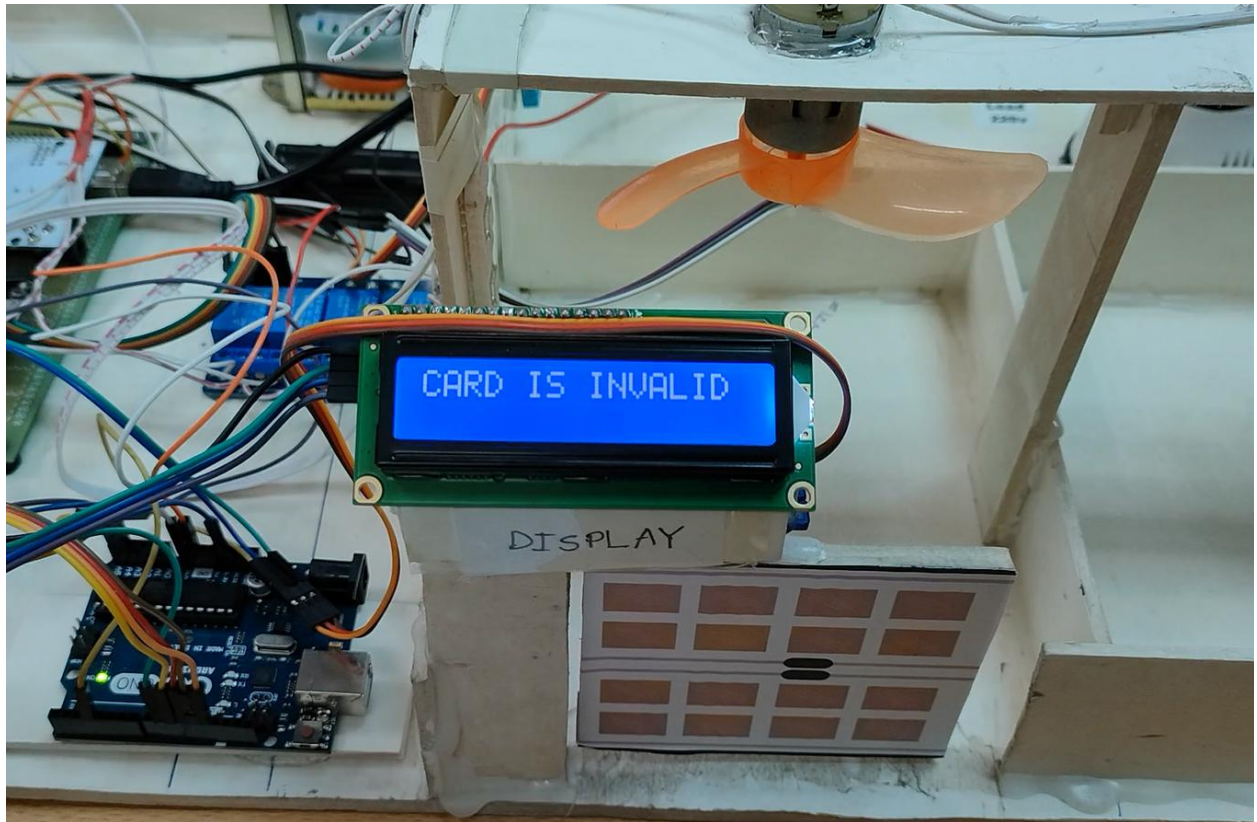
In our project, we focus on smart security. For this, we use an RFID card which is mainly used to open a door. Nowadays it's more popular. In our project, we use the RC-522 RFID card. To enter the house users need a valid card. When a user puts his/her card into the reader the gate will open automatically. So only a valid card holder can enter the house.



**Figure 85: Circuit Diagram for RFID**

Some pictures of the RFID door opener system.

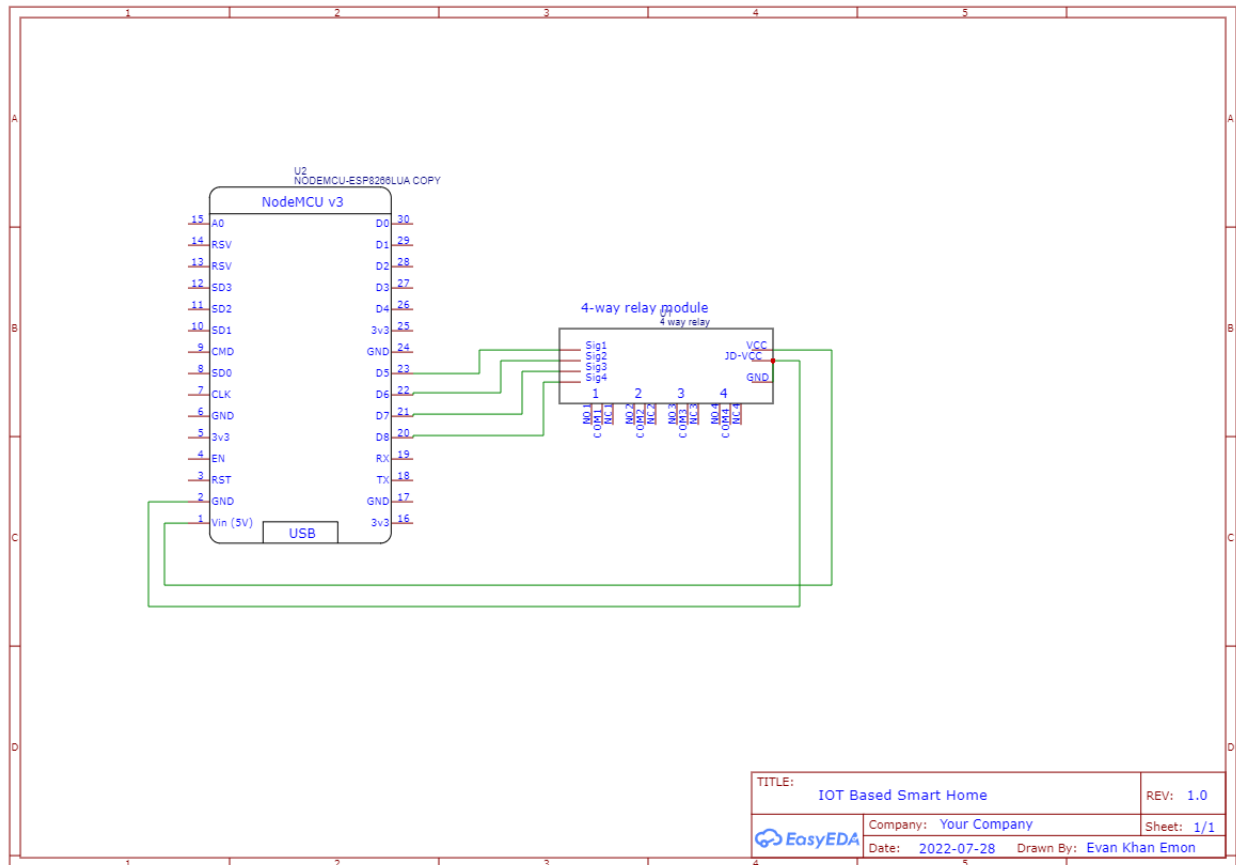




**Figure 86: RFID card Working**

## 4.4.4 Home Automation System

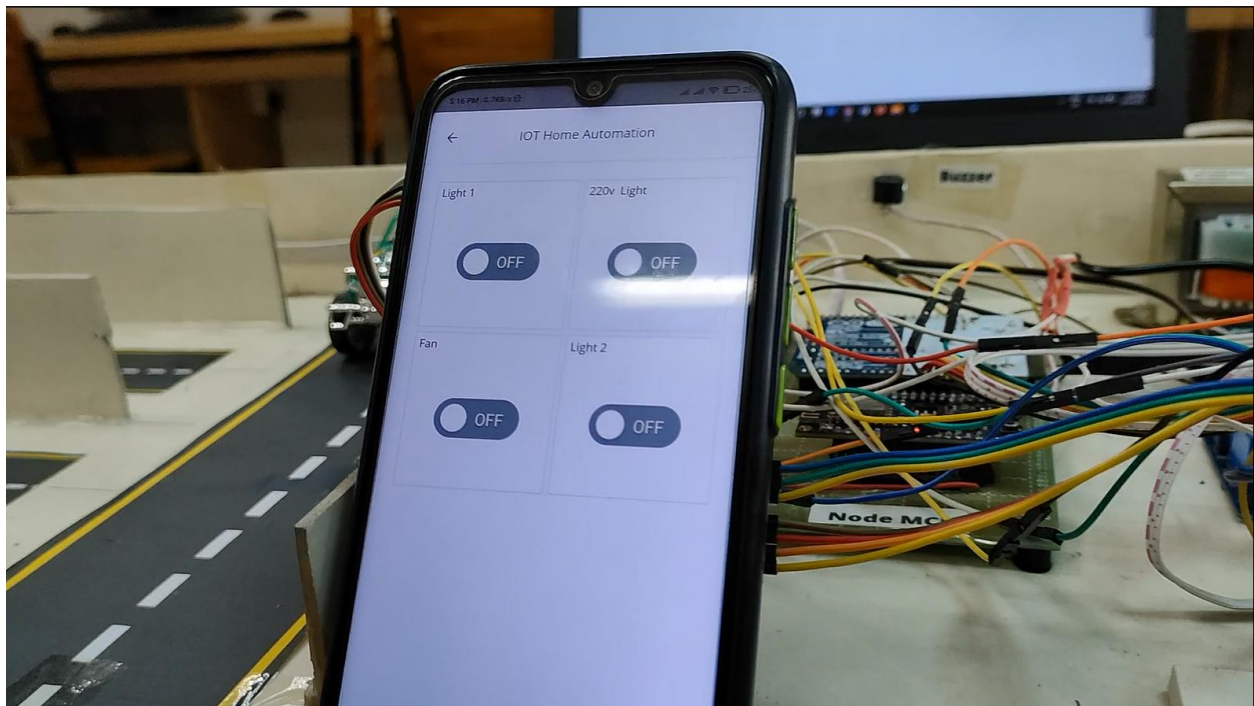
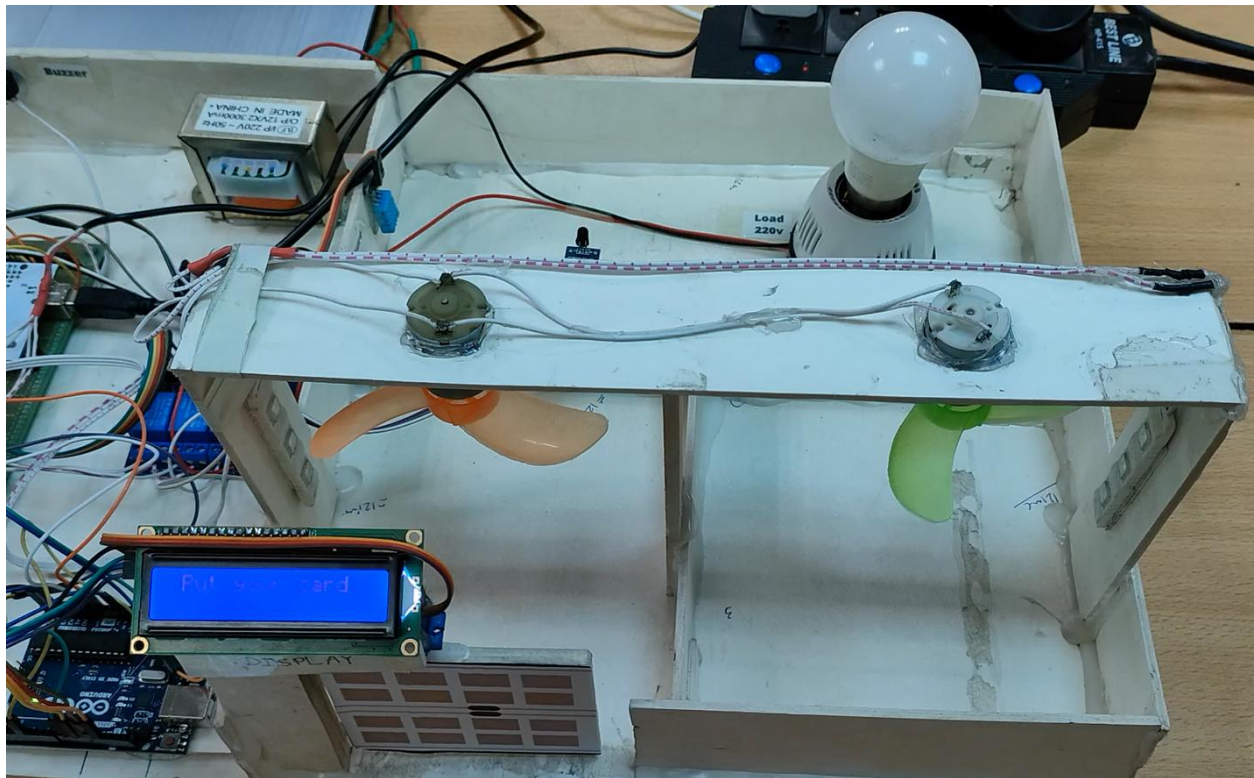
For home automation, we used a 4-way relay shield. Using this relay we can control 4 loads. In our project, we will be able to on-off light fan via web and mobile applications. We use Arduino IoT cloud to control the NODE MCU which is a wifi module. First, we have to create things in the IoT cloud and then upload the sketch to the NODE MCU.

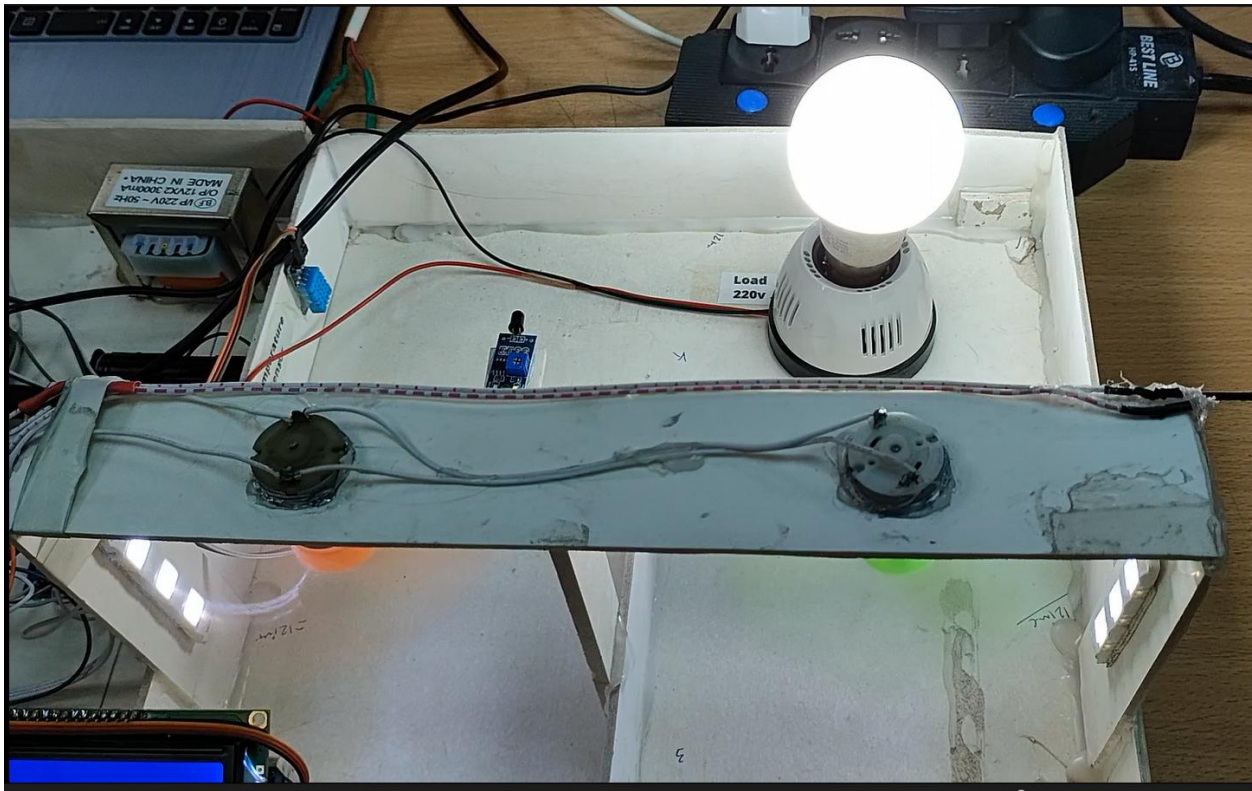


**Figure 87: Home Automation Diagram**

This is the circuit diagram for home automation. Here we take one Node MCU and one 4-way relay shield.

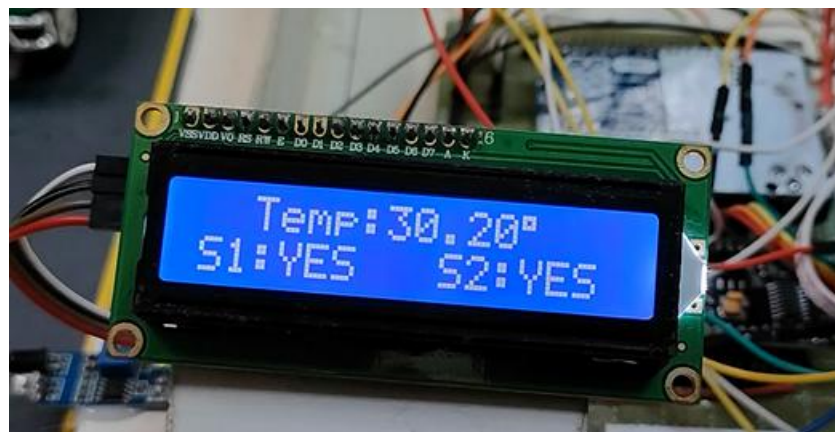
Some pictures of the Home Automation system.



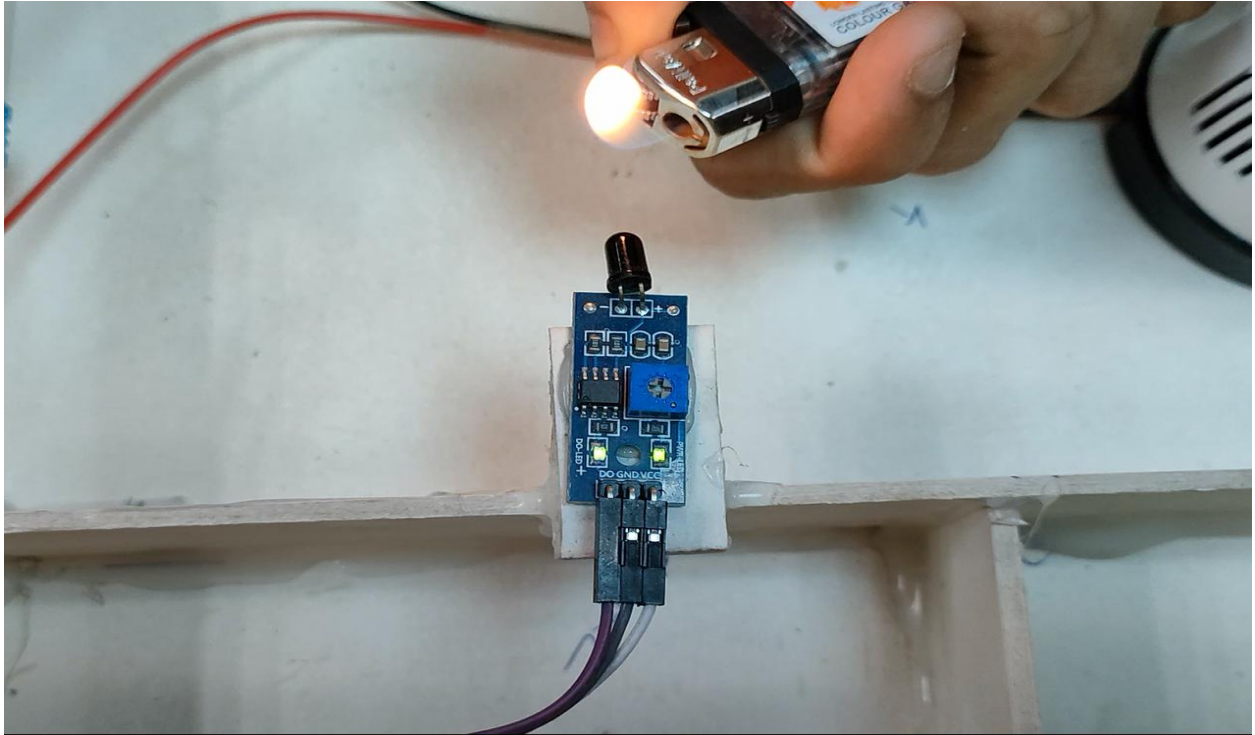


**Figure 88: Home automation**

We also use a temperature sensor to measure temperature and a flame sensor to detect the fire and get alarmed via a buzzer. We can see Temperature on the LCD screen. A0 pin used in Arduino UNO for temperature. A1, A2 these two pin is used for flame sensor and buzzer.



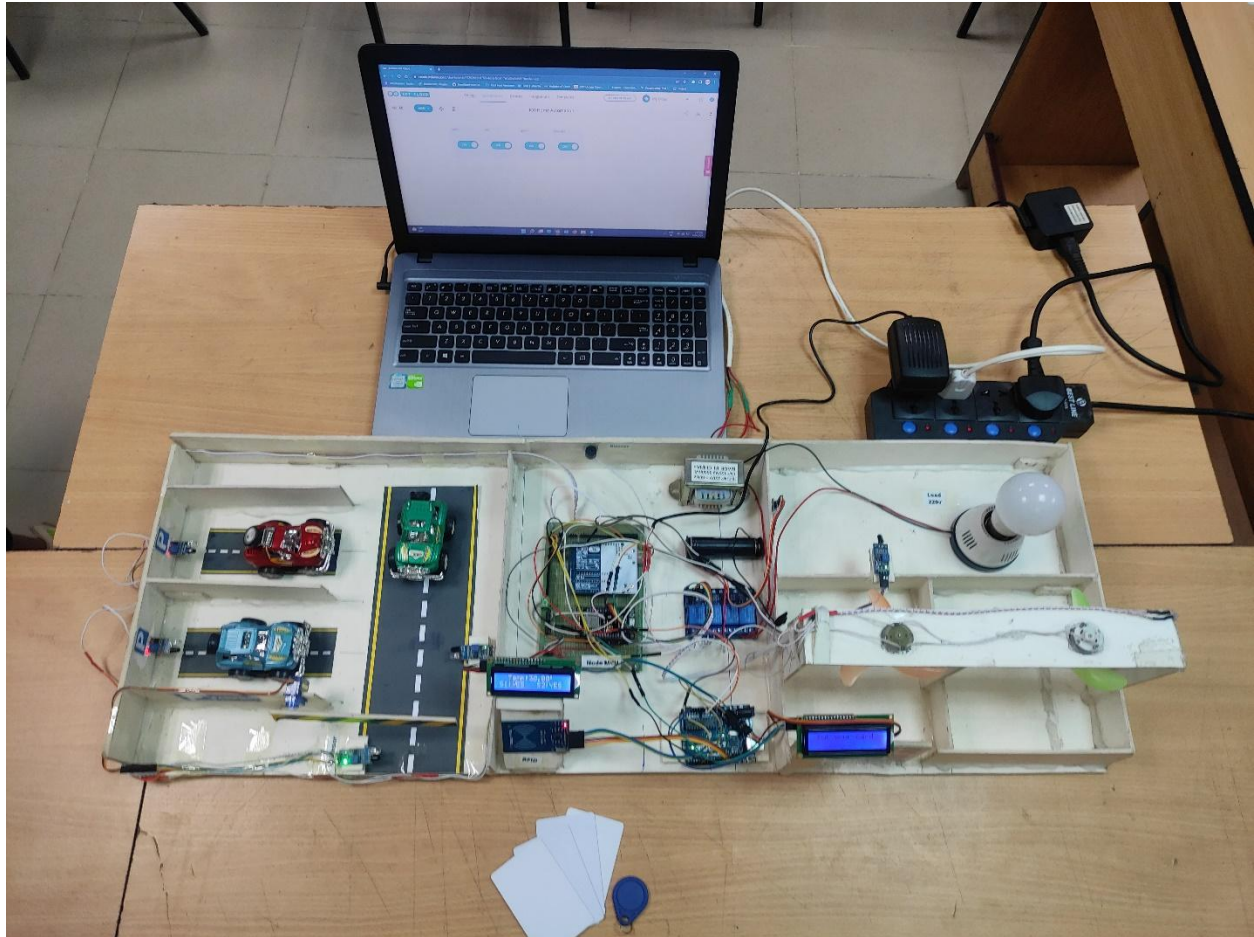
**Figure 89: Temperature Details**



**Figure 90: Flame Sensor and Buzzer**

## **Chapter 5: Result**

## 5.1 Result of the Project



**Figure 91: Full System Set up Picture (1)**

Managed to successfully apply the IOT BASED SMART HOME. It was user-friendly and cost-effective. User-friendly as in anyone can use just a click of a button on an android screen and everything works. And it is cost-effective as it will cost exactly as the project requires (optimum price).

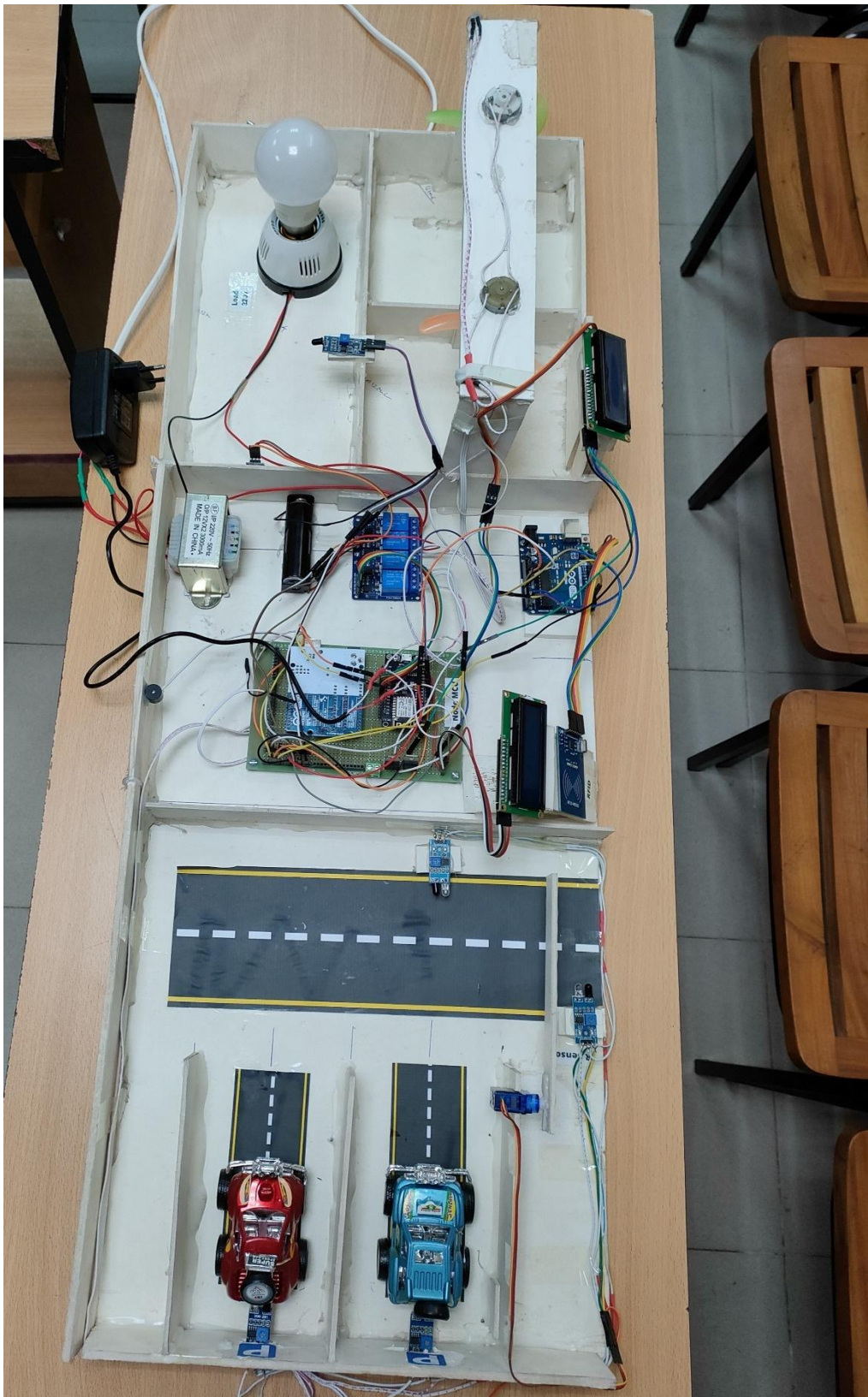


Figure 92: Full System Set up Picture (2)

## **Chapter 6: Conclusion and Future Work**

## 6.1 Conclusion

The journey of creating the IoT-Based Smart Home has been both challenging and rewarding. This project has provided invaluable practical experience in developing a system that integrates technology to improve daily life. The objectives set at the beginning were successfully achieved, including enhanced security features, efficient parking management, and user-friendly home automation controls. These innovations are cost-effective and align with the growing global trend of adopting Internet of Things (IoT) solutions.

The project allowed the team to delve into diverse areas such as hardware setup, software programming, and network communication. From designing the circuit board to configuring the Arduino IoT cloud and implementing the system in real-world scenarios, every step has been a learning opportunity. The final output, a smart home model, showcases several features: automated parking with real-time slot availability, RFID-based door access, and remote control of household appliances through a mobile app.

This system contributes significantly to making life more convenient and secure. The parking system addresses urban challenges like traffic congestion due to roadside parking. The RFID door access provides an additional layer of security by ensuring that only authorized individuals can enter. Moreover, the home automation system not only adds convenience but also promotes energy efficiency by preventing unnecessary power usage.

While the project has achieved its primary goals, it also laid the foundation for future developments and provided insights into the potential of IoT in everyday life. The team's efforts have resulted in a robust prototype that has the potential to be scaled for commercial and industrial applications.

## 6.2 Future Work

In future, we can add more features to our project. We can add more security to the parking system we can use the app to open the gate or car number tracking system. To open the gate we can use fingerprint sensors or keypad lock systems. We can add more facilities in home automation like we can give water in a garden. We can give food to the aquarium at a selected time. We can add a lot of features in future. The future of smart homes is bright and low-power, long-range, and cost-effective. IoT-powered devices are making this more likely to happen.

The current prototype offers significant functionality, but there is ample room for enhancements and additional features. Some of the possible improvements and future work include:

### **Enhanced Parking System:**

- Introduce license plate recognition using cameras to automate gate opening for registered vehicles.
- Develop an app-based parking management system to allow users to book slots in advance.
- Implement advanced sensors for real-time monitoring of parking slot availability with higher accuracy.

**Improved Home Security:**

- Integrate biometric systems such as fingerprint or facial recognition for door access, complementing the RFID system.
- Develop a mobile app with live security camera feeds to monitor home activity remotely.
- Add a panic alarm feature connected to emergency services in case of intrusion.

**Expanded Home Automation:**

- Introduce features for garden maintenance, such as automated watering systems controlled via the app.
- Implement smart kitchen appliances for scheduled tasks, like feeding fish in an aquarium or managing cooking devices.
- Add voice control capabilities by integrating virtual assistants like Alexa or Google Assistant.

**Energy Efficiency and Sustainability:**

- Use renewable energy sources, such as solar panels, to power the system.
- Introduce real-time energy consumption monitoring and alerts for high usage.
- Add automated blinds and windows that respond to temperature or time of day, reducing cooling or heating costs.

**Scalability and Integration:**

- Develop a modular design that allows easy integration of additional features without redesigning the entire system.
- Create a framework for integrating third-party IoT devices for more comprehensive smart home solutions.

**IoT Security Enhancements:**

- Implement end-to-end encryption to secure data transmitted between devices.
- Add multi-factor authentication for app access and control.
- Introduce periodic firmware updates to address emerging vulnerabilities.

**User Experience Improvements:**

- Redesign the mobile application for a more intuitive and visually appealing interface.
- Add detailed usage logs and analytics for users to track their energy savings and system usage.

**Commercial Viability:**

- Work on cost optimization to make the system affordable for a broader audience.
- Conduct pilot testing in real homes or small communities to gather user feedback.
- Develop a marketing strategy for scaling up production and distribution.

### **6.3 Closing Remarks**

The future of smart homes lies in leveraging IoT-powered devices to provide convenience, security, and efficiency. With rapid advancements in technology, this project can be transformed into a fully market-ready product. The team is optimistic about the possibilities and committed to contributing to the vision of smarter, more connected homes. This project has not only demonstrated the potential of IoT but has also encouraged further exploration into its endless applications.

# References

1. Nitu, A. M. ( 2019). Wireless home automation system using IoT and PaaS. *2019 1st International Conference on Advances in Science, Engineering and Robotics Technology (ICASERT)* (pp. 1–6). IEEE. doi:10.1109/ICASERT.2019.8934676
2. Orji, E. Z. (2018). Automatic access control system using Arduino and RFID. 5.4, pp. 333–340. *The Journal of Scientific and Engineering Research*. doi:05(04):333-340
3. Ramasamy, M. S. (2018). IoT-based smart parking system for large parking lot. *2018 IEEE 4th International Symposium in Robotics and Manufacturing Automation (ROMA)* (pp. 1–4). IEEE. doi:10.1109/ROMA46407.2018.8986731
4. Reddy, P. R. (2016). An IoT based Home Automation Using Android Application. *International Conference on Signal Processing, Communication, power & Embedded System*. Odissa. doi:10.1109/SCOPE.S.2016.7955836
5. Vakula, D. &. (2017). Low-cost smart parking system for smart cities. *2017 International Conference on Intelligent Sustainable Systems (ICISS)* (pp. 280–284). IEEE. doi:10.1109/ISS1.2017.8389415
6. Wibowo, F. W. (2017). A low-cost entry door using database based on RFID and microcontroller. *ARNP Journal of Engineering and Applied Sciences*, (pp. 5073–5077). Retrieved from [https://www.researchgate.net/publication/320134652\\_A\\_low-cost\\_entry\\_door\\_using\\_database\\_based\\_on\\_RFID\\_and\\_microcontroller](https://www.researchgate.net/publication/320134652_A_low-cost_entry_door_using_database_based_on_RFID_and_microcontroller)